

**A WEB-BASED ATTENDANCE SYSTEM WITH FACIAL RECOGNITION,
NOTIFICATION AND DATA REPORT FOR GRADE SCHOOL STUDENTS AT
SYSTEMS PLUS COMPUTER COLLEGE CALOOCAN**

**A Capstone Project
Presented to the Faculty of the
Information and Communications Technology Program STI College
Caloocan**

**In Partial Fulfilment
of the Requirements for the
Degree Bachelor of Science in Information Technology**

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OCTOBER 2025

ENDORSEMENT FORM FOR ORAL DEFENSE

TITLE OF RESEARCH: A Web-Based Attendance System with Facial Recognition, Notification and Data Report for Grade School Students at SYSTEMS PLUS COMPUTER COLLEGE CALOOCAN

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ABSTRACT

Title of Research: A WEB-BASED ATTENDANCE SYSTEM WITH FACIAL RECOGNITION, NOTIFICATIONS, AND DATA REPORT FOR GRADE SCHOOL STUDENTS AT SYSTEMS PLUS COMPUTER COLLEGE CALOOCAN

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This capstone project, "A Web-Based Attendance System with Facial Recognition, Notifications, and Data Report for Grade School Students at Systems Plus College Caloocan," addresses the inefficiencies of traditional, manual attendance tracking by introducing a secure and automated digital solution. To overcome the challenges of time-consuming roll calls, inaccurate records, and delayed parent communication, the researchers developed a comprehensive web-based system. The system utilizes client-side facial recognition via face-api.js and standard laptop webcams to automatically verify student presence, instantly classifying status as Present, Late, or Absent. It then triggers real-time email notifications to parents and provides dedicated portals for teachers and administrators. Developed using the MERN stack (MongoDB, Express.js, React, Node.js) and enhanced with Socket.IO for real-time updates, the system features a multi-role architecture. Teachers can monitor attendance live, generate exportable reports, and manage digitally submitted absence requests from parents. Parents, in turn, have a secure portal to view their child's attendance history and formally submit absence requests with supporting documentation. This integrated approach makes attendance tracking faster, contactless, and significantly more reliable, thereby reducing the administrative burden on teachers, fostering transparency between the school and home, and promoting a more accountable and efficient educational environment.

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List of Notations

Acronym	Description
SPCC	Systems Plus Computer College Caloocan
TAM	Technology Acceptance Model
PU	Perceived Usefulness
PEOU	Perceived Ease of Use
API	Application Programming Interface
UAT	User Acceptance Testing
MERN	MongoDB, Express.js, React, Node.js
VS Code	Visual Studio Code
Git	Version control system
Postman	API testing tool
MongoDB	NoSQL document database
Express.js	Web framework for Node.js
React	JavaScript library for user interfaces
React Router	Routing library for React applications
Material-UI (MUI)	React UI component framework
face-api.js	JavaScript library for facial recognition
react-webcam capture	React component for webcam
Node.js applications	JavaScript runtime for server-side
Mongoose	MongoDB ODM library for Node.js
Chart.js	JavaScript charting library
Socket.IO communication	JavaScript library for real-time
Nodemailer	Node.js module for sending emails
Multer	Middleware for handling file uploads

INTRODUCTION

Project Context

Many institutions, including the grade school department of Systems Plus Computer College Caloocan, continue to use antiquated attendance monitoring techniques like manual roll calls in an era where digital tools are revolutionizing how we work and learn. Although these conventional approaches have been used for many years, they introduce a number of inefficiencies that eventually affect the quality of education, including time loss, human error, and inaccurate records. Taking attendance takes up valuable class time, which detracts from teaching and student participation. Parents find it difficult to stay informed, and administrators encounter delays in handling attendance-related issues. These difficulties may jeopardize school decision-making and academic oversight. To address these issues, this capstone project introduces a Web-Based Attendance System with Facial Recognition, Notifications, and Data Reporting, designed specifically for grade school students. The system automates attendance tracking through facial recognition technology, using standard laptop webcams to verify each student's presence. The system classifies attendance as "Present," "Late," or "Absent" with accurate timestamps and sends real-time email notifications to parents or guardians, keeping them promptly informed.

The platform includes user-friendly portals for both teachers and parents. Teachers can monitor attendance in real time, review and approve absence requests with supporting documentation, and generate detailed reports for record-keeping. Parents can securely view their child's attendance history, submit formal absence requests, and access up-to-date information. The system even accommodates families with multiple children, ensuring ease of use across all students. All data is processed securely, with full compliance to privacy

regulations, including signed parental consent forms.

Accessible on any web-enabled device with internet access, this solution is designed for classroom use and aims to streamline school operations, improve communication, and create a more accountable, transparent environment for both students and parents. By embracing automation and facial recognition technology, the system not only enhances attendance tracking but also sets the stage for future digital innovations in education.

Purpose and Description

The Web-Based Attendance System with Facial Recognition, Notifications, and Data Report for Grade School Students at Systems Plus Computer College Caloocan aims to modernize the traditional attendance process by introducing a secure, automated solution. Designed specifically for classroom use, the system uses facial recognition technology via standard laptop webcams to accurately record student attendance with precise timestamps, automatically classifying students as Present, Late, or Absent while minimizing errors and preventing proxy attendance. Real-time notifications are sent to parents or guardians via email immediately upon attendance recording, and a comprehensive parent portal allows for viewing attendance history and submitting absence requests with document uploads. For teachers, the system provides dedicated portals for monitoring class attendance and generating exportable reports. As a web-based platform, it is accessible on any internet-connected device and intended for on-campus use, requiring a stable Wi-Fi connection and webcam-equipped devices in each classroom. This system reduces administrative workload, promotes transparency between school and home, and ensures reliable and efficient attendance tracking aligned with the school's digital transformation.

The Web-Based Attendance System with Facial Recognition, Notifications, and Data Report for Grade School Students at Systems Plus Computer College Caloocan aims to

modernize the traditional attendance process by introducing a secure, automated solution. Designed specifically for classroom use, the system uses facial recognition technology via standard laptop webcams to accurately record student attendance with precise timestamps, automatically classifying students as Present, Late, or Absent while minimizing errors and preventing proxy attendance. Real-time notifications are sent to parents or guardians via email immediately upon attendance recording, and a comprehensive parent portal allows for viewing attendance history and submitting absence requests with document uploads. For teachers, the system provides dedicated portals for monitoring class attendance and generating exportable reports. As a web-based platform, it is accessible on any internet-connected device and intended for on-campus use, requiring a stable Wi-Fi connection and webcam-equipped devices in each classroom. This system reduces administrative workload, promotes transparency between school and home, and ensures reliable efficient attendance tracking aligned with the school's digital transformation.

Statement of the Problem

Many educational institutions, including Systems Plus Computer College Caloocan, rely on manual methods for tracking student attendance. These traditional processes are inefficient, prone to errors, and fail to provide timely information to key stakeholders. There is a critical need for a more secure, automated, and intelligent system that can accurately record attendance, instantly communicate with parents, and provide reliable data for school administration, thereby fostering a more accountable and efficient educational environment.

General Problem

Systems Plus College Caloocan (SPCC) currently experiences inefficiencies, inaccuracies, and a lack of timely communication due to its manual attendance tracking process for grade school students.

At Systems Plus College Caloocan (SPCC), the school still uses manual methods to track attendance for grade school students. These traditional methods take up a lot of time, can lead to mistakes. As a result, teachers lose valuable instructional time, the administrative workload increases, attendance records can be inaccurate, and it delays addressing absenteeism. This all leads to problems with student accountability and keeps parents out of the loop.

Specific Problems

- 1. Traditional attendance tracking reduces both instructional time and overall classroom efficiency.**

Traditional attendance methods, such as roll calling or using sign-in sheets, can take up a significant amount of time, especially in classes with many students. Teachers often spend several minutes at the beginning of each lesson just checking who is present, which takes away from valuable learning time.

- 2. Attendance records are often inaccurate and unreliable due to human error and misuse of the traditional tracking process.**

Traditional attendance methods often result in errors, such as incorrectly marking students or losing records. In grade schools, these methods are also prone to misuse, with students sometimes marking attendance for absent classmates. Such issues lead to inaccurate records, making it challenging for teachers to track actual student presence and effectively manage their classrooms.

3. Lack of real-time communication between the school and parents regarding student attendance.

The school faces a significant challenge due to the lack of real-time communication with parents about student attendance. Without timely updates, the school struggles to engage parents promptly when issues like absenteeism or tardiness arise. This communication gap makes it difficult for the school to work closely with families in supporting students' consistent attendance and overall academic success.

General Objective

To develop A Web-Based Attendance System with Facial Recognition, Notification, and Data Report to address the inefficiencies, inaccuracies, and communication delays in the current attendance tracking process at Systems Plus College Caloocan (SPCC).

This system aims to address the challenges associated with traditional attendance methods such as roll calls, which are time-consuming, prone to errors, and susceptible to fraud. By integrating facial recognition technology, the system will provide a secure, accurate, and automated method for tracking student attendance. Additionally, it will include a real-time notification system that updates parents and guardians on their child's attendance, promoting better communication and accountability. This web-based system will streamline the attendance process, enhance efficiency, reduce administrative workload, and ensure that attendance records are always accurate and up to date. It ultimately aims to improve overall school operations and foster greater engagement between students, teachers, and parents.

Specific Objectives

- 1. To design a facial recognition feature that can accurately identify students during attendance checking**

The system will utilize facial recognition technology to accurately and securely verify student identities during attendance checks. This automation will eliminate the need for manual roll calls, which are often prone to human error and misuse. The module will ensure that only the registered student can mark their own attendance, improve the accuracy of the data, and reduce administrative burdens on teachers and staff. The system will also provide quick and efficient attendance records, saving valuable classroom time.

- 2. To design a structured reporting system that organizes attendance data with comprehensive time logs, attendance status classifications (present, late, absent).** This objective aims to establish a systematic approach for capturing and organizing attendance data. By incorporating detailed time logs and clearly defined attendance status categories, the system will provide a comprehensive view of student attendance. This supports efficient record-keeping and facilitates analysis for both administrators and parents.

- 3. To develop a system that will send attendance notifications to parents or guardians.**

The notification module sends real-time attendance updates to parents and guardians via email, informing them whether their child is marked present, late, or absent. This feature enhances communication between the school and home, ensuring parents are consistently informed and able to respond promptly when needed. By providing immediate notifications, the system promotes student accountability, supports better attendance habits, and fosters transparency and collaboration between the school and families.

Scope and Limitations

Scope

This system aims to digitize and automate the attendance-taking process using facial recognition and to notify parents or guardians using email. Designed specifically for grade school students at Systems Plus Computer College Caloocan, the system utilizes modern web technologies to improve efficiency, transparency, and data accuracy.

Back-end Features

1. Admin Features

User Management: Handles the creation, updating, and deletion of user accounts for students, teachers, and parents. Each parent account is securely linked to their child's profile. Also manages the upload and maintenance of facial recognition data for students.

Attendance and Report Generation: Manages all attendance records and allows the generation of attendance reports on daily, weekly, and monthly bases.

Notification System: Configures and sends attendance notifications via email to parents or guardians, keeping them informed in real- time about their child's attendance status. Also provides access to delivery logs for transparency.

1. Teacher Functions

Monitor Real-Time Attendance: Allows teachers to view live attendance updates for their assigned classes.

Attendance Record: Enables teachers to review past attendance records and generate reports for their students.

Generate Reports: Teachers can generate attendance reports and export them, for example in Excel format.

Absence Validation: Teachers can review and approve or decline absence requests submitted by parents, such as medical certificates, ensuring valid reasons are properly acknowledged.

2. Parent Functions

Secure Login and Child Access: Parents can log in through individual accounts tied to their child's profile. They only have access to their child's attendance data and related features.

Absence Request Submission: Parents can submit absence applications for specific dates and upload relevant documents such as medical certificates or excuse letters.

View Attendance Record: Parents can view their child's attendance records, including absences and the approval status of submitted absence requests.

Notification Log: Parents can view a history of attendance notifications received via email.

Facial Recognition Identity Verification: The system uses facial recognition technology to automatically record attendance, making the process faster, more accurate, and contactless. As a web-based system, this feature requires an active internet connection and functions without direct interaction from students

Front-End Features

1. Admin Portal

Dashboard: Provides an overview of system status including attendance statistics and notification summaries.

User Management: Admin can manage student, teacher, and parent accounts, and upload facial recognition data.

Report Generation: Enables the admin to generate and export attendance reports in various formats.

Email Configuration: Allows configuration of SMTP settings to send notifications via email to parents or guardians.

2. Teacher Portal

Dashboard: Displays real-time attendance information for teachers assigned classes.

Attendance Tracking: Enables teachers to view attendance records and generate reports for their students.

Absence Request Management: Teachers can view absence applications submitted by parents and approve or decline them based on validity.

3. Parent Portal

Login and Dashboard: Provides a secure login for parents to access a personalized dashboard linked to their child.

Submit Absence Requests: Parents can file absence applications and attach valid documentation (PDF, image, etc.).

View Attendance Record: Displays their child's attendance records and the status of absence submissions (e.g., pending, approved, declined).

Notification Log: View a timeline of received email alerts related to attendance events.

4. Student Integration

Facial Recognition Attendance: Attendance is automatically recorded through facial recognition technology, with no direct system interaction needed from students. This feature requires internet connectivity, as it relies on cloud-based processing and data synchronization.

5. Reports and Notifications

Attendance Reports: Both admins and teachers can generate detailed attendance reports and export them for analysis or record-keeping.

Automated Notifications: The system automatically sends real-time attendance notifications to parents or guardians via email, informing them when their child arrives at or leaves school. This feature ensures timely updates and helps strengthen communication and trust between the school and home.

Limitations

- **Internet Dependency on Notification Delivery (Email):** The system's ability to send notifications to parents or guardians via email depends on a stable internet connection. email notifications rely on correct SMTP settings. Any disruption in internet connectivity may delay or prevent these notifications from being sent.
- **Classroom Access Requirements:** Although the system is deployed as a web-based platform and can be accessed through any internet-connected device, it is intended for use within classrooms. Each classroom must have stable Wi-Fi connection and a compatible device (such as laptop, desktop, or smartphone with a webcam). Reliable internet access and proper equipment in each room are essential to ensure the smooth operation of facial recognition and accurate, real-time attendance logging.

- **Core Feature Internet Dependency:** Essential system features such as facial recognition, real-time data synchronization, and live access to attendance reports require an active internet connection. Without internet access, these features will be temporarily unavailable, limiting the system's full functionality.
- **Fallback to Traditional Attendance:** In the event of system downtime or internet outages, teachers can temporarily use traditional (manual) attendance recording methods. These records can be manually encoded into the system once normal operations resume.
- **Data Privacy and Security Limitations:** While the system complies with the Data Privacy Act and employs encryption and access controls, full data security cannot be fully guaranteed due to the wide scope of sensitive biometric data (e.g., facial recognition). Implementation requires signed parental consent forms (waivers), and security measures are continually monitored, but risks associated with large-scale data handling remain.
- **On-Campus Use Only:** The system is designed exclusively for on-campus use and does not support attendance tracking for online classes or remote learning. All attendance logging must be conducted within the school premises.

Significance of the Study

The proposed system provides a significant technological advancement in how attendance is managed within the academic environment of Systems Plus Computer College, particularly for grade school students.

For the School Administration:

The system provides administrators with a reliable, real-time platform for monitoring student attendance. It reduces paperwork, prevents manipulation of records, and offers data-driven insights that can support policy development and decision-making. Attendance trends can be easily analyzed through generated reports, enabling early detection of irregularities and absenteeism patterns.

For Teachers:

Teachers will benefit from a streamlined attendance process that minimizes class disruption. The automation through facial recognition reduces time spent on roll calls, allowing educators to focus more on instruction and student engagement. Teachers can also access organized records quickly, supporting more efficient classroom management and communication with parents.

For Parents/Guardians:

The real-time email notification feature ensures that parents are immediately informed about their child's attendance status. This promotes accountability, fosters transparency, and strengthens the partnership between the school and home. Parents are empowered to respond to issues proactively, such as when a child is unexpectedly absent.

For Students:

By integrating facial recognition, students are encouraged to attend regularly and on time, knowing their attendance is monitored securely and in real-time. This promotes responsibility and discipline. Additionally, the non-intrusive, contactless nature of facial recognition supports a safer, more hygienic environment particularly important in a post-pandemic setting.

For Future Developers and Researchers:

This study serves as a foundational framework for integrating biometric technology with web-based systems in the field of educational management. It may inspire future research or system enhancements, such as mobile-based access, multi-factor authentication, or integration with other school systems like grading and behavior monitoring.

Review of Related Literature/Studies/Systems

This chapter examines the essential aspects and potential challenges of a web-based attendance system that utilizes facial recognition and notifications data report for grade school students at Systems Plus Computer College Caloocan. It discusses the system's advantages and limitations while ensuring a clear understanding of its design and functionality. Furthermore, the chapter presents an overview of key concepts and findings, backed by relevant studies and expert literature in the field.

Foreign Literature

In **“Web-based University Classroom Attendance System Based on Deep Learning Face Recognition”**, Leng (2022) explores the implementation of facial recognition technology in university-level attendance systems, specifically using deep learning algorithms to improve the accuracy of facial identification. Leng's system operates on a client-server model with a web-based database that securely stores and processes facial data, which is then used to automate attendance marking. The study highlights how deep learning models, specifically pre-trained facial recognition algorithms, can optimize the efficiency and precision of attendance systems by reducing human error and fraud.

The relevance of Leng's research to our project, **“A Web-Based Attendance System with Facial Recognition, Notification, and Data Report for System Plus College Caloocan”**, is clear. Just like Leng's system, our proposed system at Systems Plus College will leverage facial recognition technology to automate the attendance process for grade school students, ensuring that attendance is recorded accurately and in real-time. Additionally, Leng's study demonstrates the importance of creating a secure, web-based environment for managing student data, which is essential for handling sensitive attendance records. The recommendation in Leng's work to collect facial data

under optimal conditions such as proper lighting provides valuable insights that can be applied to ensure accurate and efficient data collection for our own system. Furthermore, Leng's use of a server-client model to handle facial recognition data will inform the backend architecture of our web-based system, contributing to the overall design of our solution.

A Web-Based Mobile Fauzi (2021) introduces a **“Web-Based Mobile Attendance System with Facial Recognition Feature”**, which integrates mobile technology with facial recognition for automated attendance marking in educational settings. The study emphasizes the use of mobile devices, such as smartphones and tablets, to capture and process facial images, making the system more accessible and easier to implement in classrooms. Fauzi’s research showcases how mobile-based systems can reduce the administrative burden on teachers by automating the attendance process, while also improving accuracy by reducing human errors associated with traditional methods.

This study is directly connected to our project, “A Web-Based Attendance System with Facial Recognition, Notification, and Data Report for System Plus College Caloocan”, as it highlights the potential for using mobile technology to facilitate attendance marking. For grade school students at Systems Plus, a mobile-based system would be especially practical, as teachers and administrators can easily implement and manage it. By integrating facial recognition with mobile devices, our system will automate the attendance process, ensure real-time data updates, and provide teachers with more time to focus on classroom instruction rather than administrative tasks. Additionally, Fauzi’s mobile approach will allow parents to receive real-time attendance notifications via email, a feature that will be central to our project’s goal of improving communication between the school and parents.

Hegde (2021) discusses the widespread adoption of facial recognition technology for attendance management, particularly in educational and workplace settings. **“Face Recognition Based Attendance Management System”** explores the advantages of using facial recognition over traditional methods such as sign-in sheets or roll calls, which are prone to errors, time consumption, and potential fraud (e.g., proxy attendance). Hegde’s research emphasizes that facial recognition systems are non-invasive and contactless, making them especially beneficial in environments that prioritize hygiene, such as schools. For our capstone project, “A Web-Based Attendance System with Facial Recognition, Notification, and Data Report for System Plus College Caloocan”, this study is highly relevant because we are creating a system specifically designed to be hygienic, efficient, and accurate for grade school students. The non-contact nature of facial recognition is particularly advantageous in preventing the spread of germs, a consideration that is critical in educational environments. Hegde’s focus on automation aligns perfectly with our system’s goal of eliminating manual attendance taking, which is often prone to human error. Additionally, Hegde’s study stresses the importance of accurate real-time attendance records, which is a key feature we aim to incorporate into our system. We will use similar machine learning models to ensure the accuracy and speed of our facial recognition technology, especially for younger students whose appearance may change over time.

Mahmood (2023) examines a **“Class Attendance System Based on Face Recognition”** that uses a large, diverse dataset of facial images to train machine learning models for improved recognition accuracy in a classroom environment. Mahmood’s system is designed to handle various lighting conditions and student positions, which are common challenges in classroom settings. The study details the use of advanced algorithms, including Haar cascaded classifiers and the FaceNet network, to detect and recognize faces

in real-time. This approach ensures that attendance is recorded quickly and accurately, even in challenging conditions.

Mahmood's work is directly connected to our capstone project, "A Web-Based Attendance System with Facial Recognition, Notification, and Data Report for System Plus College Caloocan", because it highlights the importance of real-world testing and data collection to improve facial recognition accuracy. For our system at Systems Plus College, we will need to ensure that the facial recognition system can handle variations in lighting, student positioning, and environmental factors. Mahmood's insights on using a large dataset to train the system will inform the data collection and training processes for our facial recognition system, ensuring that it works effectively for grade school students. Moreover, his use of the FaceNet model for facial recognition will provide a strong foundation for our system's backend architecture, ensuring fast and reliable student identification.

Mohd Fauzi (2020) proposes a **"Mobile-based Attendance Monitoring System Using Face Tagging Technology"**, which combines mobile devices with facial recognition to automate attendance monitoring in educational institutions. This study highlights how mobile-based systems can streamline attendance tracking, reduce administrative workload, and prevent issues such as proxy attendance by accurately identifying students using facial recognition. The system uses mobile phones to capture student faces, automatically logging their attendance and generating real-time reports that can be accessed by teachers and administrators. This study is highly relevant to our capstone project, "A Web-Based Attendance System with Facial Recognition, Notification, and Data Report for System Plus College Caloocan", because it demonstrates the potential of using mobile technology to simplify attendance processes. As our system will also focus on grade school students, the use of smartphones and tablets to capture facial images will make it easier for teachers to

manage attendance without needing expensive or specialized equipment. Fauzi's use of mobile devices for face tagging will be applied to our system's design to ensure ease of use and accessibility. The real-time reporting feature will also be a central component of our system, allowing us to send attendance data to parents via email notifications, improving communication between school and home.

Local Studies and/or Systems

In **“Ilocos Norte School Adopts e-ID to Check Attendance,”** Adriano (2019) discusses the adoption of an electronic ID (e-ID) system in Badio Elementary School, Pinili, Ilocos Norte, to automate and modernize the student attendance process. The system assigns each student a unique barcode linked to their record, and upon arrival at school, students simply present their e-ID, which is scanned by the teacher to automatically register their attendance. This approach eliminates the need for traditional manual methods, offering a faster, more accurate, and efficient way of managing attendance. Additionally, the system automatically notifies parents via text message, while this study utilizes SMS, our proposed system will adapt this concept of real-time communication by using email, which is the preferred and more sustainable channel for our client institution, allowing for more detailed messages and easier integration with the web-based platform.

The relevance of Adriano's study to our project, **A Web-Based Attendance System with Facial Recognition, Notification, and Data Report for System Plus College Caloocan**, lies in its focus on automating attendance tracking and improving communication with parents. Like the e-ID system in Adriano's study, our proposed system also emphasizes real-time attendance tracking and the use of automated notifications, albeit through email. The automation of attendance logging in Adriano's system can be extended to our facial recognition-based solution, which offers a more secure and personalized method of

attendance tracking. Furthermore, the real-time parent notifications in Adriano's system are directly aligned with our project's goal of improving communication and engagement between the school and students' families.

The relevance of Adriano's study to our project, **A Web-Based Attendance System with In "Employee Attendance and Performance Monitoring System with Facial Recognition Technology and Data Analytics,"** Arreglado (2023) presents a comprehensive facial recognition system, FRaDA, used to automate faculty and staff attendance tracking at Talisay Elementary School. The system employs live facial recognition technology to scan and log attendance in real-time, eliminating the need for traditional timecards or manual entries. The system also provides detailed reports that track attendance trends, facilitating better administrative decision-making and performance monitoring. Furthermore, Arreglado highlights the use of data analytics to monitor trends in attendance, providing insights into faculty punctuality and overall performance. The relevance of Arreglado's system to our project is clear, especially with its use of facial recognition technology to automate the attendance process. Just like Arreglado's approach, our proposed system at Systems Plus College will leverage facial recognition to improve the accuracy and efficiency of attendance tracking for grade school students. Additionally, the use of data analytics in Arreglado's system could be applied to monitor student attendance trends, helping identify patterns such as chronic absenteeism. This aligns with our project's goal to not only automate attendance tracking but also provide valuable insights to administrators and parents regarding student behavior and attendance.

In **"Face Recognition Approach in the Observance of Faculty Attendance System with Voice Command,"** Dalisay (2021) introduces a paperless attendance system that combines facial recognition technology with voice command functionality to automate faculty

attendance tracking. The system scans faculty members' faces upon arrival and automatically logs their attendance, eliminating manual data entry and minimizing errors. Voice commands are also incorporated, allowing users to interact with the system without physical touchpoints, offering a hands-free and more user-friendly experience. The system also generates digital time records, which are easily accessible for administrative use, ensuring efficient and accurate record-keeping. The relevance of Dalisay's study to our project lies in its integration of facial recognition for automated attendance logging, which directly mirrors the core functionality of our proposed Web-Based Attendance System for grade school students at Systems Plus College. Additionally, the voice command functionality demonstrated in Dalisay's system could be a valuable addition to our system's user interface in the future, particularly for enhancing the experience of school staff and administrators. This hands-free feature can further streamline interactions with the system, improving usability for both students and school personnel. In **“Public High School Web-Based Attendance Monitoring System with SMS Notification,”** Maligang (2021) discusses the implementation of a web-based attendance system at Estancia National High School, designed to automate attendance tracking and improve communication with parents. The system enables teachers to monitor student attendance through an online platform, which updates attendance records in real-time. One of the key features is the integration of SMS notifications, which automatically inform parents when their child is absent from school. This real-time communication helps ensure that parents are kept up to date with their child's attendance, promoting collaboration between the school and families. In **“Public High School Web-Based Attendance Monitoring System with SMS Notification,”** Maligang (2021) discusses the implementation of a web-based attendance

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In **“Mobile-based Attendance Monitoring System Using Face Tagging Technology,”**

Cabrillas (2021) presents a mobile-based solution that enables teachers to capture and authenticate student attendance through facial recognition using smartphones. The system uses face tagging technology to scan students’ faces and log their attendance automatically. The system also generates attendance reports in real-time, which are accessible to both teachers and administrators. Cabrillas emphasizes the advantages of mobile attendance tracking, such as portability, ease of use, and the ability to operate without specialized equipment. However, the system also acknowledges that factors like camera quality and lighting conditions can impact the accuracy of face recognition.

Cabrillas’ mobile-based system is particularly relevant to our project because it highlights the use of facial recognition for automated attendance tracking in an educational setting. While Cabrillas’ system utilizes smartphones for attendance capture, our proposed Web-Based Attendance System for grade school students at Systems Plus College will use a more centralized, web-based platform for attendance tracking. Nevertheless, Cabrillas’ emphasis on the need for proper lighting and camera quality provides valuable insights into the technical considerations for ensuring accurate facial recognition in our system. This connection will be important for ensuring the reliability and effectiveness of facial recognition data capture in varying classroom environments.

METHODOLOGY

Technical Background

This chapter presents an in-depth discussion of the system's design and implementation, along with its key functionalities. It provides a comprehensive overview of the system architecture and the various technologies integrated to develop a robust, scalable, and efficient Web-Based Attendance System with Facial Recognition for Systems Plus Computer College Caloocan.

Technologies to be Used in the System

To ensure optimal system performance and user experience, the proponents utilized the following technologies, frameworks, and libraries for the proposed system:

Frontend Technologies:

- **React** - A powerful JavaScript library for building dynamic and responsive user interfaces. React's component-based architecture allows for the creation of reusable UI elements, making the application modular, maintainable, and efficient for both the admin and parent portals.
- **React Router** - A standard routing library for React that enables navigation between different components and pages in a single-page application (SPA) without refreshing the browser, providing a seamless user experience.
- **Material-UI (MUI)** - A popular React UI framework that implements Google's Material Design. It provides a comprehensive collection of pre-built, customizable components such as buttons, dialogs, and menus, ensuring a consistent, modern, and accessible interface.
- **face-api.js** - A JavaScript library for face detection and recognition directly in the browser. It enables real-time face detection from the webcam feed, allowing for contactless and

automated student identification during attendance logging.

- **React-Webcam** - A React component that provides access to the user's camera. It is integrated with the face-api.js library to capture live video streams for facial recognition processing.
- **Socket.IO-client** - The client-side library of Socket.IO that enables real-time, bidirectional communication between the web client and the server. It is used to instantly update attendance records and notifications across all connected devices.
- **Chart.js** - A flexible JavaScript charting library that is used to visualize attendance data through various chart types, such as bar charts and line graphs, on the administrator and teacher dashboards for easy analysis of attendance trends.

Backend Technologies:

- **Node.js** - A JavaScript runtime built on Chrome's V8 engine that allows for the development of scalable network applications. It is used to build the server-side logic of the system, handling multiple concurrent client requests efficiently.
- **Express.js** - A minimal and flexible web application framework for Node.js that provides a robust set of features for building web and mobile applications. It is used to create the RESTful API endpoints that handle data processing, user authentication, and business logic.

Backend Technologies:

- **MongoDB** - A document-oriented NoSQL database program that uses JSON-like documents with optional schemas. It is chosen for its flexibility and scalability in storing diverse data types, including user profiles, attendance records, and facial encoding data.
- **Mongoose** - An Object Data Modeling (ODM) library for MongoDB and Node.js. It

provides a straightforward, schema-based solution to model application data, manage relationships between data, and enforce validation at the application layer.

- **Nodemailer** - A module for Node.js applications that allows easy email sending. It is integrated to handle the automated dispatch of real-time email notifications to parents and guardians regarding their child's attendance status.
- **Socket.IO** - The server-side library that enables real-time, event-based communication between the server and clients. It works in tandem with the client-side library to push immediate updates for attendance logging and notification alerts.
- **Multer** - A middleware for handling multipart/form-data in Node.js, which is primarily used for uploading files. It manages the secure upload of documents, such as medical certificates or excuse letters, submitted with absence requests.

Development Tools:

- **Visual Studio Code** - A lightweight but powerful source code editor that supports debugging, task running, and version control. It provides excellent support for JavaScript and Node.js development, enhancing developer productivity.
- **Git** - A distributed version control system used to track changes in the source code during development, allowing for collaboration among team members and maintaining a history of project evolution.
- **Postman** - An API platform for building and testing APIs. It was used to test all backend API endpoints to ensure they function correctly and return the expected responses before integration with the frontend.

Calendar of Activities

MONTH	JANUARY	FEBRUARY	MARCH	APRIL	MAY	JUNE	JULY	AUGUST	SEPTEMBER	OCTOBER	NOVEMBER
ACTIVITY											
Introduction Phase											
Group Creation/ Finding Adviser											
Planning											
Title Proposal/Revision											
Finding Client											
RRL Gathering											
Writing & Revision Chapter 1											
Creating Questionnaires /Client Interview											
Writing & Revision Chapter 2											
Writing & Revision Chapter 3											
Wireframing											
System Prototyping											
System Development											

Resources

Hardware Specifications

Development & Server Hardware:

- **Processor:** Intel Core i5 (or equivalent AMD Ryzen) or better.
 - **RAM:** 8 GB (minimum).
 - **Storage:** 256 GB SSD (or larger).
 - **Graphics Card:** Integrated graphics are sufficient for basic facial recognition model processing.
 - **Operating System:** Windows 10 or 11, macOS, or Linux distribution.
- #### Classroom Hardware:

- **Device:** Laptops or desktop computers with a modern web browser (Google Chrome recommended).
- **Camera:** Built-in or external HD webcam (720p or 1080p) for facial recognition capture.
- **Network:** A stable, high-speed internet connection to support real-time data processing and communication.

Software & Technologies

Frontend Development:

- **React:** A JavaScript library for building dynamic and responsive user interfaces.
- **React Router:** For handling navigation within the single-page application.
- **Material-UI (MUI):** For pre-built, customizable React UI components and styling.
- **face-api.js:** A JavaScript API for client-side face detection and recognition.

- **React-Webcam:** A React component to capture live video stream from the webcam.
- **Socket.IO-client:** Enables real-time, bidirectional communication with the server.
- **Chart.js:** For data visualization and generating attendance charts and graphs.

-

Backend Development:

- **Node.js:** A JavaScript runtime for building scalable server-side applications.
- **Express.js:** A web application framework for Node.js used to create RESTful APIs.
- **MongoDB:** A NoSQL document database for storing user profiles, attendance records, and system data.
- **Mongoose:** An Object Data Modeling (ODM) library for MongoDB and Node.js.
- **Nodemailer:** A module for Node.js to send automated email notifications.
- **Socket.IO:** Enables real-time, bidirectional event-based communication on the server.
- **Multer:** A middleware for handling file uploads, such as documents for absence requests.

Development Tools:

- **Visual Studio Code:** The primary source code editor.
- **Git:** A version control system for tracking changes and collaborating.
- **Postman:** An API platform for testing backend endpoints.
- **Third-Party Services**
 - **Cloud Database:** MongoDB Atlas for cloud-based database hosting.
 - **Email Service:** A transactional email service (e.g., Gmail SMTP, SendGrid) for Nodemailer.
- **Hosting/Deployment:** A platform for hosting the frontend (e.g., Netlify, Vercel) and the backend (e.g., Heroku, AWS).

Requirements Analysis

The People Involved

These are the key individuals and entities involved in the development and use of the Web-Based Attendance System:

- **Administrators** – Manage all system users (students, teachers, parents), configure system settings, upload facial data, and oversee platform operations via the admin portal.
- **Teachers** – Monitor real-time student attendance for their assigned classes, review and approve absence requests, and generate attendance reports via the teacher portal.
- **Parents/Guardians** – Receive real-time email notifications of their child's attendance, view attendance history, and submit absence requests with supporting documentation via the parent portal.
- **Students** – Interact with the system passively through facial recognition check-ins using classroom webcams, with no direct login required.

The System Activity

The Web-Based Attendance System functions as an automated platform for student attendance tracking, parental notification, and attendance data reporting. It uses facial recognition technology to automatically verify and record student attendance with precise timestamps, then immediately sends email notifications to parents indicating their child's status (Present, Late, or Absent). The system provides comprehensive reporting tools for teachers and administrators to monitor attendance patterns and generate exportable reports.

The Timing

The system operates continuously during school hours, automatically recording attendance as students enter classrooms. Real-time email notifications are sent to parents immediately after each attendance recording. Attendance reports can be generated by teachers and administrators at any time, with data available for current and historical periods. The system is designed for immediate implementation following the completion of development and testing phases.

The Environment Where the System Works

The system is implemented at Systems Plus Computer College Caloocan, specifically within the grade school department:

- **Administrators and Teachers** access the system through web browsers on school computers or personal devices with internet connectivity.
- **Parents/Guardians** access the system through web browsers on any internet-connected device.
- **Students** interact with the system through classroom-based devices equipped with webcams for facial recognition.
- All attendance recording occurs within classroom environments during school hours.

The System

Administrators use the web portal to manage user accounts, system configuration, and generate institutional reports. Teachers use the web portal to monitor class attendance in real-time, review and approve absence requests, and generate class-specific attendance reports. Parents use the web portal to view their children's attendance history and submit formal absence requests. The facial recognition system automatically captures and verifies student identities, while the notification system instantly communicates attendance status

to parents via email. All attendance data is securely stored and organized for comprehensive reporting and analysis. Administrators use the web portal to manage user accounts, system configuration, and generate institutional reports. Teachers use the web portal to monitor class attendance in real-time, review and approve absence requests, and generate class-specific attendance reports. Parents use the web portal to view their children's attendance history and submit formal absence requests. The facial recognition system automatically captures and verifies student identities, while the notification system instantly communicates attendance status to parents via email. All attendance data is securely stored and organized for comprehensive reporting and analysis.

Requirements Documentation

This document establishes the formal agreement between the client (Systems Plus Computer College Caloocan) and the developers regarding the software product's functionality, performance, and technical specifications. These requirements were refined iteratively throughout the Agile development process.

1. Functional Requirements

Authentication and Access Control

Role-Based User Sign-In

- Authorized users (Administrators, Teachers, Parents) must securely sign in to their respective portals using unique credentials. The system must implement password hashing and token-based session management to ensure data protection and prevent unauthorized access, with each role redirecting to a specialized dashboard.

Student Features

Passive Facial Recognition Check-In

- Students must be able to check in for attendance passively through a classroom webcam. The system must use facial recognition to automatically identify the student, record their attendance with a timestamp, and classify their status as "Present," "Late," or "Absent" without requiring direct student interaction with a device.

Parent/Guardian Features

Real-Time Email Notifications

- The system must automatically send an email notification to the parent or guardian immediately after their child's attendance status is recorded. The notification must clearly state the student's name, date, time, and attendance status.

Attendance History View

- Parents must be able to log in to a secure portal to view a history of their child's attendance records, including dates, times, and statuses.

Absence Request Submission

- Parents must be able to submit a formal absence request for their child through the parent portal. The system must allow them to specify the date(s) and reason for absence, and provide functionality to upload supporting documents (e.g., medical certificates).

Teacher Features

Real-Time Attendance Monitoring

- Teachers must be able to view a live dashboard showing the attendance status (Present, Late, Absent) of all students in their currently active classes.

Attendance Report Generation

- Teachers must be able to generate and export detailed attendance reports for their classes. The system must allow filtering by date range and class section, and support export to Excel format.

Absence Request Management

- Teachers must be able to view a list of absence requests submitted by parents for students in their classes. The system must provide an interface to approve or decline these requests, with the status updated for the parent to see.

Administrator Features

Comprehensive User Management

- Administrators must have full rights to create, read, update, and disable user accounts for students, teachers, and parents. This includes linking parent accounts to their respective children.

System Configuration

- Administrators must be able to configure system-wide settings, including the facial recognition model parameters and the SMTP settings for the email notification system.

Master Report Generation

- Administrators must be able to generate and export attendance reports across all classes, grade levels, and time periods for administrative analysis.

2. Non-Functional Requirements

- **Performance:** The facial recognition process must identify a student and record attendance within seconds. The system must support concurrent attendance logging across multiple classrooms without significant latency.
- **Security:** All data, especially biometric facial encodings, must be encrypted both in transit and at rest. The system must comply with the Data Privacy Act of 2012 (RA 10173). Access to data must be strictly controlled by user roles.
- **Usability:** The user interface for Teachers and Parents must be intuitive and require minimal training. The facial recognition process for students must be completely effortless.

- **Reliability:** The system must maintain high uptime during school hours. A fallback procedure for manual attendance logging must be established for use during system or internet outages.
- **Compatibility:** The web application must be fully functional on modern browsers (Chrome, Firefox) and accessible on various devices, including laptops, tablets, and smartphones.
- **Scalability:** The system architecture must be capable of accommodating an increasing number of students and classes without a degradation in performance.

3. Technical Requirements

- **Front-End Technologies:**
 - React with React Router for building a dynamic and navigable single-page application.
 - Material-UI (MUI) for a consistent, modern, and responsive user interface.
 - face-api.js integrated with react-webcam for client-side facial recognition processing.
 - Chart.js for data visualization in reports.
 - Socket.IO-client for receiving real-time updates on the teacher dashboard.
- **Back-End Technologies:**
 - Node.js with Express.js framework to build a scalable server and RESTful API.
 - Socket.IO for enabling real-time, bidirectional communication between the server and clients (e.g., live attendance updates).
- **Database:**
 - MongoDB with Mongoose ODM for flexible and scalable data storage, handling user profiles, facial encodings, attendance records, and notifications.
- **Third-Party Services & APIs:**
 - Nodemailer for constructing and sending real-time email notifications via an SMTP service (e.g., Gmail, SendGrid).

- Multer middleware for handling file uploads of absence request documents.

Development Tools & Methodology:

- Agile Methodology was employed, with requirements being prioritized in a product backlog and implemented over iterative two-week sprints.
- Visual Studio Code as the primary development environment.
- Git for version control and team collaboration.
- Postman for API testing and development.

4. Development Methodology

- The development of the Web-Based Attendance System followed the Agile methodology, enabling iterative progress, continuous stakeholder feedback, and adaptive planning throughout the project lifecycle.

Theoretical Framework

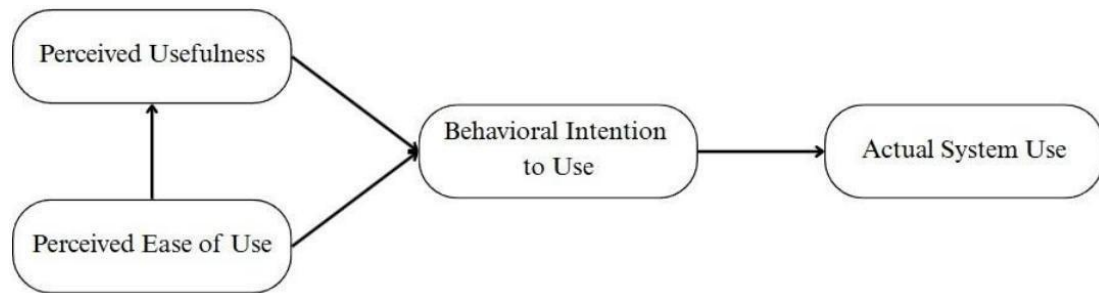


Figure 1. TAM Theory

The Technology Acceptance Model (TAM) serves as the theoretical foundation for the development of the Web-Based Attendance System with Facial Recognition. TAM explains how users adopt new technology based on two key beliefs: Perceived Usefulness (PU) and Perceived Ease of Use (PEOU). This framework is ideal for evaluating the acceptance of an automated attendance system among teachers and parents of grade school students at Systems Plus Computer College Caloocan, as it highlights the importance of both performance benefits and usability in successful technology adoption.

1. Perceived Usefulness (PU) refers to the extent to which users believe that using a system will improve their performance in specific tasks. For teachers, the Web-Based Attendance System addresses the inefficiency of manual roll calls by automating the entire process, thereby restoring valuable instructional time each day. The system enhances their performance by providing instant, accurate attendance records and enabling the generation of detailed reports with a single click, which simplifies academic monitoring and reporting. For parents, the system's real-time email notifications provide immediate knowledge of their child's school attendance, allowing them to respond proactively to any issues like

unexpected absences or tardiness. This empowers them to better support their child's punctuality and academic consistency, thereby establishing the system's overall usefulness in fostering a collaborative and accountable educational environment.

2. Perceived Ease of Use (PEOU), on the other hand, refers to the degree to which users believe that a system is simple and effortless to operate. The Web-Based Attendance System is intentionally designed to be intuitive and accessible. For students, the process is entirely passive and requires no effort; they simply look at a classroom webcam for a moment to be marked present. For teachers, the platform features a clean, straightforward dashboard that displays student attendance status at a glance. Generating reports or reviewing absence requests is designed to be a simple, guided process with clear options and filters. For parents, the system is accessed through a standard web browser, and the portal presents a clear view of their child's attendance history. Submitting an absence request is as simple as filling out a single form and uploading a file. This focus on simplicity for all user roles reduces the learning curve, enhances confidence, and encourages consistent use of the platform.

Agile Methodology

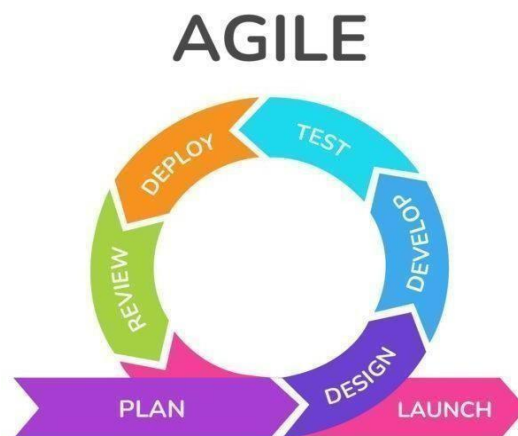


Figure 2. Agile Methodology

Phase 1 – Requirements

The first phase under the Agile methodology focused on gathering and prioritizing user needs through collaborative engagements with stakeholders from Systems Plus Computer College Caloocan. Through interviews and consultations with administrators, teachers, and parents, the researchers identified key challenges such as time-consuming manual roll calls, inaccurate attendance records, and delayed communication with parents. From these findings, user stories were developed to represent the needs of administrators, teachers, and parents. Functional requirements were defined, including facial recognition for automated attendance logging, real-time email notifications to parents, digital attendance reporting, and an online absence request system. This phase concluded with the creation of a product backlog that outlined essential system features for development.

Phase 2 – Design

During the design phase, the researchers created the system's overall layout and interaction flow. Wireframes and user interface mockups were developed for the web application, emphasizing user-friendly navigation and role-based dashboards for administrators, teachers, and parents. Key modules were introduced, including user management, facial recognition processing, notification systems, and reporting dashboards. The system's architecture was designed using a modern web stack to ensure efficiency and responsiveness. Security and data privacy were prioritized, with encryption and access controls designed to protect sensitive student biometric and personal information, aligning with best practices and the Data Privacy Act.

Phase 3 – Development

The development phase involved building the Web-Based Attendance System using a range of tools and technologies to ensure functionality and responsiveness. Visual Studio Code served as the primary development environment. The frontend was built using React with Material-UI (MUI) for a responsive and modern design, and face-api.js with react-webcam for client-side facial recognition. Backend development was handled using Node.js with the Express.js framework for API routing, authentication, and business logic. MongoDB with Mongoose was used as the database system for managing all user data, facial encodings, attendance records, and notifications. Socket.IO was implemented for real-time features, and Nodemailer was integrated for sending automated email alerts. All components were connected through RESTful APIs to ensure smooth data exchange.

Phase 4 – Testing

In this phase, the researchers conducted comprehensive testing to ensure all system features functioned correctly and integrated smoothly. Unit testing validated individual components such as facial recognition accuracy, login authentication, and email notification delivery. Integration testing verified the consistency of data flow between the frontend, backend, and database. User Acceptance Testing (UAT) involved a select group of teachers and parents from Systems Plus Computer College Caloocan to evaluate usability, reliability, and performance under real classroom conditions. Feedback was gathered and used to resolve issues, refine features, and prepare the system for deployment.

Phase 5 – Deployment

Following successful testing, the system was deployed to a production environment. The frontend React application was hosted on Netlify, while the backend Node.js server and MongoDB database were deployed on Heroku and MongoDB Atlas, respectively, to ensure

availability and data protection. Orientation sessions were conducted for administrators, teachers, and parents to introduce key features such as the facial recognition check-in process, real-time monitoring dashboard, and the absence request portal. System monitoring tools were configured to track usage, identify technical issues, and gather early feedback to inform future updates.

Phase 6 – Review

In the final phase, the researchers established a process for ongoing monitoring, feedback collection, and system enhancement. Regular updates are planned to improve system performance, fix identified issues, and refine features based on user experience. Advanced features such as a dedicated mobile application, predictive analytics for attendance trends, and integration with other school management systems are being considered for future development. Feedback from all user groups will continue to guide these enhancements, ensuring the platform remains effective, relevant, and aligned with the school's evolving needs. Using Agile methodology for the Web-Based Attendance System project ensured that the platform remained flexible, adaptable, and aligned with the evolving needs of its users—Administrators, Teachers, and Parents. By employing Agile's iterative approach, the development team was able to deliver core features early, refine them based on continuous user feedback, and consistently enhance the system to meet user expectations. This ensures that the final product not only meets but exceeds the functional requirements and operational goals of the platform, ultimately creating a more efficient, transparent, and connected school community.

Flowcharts of WEB-BASED ATTENDANCE SYSTEM WITH FACIAL RECOGNITION, NOTIFICATIONS AND DATA REPORT FOR GRADE SCHOOL STUDENTS IN SYSTEMS PLUS COLLEGE CALOOCAN

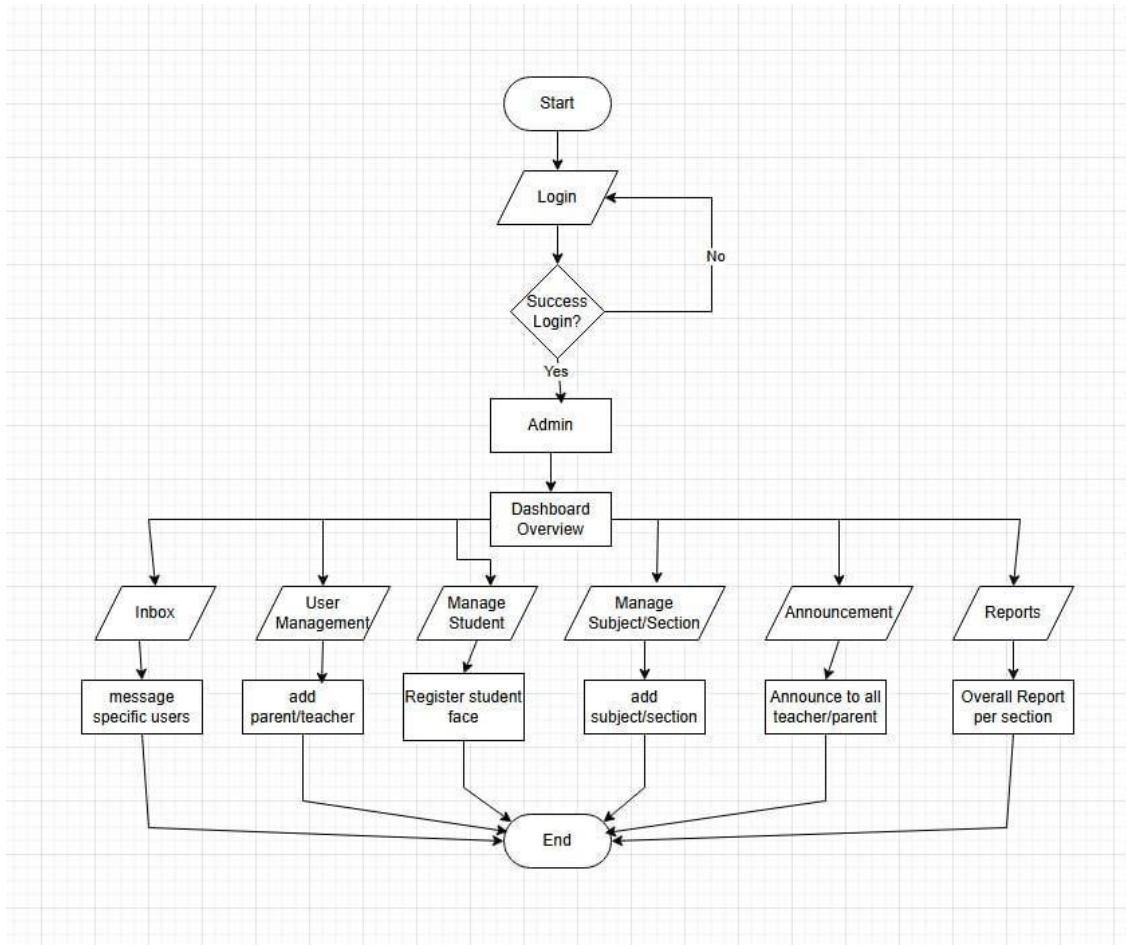


Figure 3. Flowchart of Web-based Attendance System (Admin)

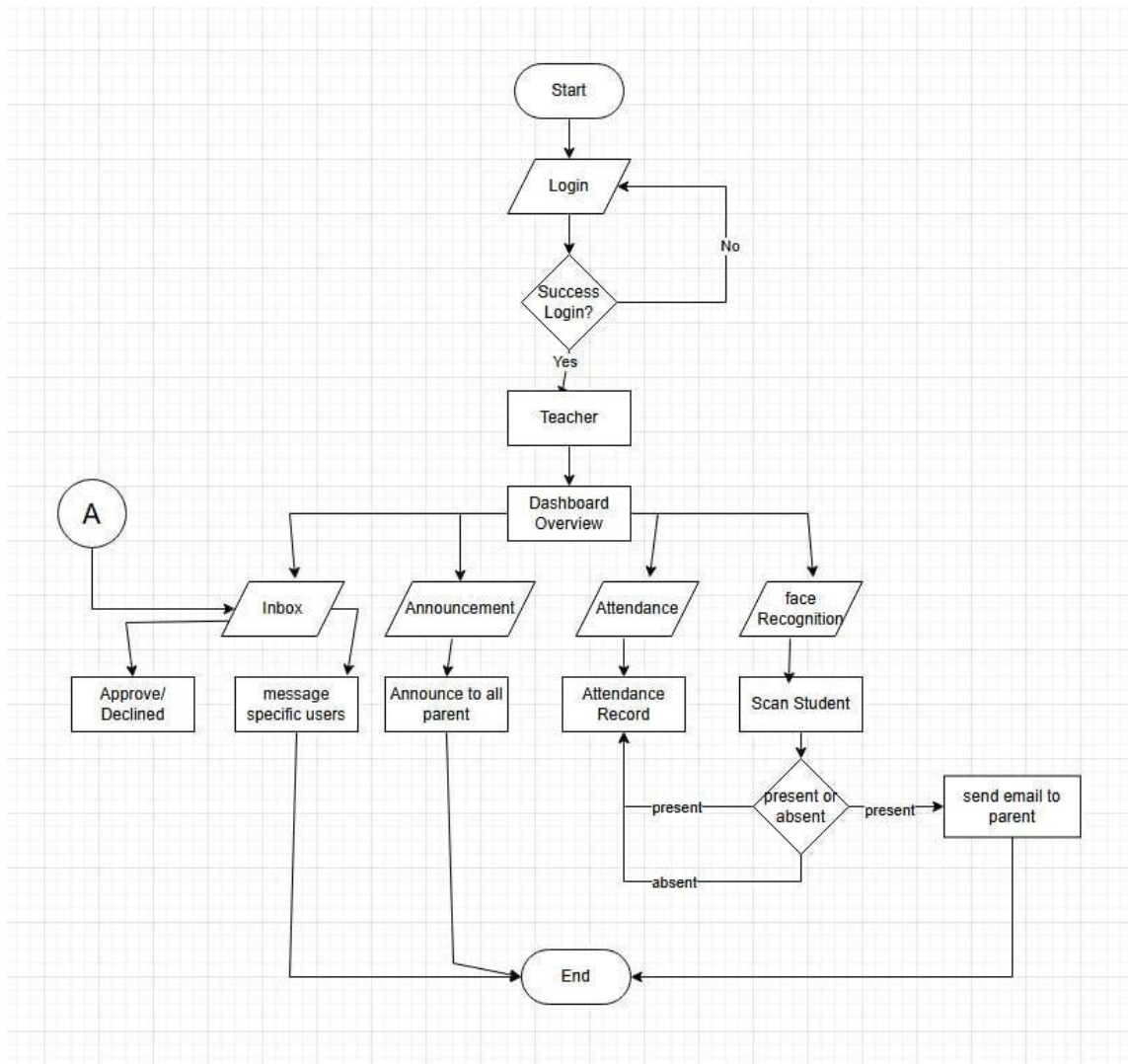


Figure 4. Flowchart of Web-based Attendance System (Teacher)

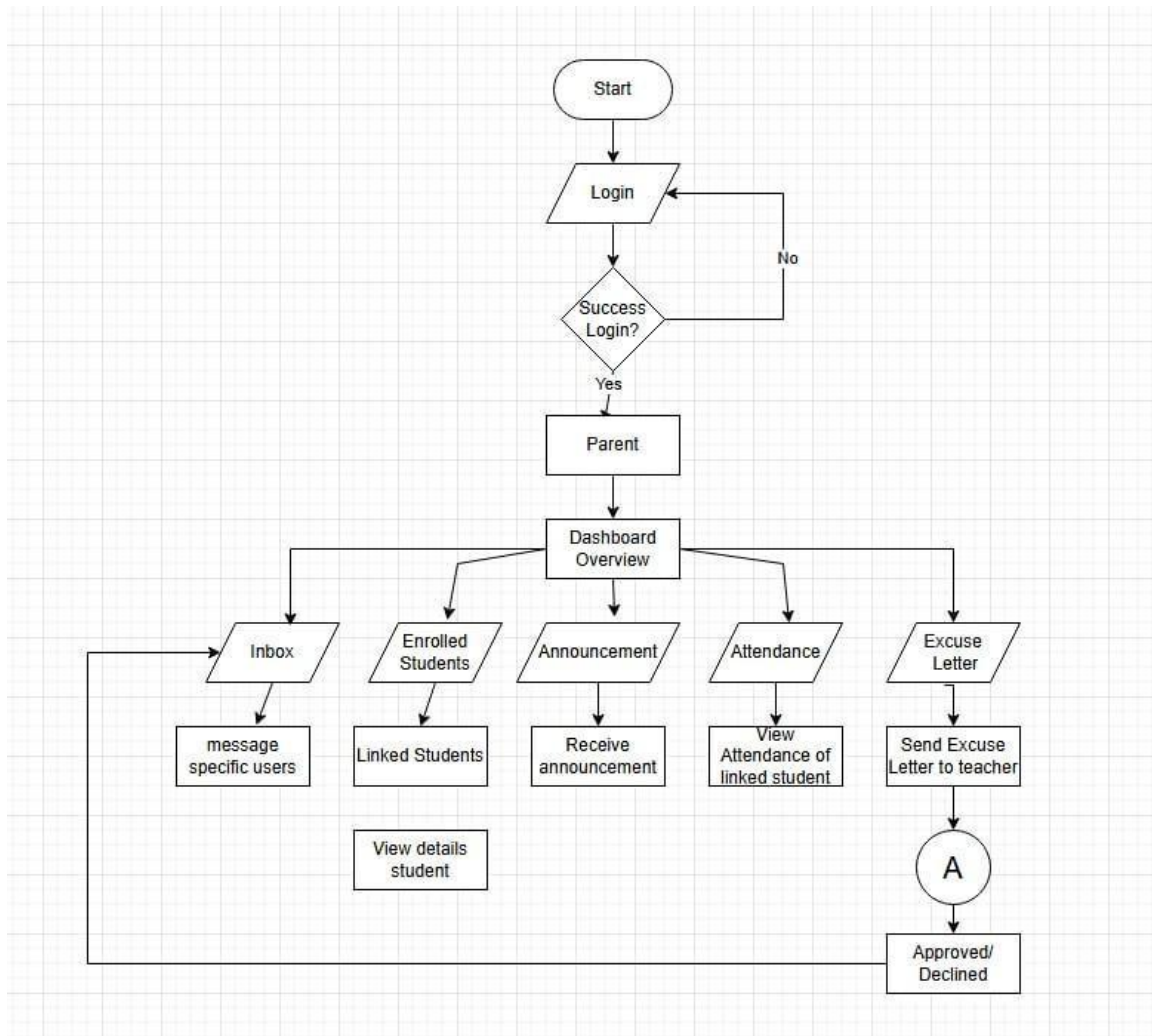


Figure 5. Flowchart of Web-based Attendance System (Parent/Guardian)

These Flowcharts illustrates the flow of the processes done in the system, showing how the admin, teacher and parent/guardian can manage the system of Web-based Attendance System in its website.

Data Flow Diagram

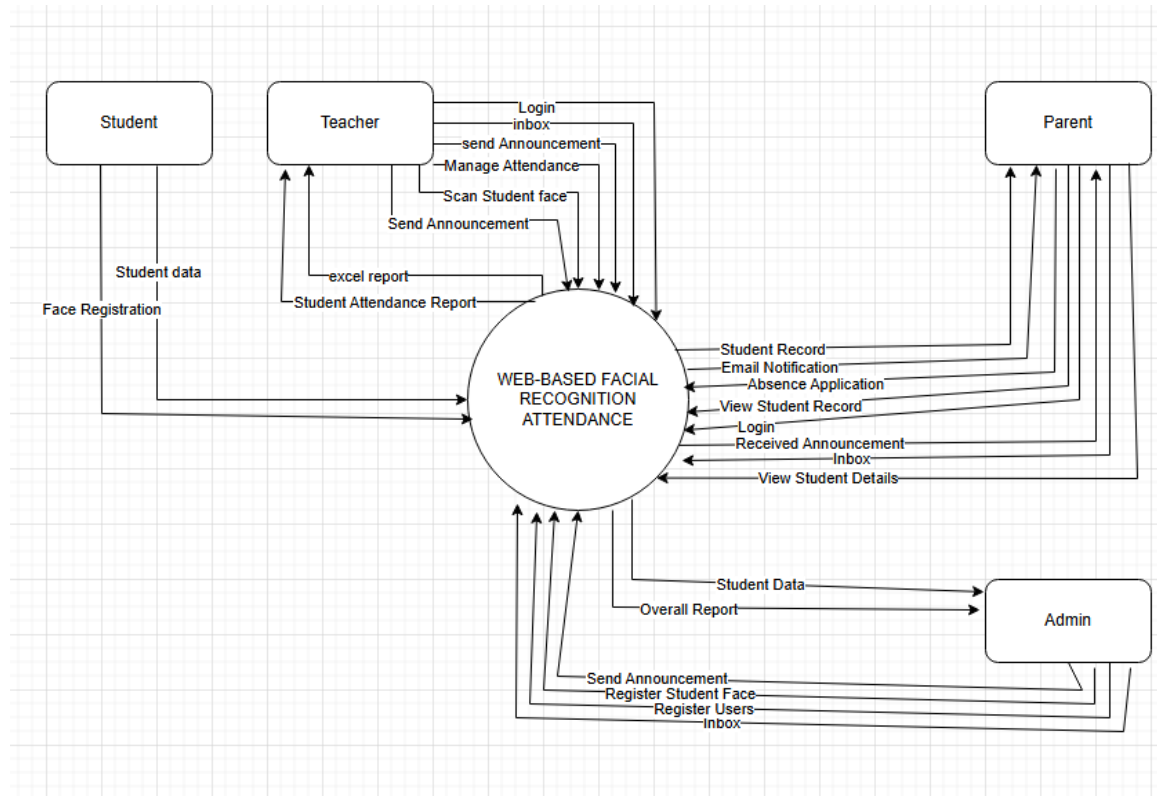


Figure 6. Data Flow Diagram of Web-based Attendance System

This Data Flow Diagram illustrates the structured data interactions within the Web-Based Attendance System. It shows how admins, teachers, and parents connect and exchange information across different system processes such as attendance tracking, facial recognition, and absence management. The diagram highlights how each user role interacts with the system—admins managing student and teacher data, teachers monitoring attendance and reviewing absence requests, and parents submitting and tracking absence applications. Overall, it demonstrates the efficient flow of data that ensures accurate attendance monitoring, clear communication, and streamlined operations within the school environment.

HIPO Chart

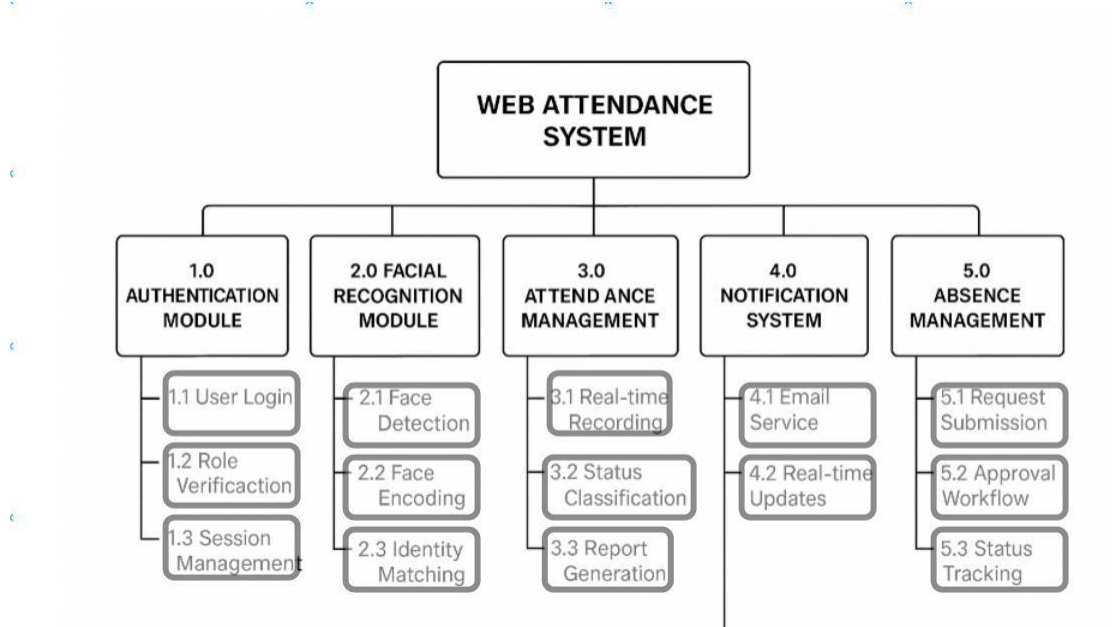


Figure 7. HIPO Chart Diagram

This HIPO Chart illustrates how each module contributes to the system's goal of automating attendance through facial recognition while improving communication between admins, teachers, and parents. It provides a clear breakdown of processes that ensure accuracy, transparency, and efficiency in attendance management within the school environment.

Design of Software, System, Product and/or Processes

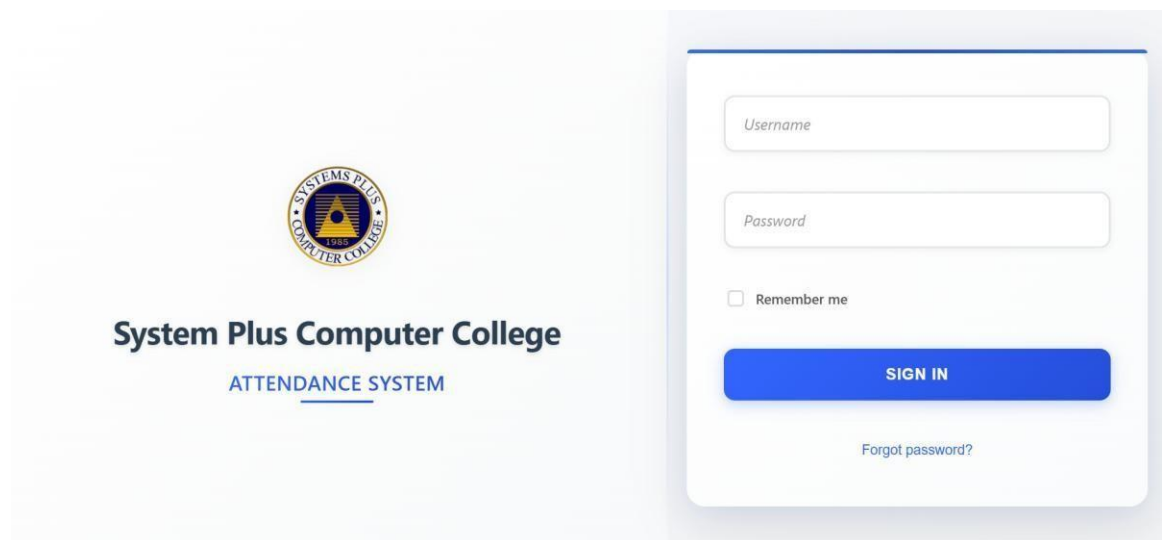


Figure 8. Log-in Module

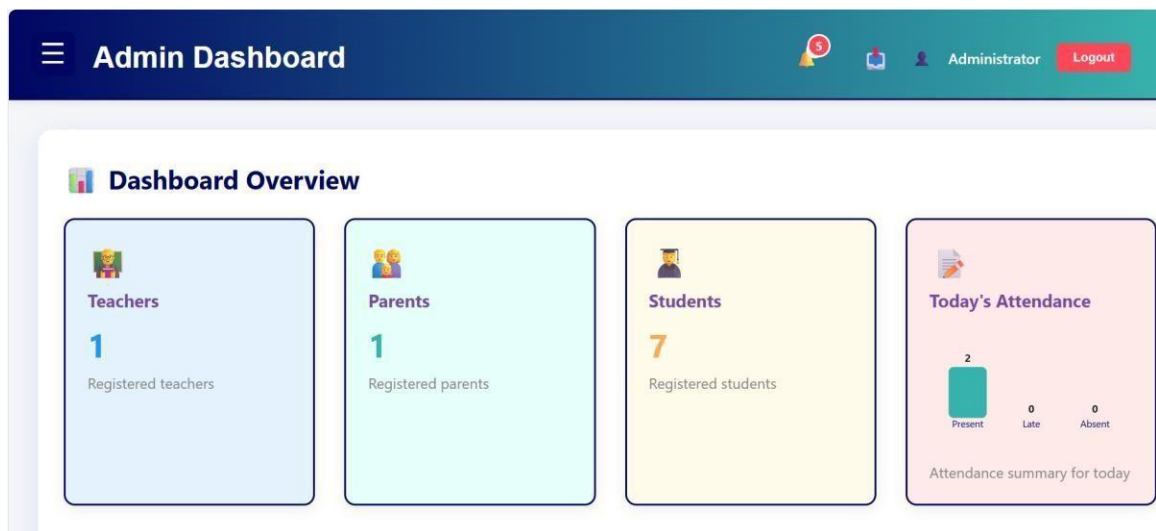


Figure 9. Admin's Dashboard

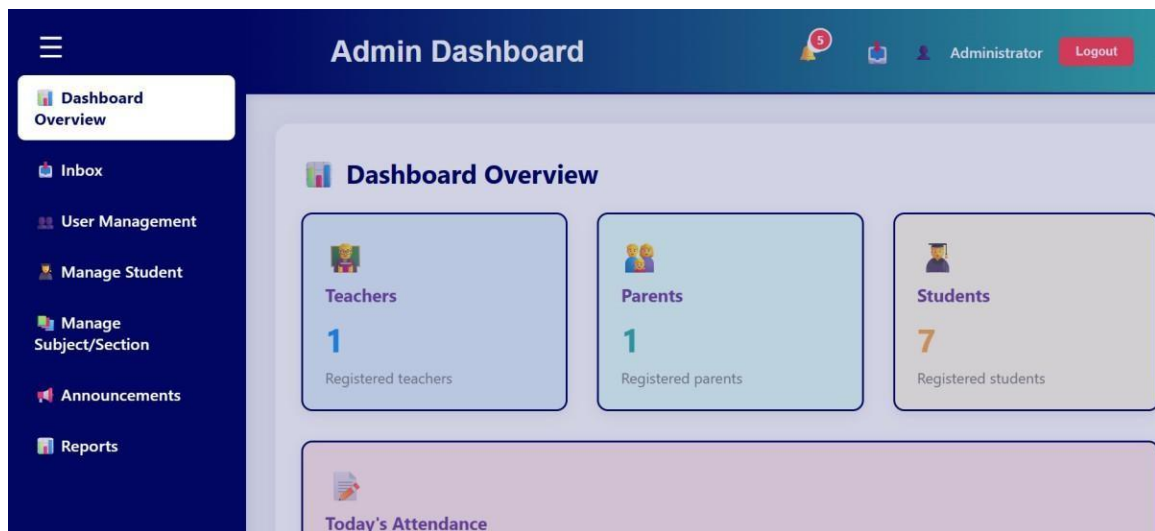


Figure 10. Admins Sidebar

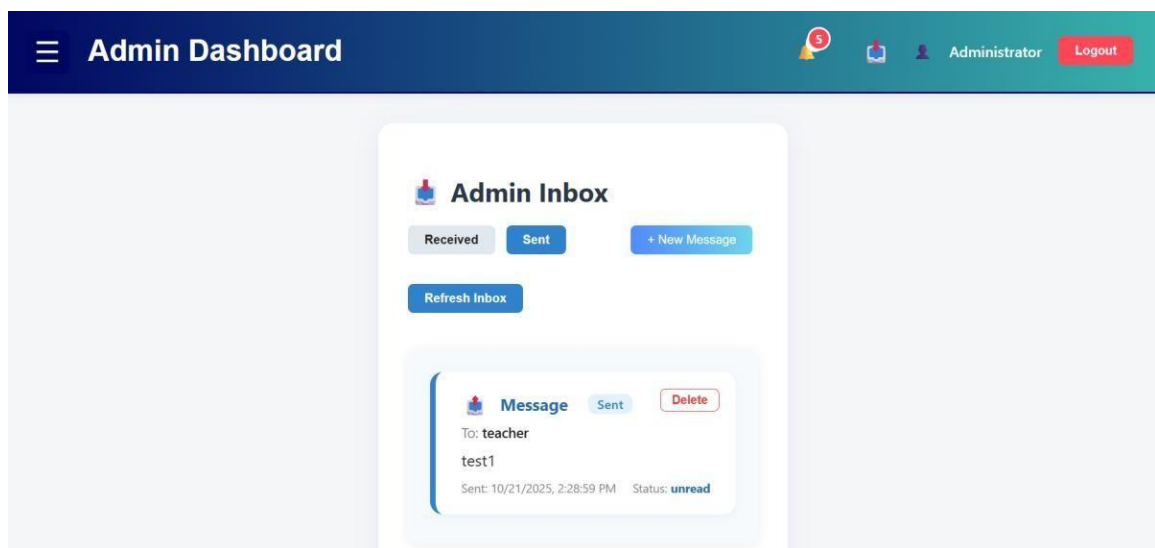


Figure 11. Admin's Inbox

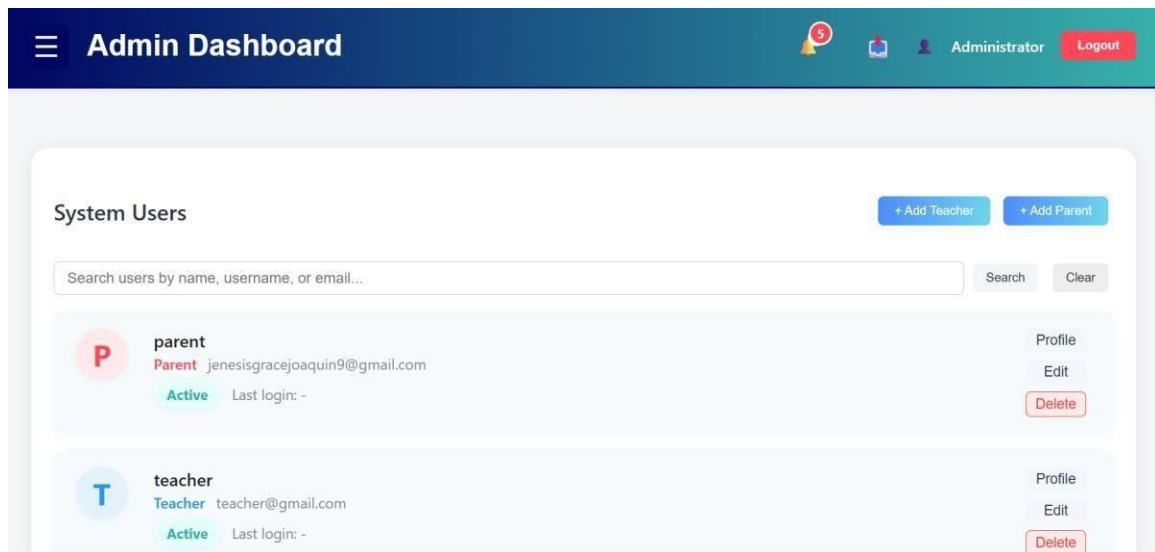


Figure 12. Admin's User Management

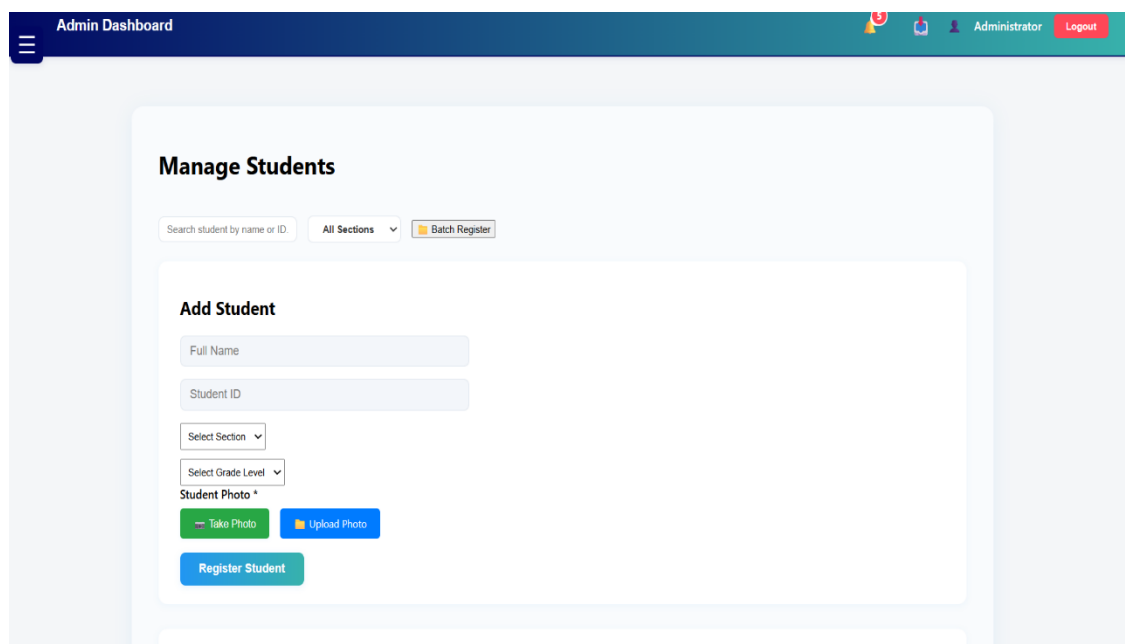


Figure 13. Student Management

Manage Subjects & Sections

Add New Subject/Section

Section (e.g. A1)

Subject (e.g. Math)

Add

Filter by Section: All Sections

Section	Subject	Actions
2	math	Edit Delete
3	english	Edit Delete
4	science	Edit Delete
1	math	Edit Delete

Figure 14. Subjects & Sections Management

Announcements

Select Audience

Announcement Title

Write your announcement here...

Announce

Announcement History

Title	Posted By	Date	Audience	Status
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Figure 15. Admin's Announcement Module

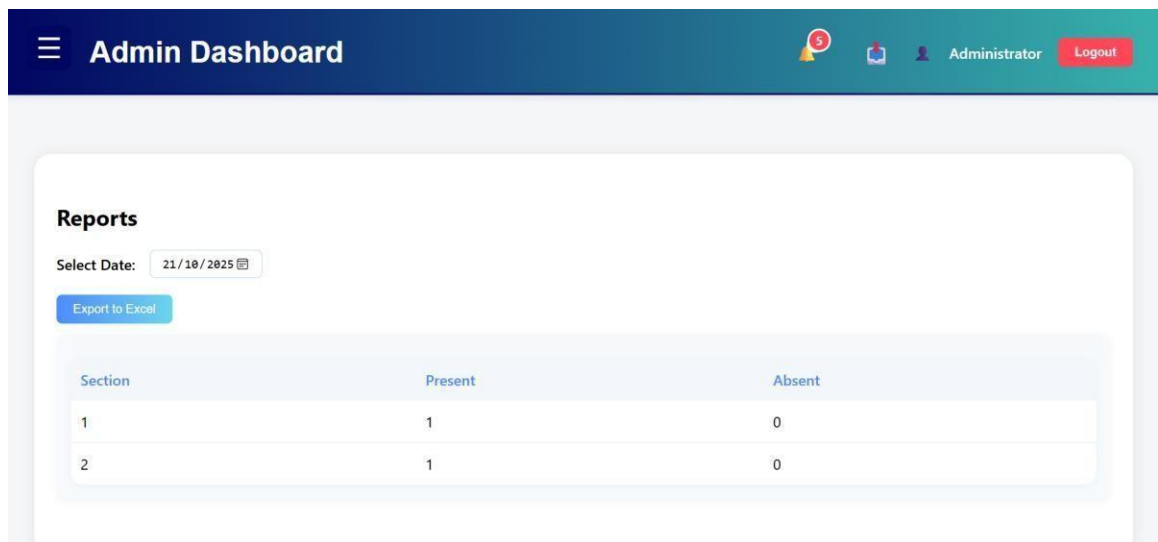


Figure 16. Admin's Report Module

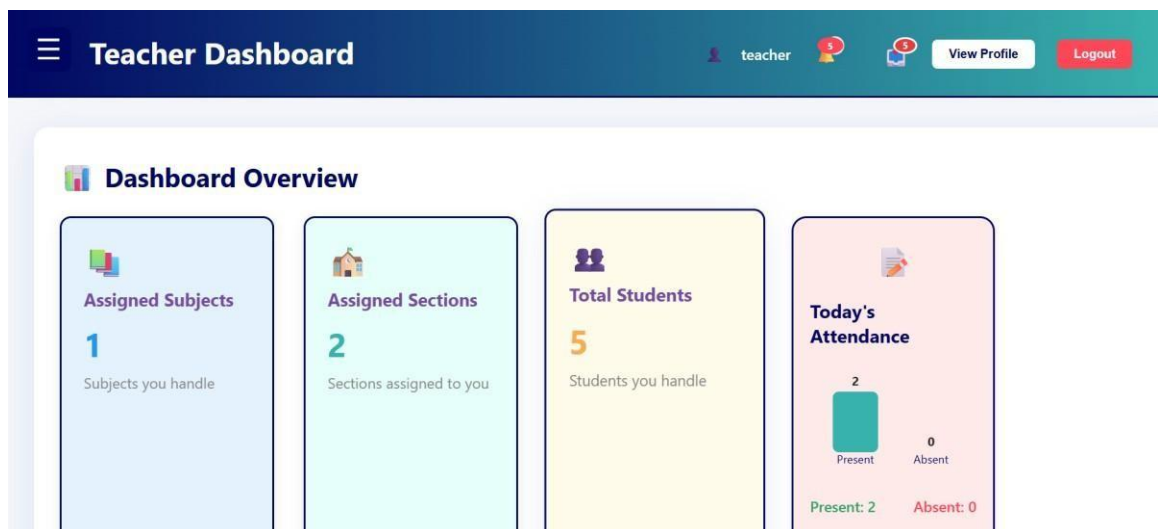


Figure 17. Teacher's Dashboard

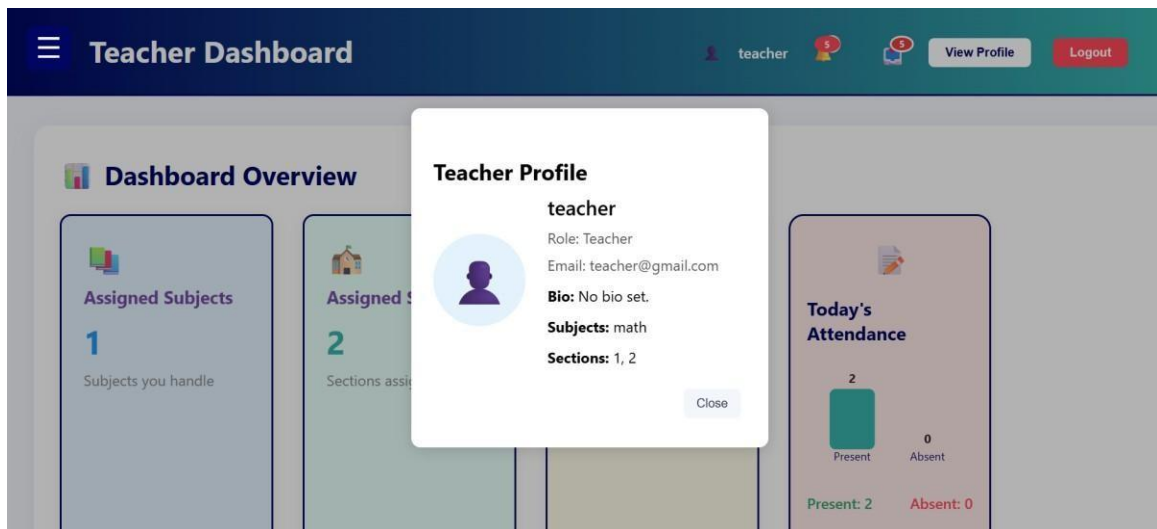


Figure 18. Teacher's Profile

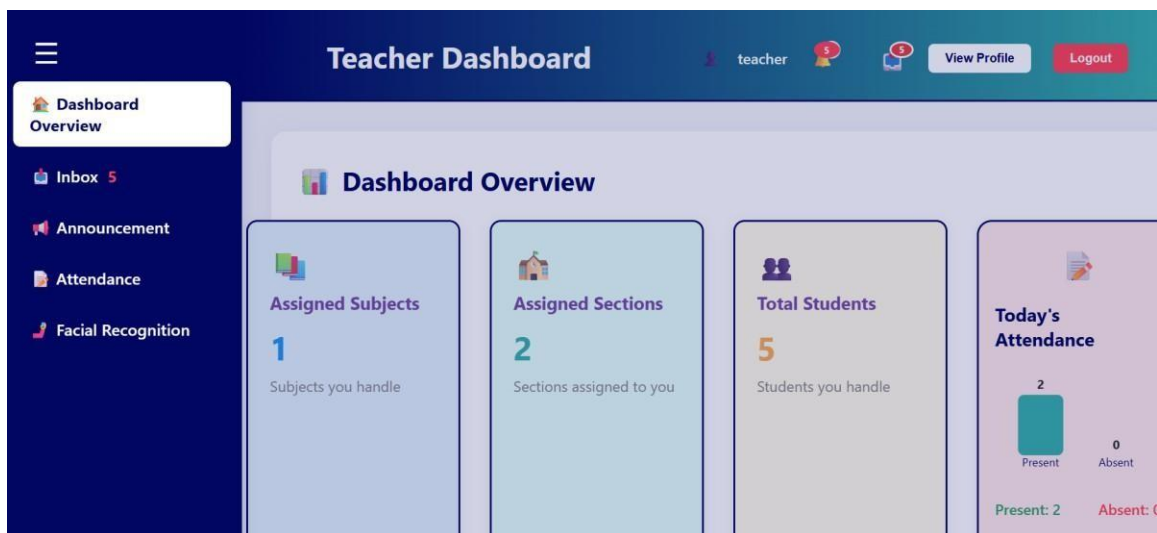


Figure 19. Teacher's Sidebar

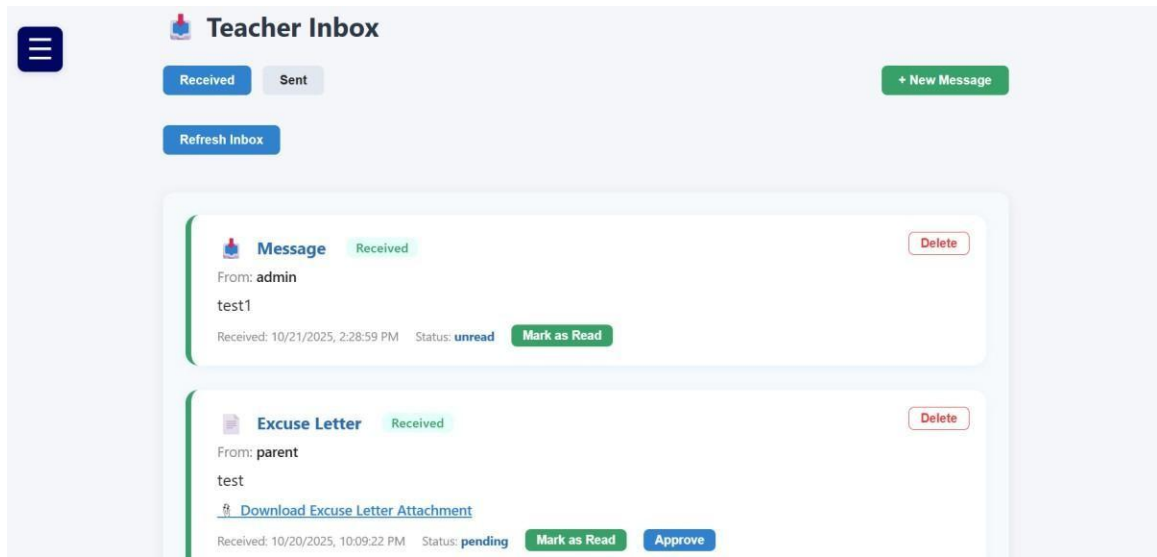


Figure 20. Teacher's Inbox Module

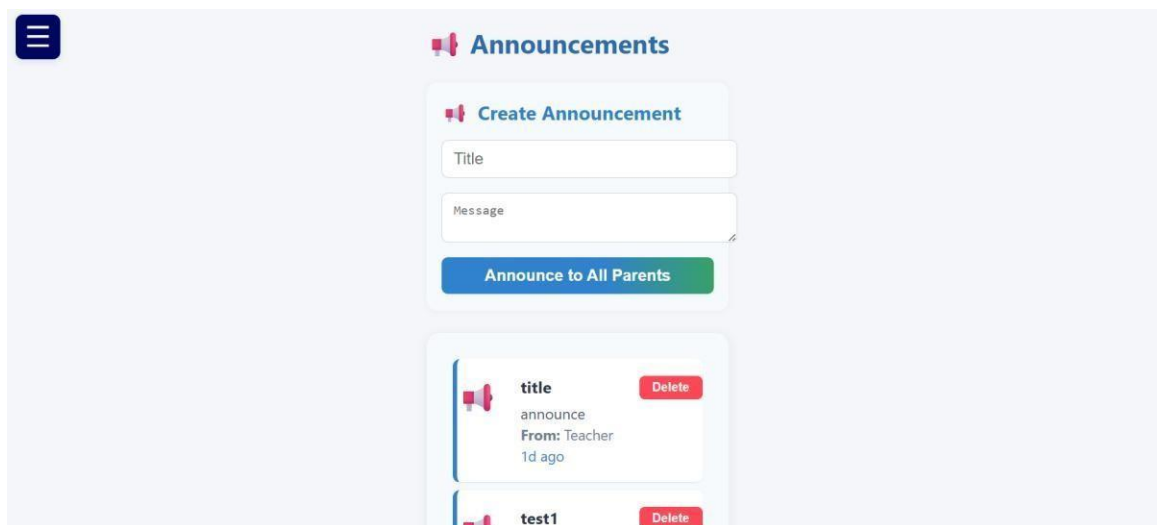


Figure 21. Teacher's Announcement Module

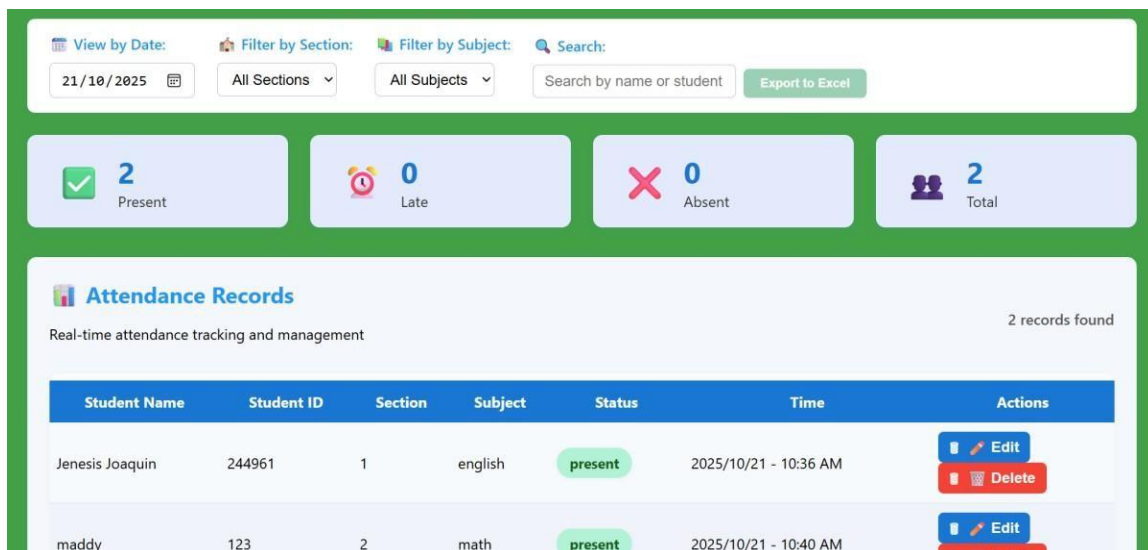


Figure 22. Teacher's Attendance Record Module

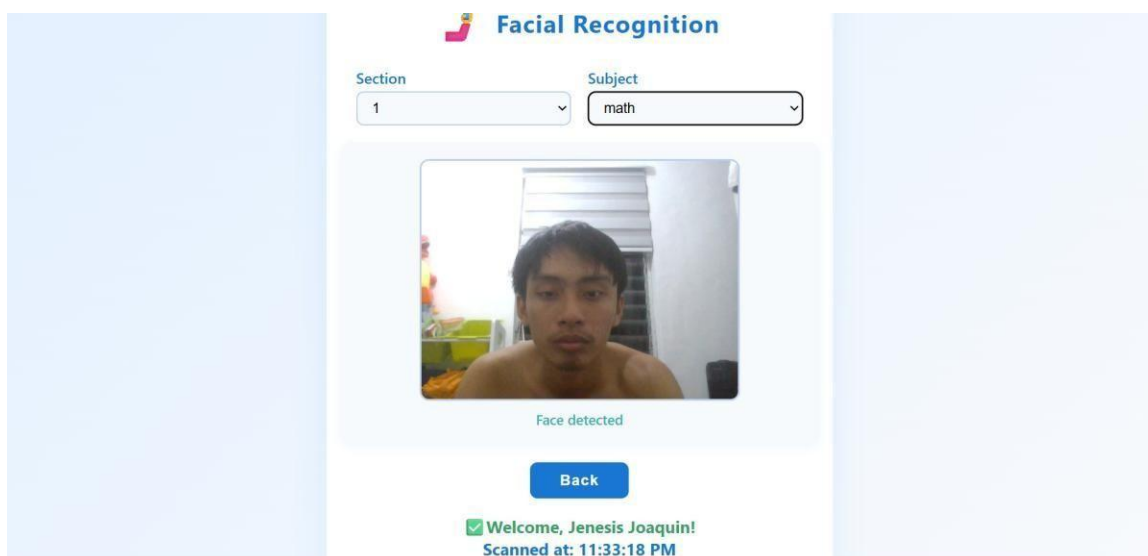


Figure 23. Teacher's Attendance using Facial Recognition

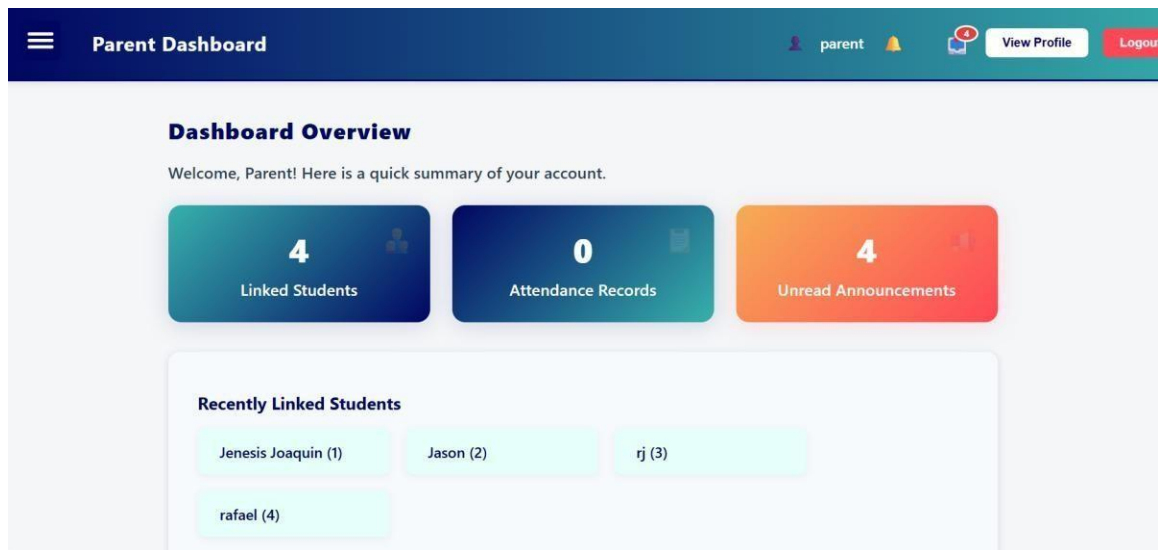


Figure 24. Parent's Dashboard

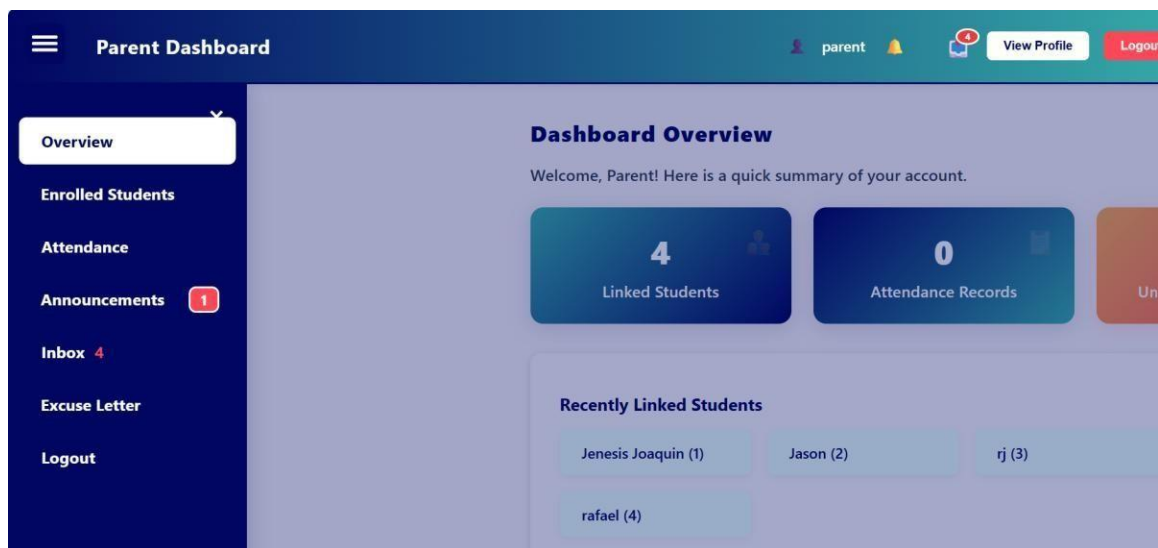


Figure 25. Parent's Sidebar

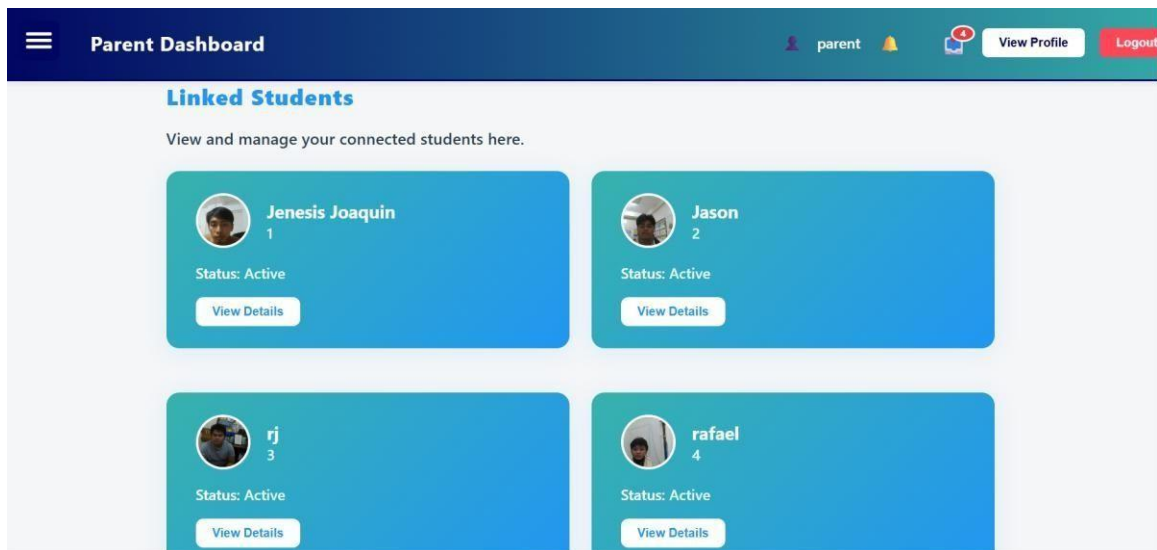


Figure 26. Parent's Linked Students Module

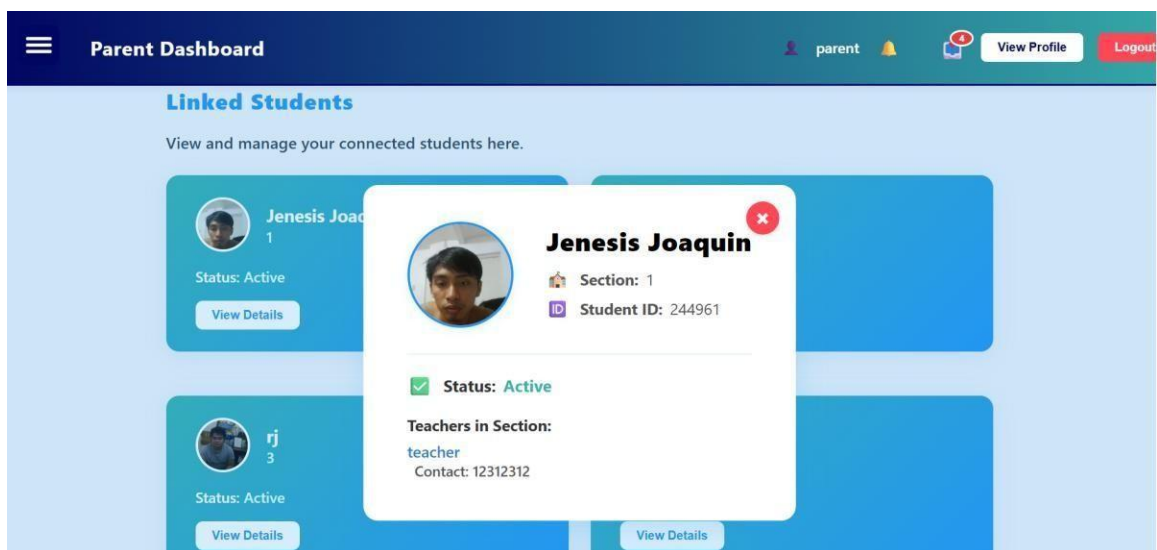


Figure 27. Parent's Student Profile

Parent Dashboard

parent

View Profile Logout

Attendance Records

Check your students' attendance history and details.

Filter by Date: Clear Refresh

Date	Time	Section	Student	Status
10/17/2025	-	1	Jenesis Joaquin	Present
10/18/2025	-	1	Jenesis Joaquin	Present
10/19/2025	-	1	Jenesis Joaquin	Present
10/20/2025	-	2	Jason	Present
10/20/2025	-	1	Jenesis Joaquin	Present

Figure 28. Parent's Attendance Record Module for Student

Parent Dashboard

parent

View Profile Logout

Announcements

Stay updated with the latest school news and announcements.

Unread Announcements: 1

title

10/20/2025, 11:22:08 PM

announce

From: Teacher

Mark as Read

test1

10/20/2025, 10:32:12 PM

From: Teacher

Figure 29. Parent's Announcement Notification Module

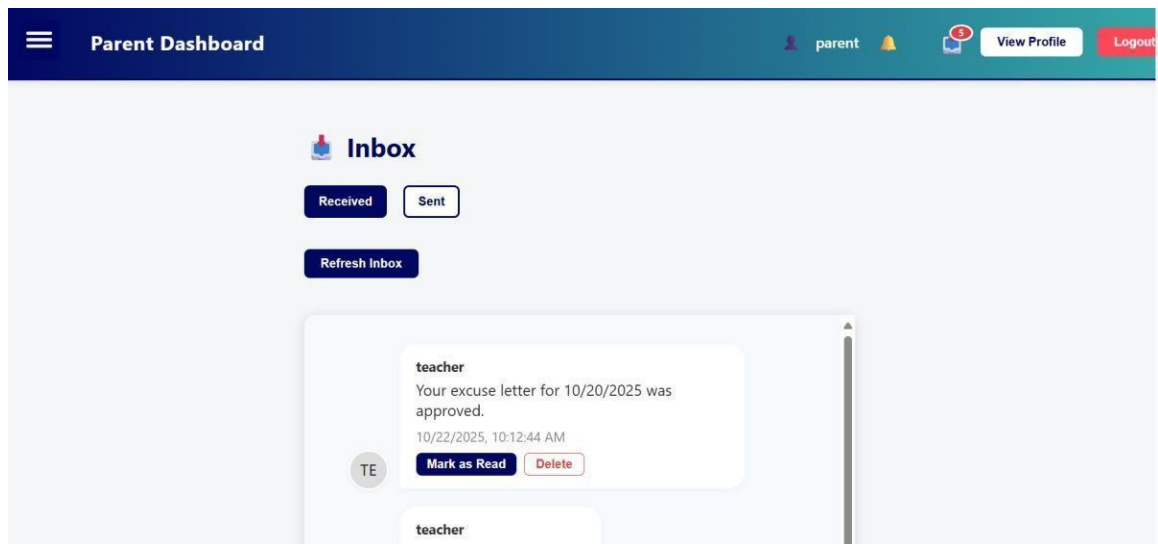


Figure 30. Parent's Inbox Module

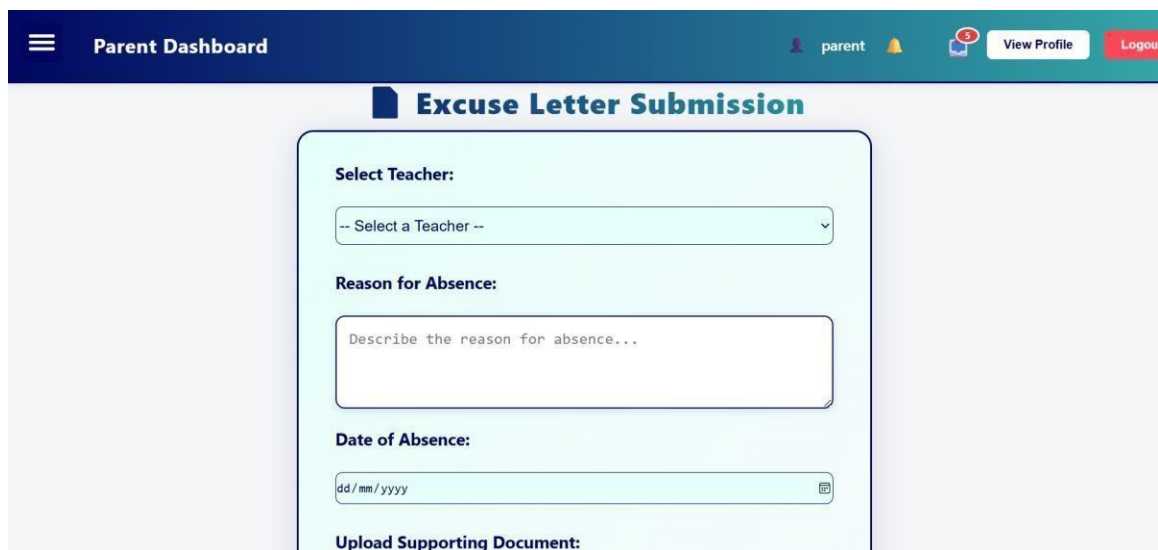


Figure 31. Absence Request Module

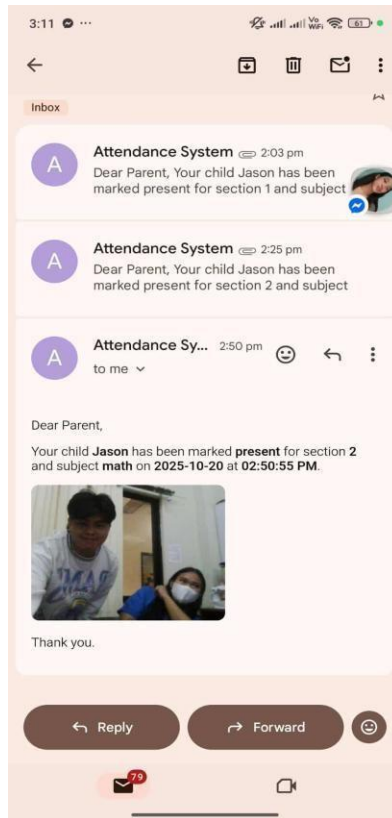


Figure 32. Attendance Email Notification Module

Development

The development of the Web-Based Attendance System with Facial Recognition for the Grade School Department of Systems Plus Computer College Caloocan focuses on building a secure, efficient, and automated attendance management environment. The platform provides a web-based system for administrators, teachers, and parents, enabling a seamless transition from manual, paper-based tracking to a digital, intelligent system enhanced by real-time notifications and comprehensive reporting.

1. Development Standards and Practices

a. Coding Standards

- **Consistent Coding Practices:** Developers maintained consistent naming conventions, a modular component structure in React, and reusable backend services in Node.js to ensure clarity, maintainability, and scalability across the entire system.
- **Version Control:** The team utilized Git with GitHub to manage versions and collaboration. Each feature (e.g., facial recognition module, notification service) was developed in separate branches, enabling efficient merging, debugging, and progress tracking throughout the Agile sprints.

b. Frameworks and Libraries

- **React with Material-UI (MUI):** Used to develop the dynamic and responsive web application for all user roles. It provided a comprehensive suite of pre-built components for a consistent, modern, and accessible user interface.
- **Node.js with Express.js:** Served as the backend runtime and web application framework, handling API routes, business logic, authentication, and integration with all other services.
- **MongoDB with Mongoose:** Served as the system's primary database, using a NoSQL

document-based structure to flexibly store and manage user profiles, facial encoding data, attendance records, and notification logs.

- **face-api.js:** This JavaScript API was the core library for implementing client-side facial recognition, enabling real-time face detection and identification directly in the web browser.
- **Socket.IO:** Facilitated real-time, bidirectional communication between the server and client, allowing for live updates on the teacher's dashboard when attendance was recorded.
- **Nodemailer:** Integrated for sending automated email notifications to parents upon attendance recording.

c. Development Methodology

- **Agile Development Approach:** The team adopted the Agile methodology, focusing on iterative progress and continuous stakeholder feedback. The project was divided into two-week sprints, each culminating in a potentially shippable increment of the system, allowing for regular refinement and reprioritization based on user feedback.

2. Feature Implementation

a. Core Features

- **Role-Based Access:** The platform defines three primary user roles with specific privileges:
 - **Administrators:** Manage all system users, configure system settings, and generate institutional reports via the web portal.
 - **Teachers:** Monitor real-time attendance for their classes, generate and export reports, and review/approve student absence requests via the web portal.

- **Parents/Guardians:** View their child's attendance history and submit absence requests with documentation via the web portal.
- **Facial Recognition Attendance:** The core feature that allows for passive, contactless student check-in. Using classroom webcams and the face-api.js library, the system automatically identifies students and records their attendance with a timestamp upon entry.
- **Real-Time Email Notification System:** Integrated using Nodemailer, this feature automatically sends an immediate email to parents when their child's attendance is recorded, specifying their status (Present, Late, or Absent).
- **Automated Reporting Dashboard:** Provides teachers and administrators with tools to generate detailed attendance reports. Reports can be filtered by date, class, and student, and exported to Excel format for further analysis.
- **Digital Absence Request Management:** A streamlined process where parents can submit absence requests online. Teachers can then review these requests and attached documents (e.g., medical certificates) to approve or decline them within the system.

b. Security Measures

The following security measures were implemented to ensure the protection of sensitive student data and maintain compliance with data privacy regulations:

- **Biometric Data Protection:** The system does not store raw facial images. The face-api.js library converts facial features into encrypted numerical vectors (facial encodings) that cannot be reverse-engineered, protecting student biometric identity.
- **Role-Based Access Control (RBAC):** Strict access permissions ensure data

confidentiality. Administrators have full system access, teachers access only their class data, and parents access only their children's information.

- **Secure Data Transmission:** All data exchanges between the web browser and server are encrypted using HTTPS (TLS), protecting facial encodings and personal information during transmission.
- **Parental Consent Framework:** System operation is predicated on signed parental consent forms, ensuring transparent and compliant processing of student biometric data in accordance with the Data Privacy Act of 2012.

3. Testing Assurance

a. Testing Strategies

- **Unit Testing:** Individual modules and functions, including the facial recognition algorithm, email notification service, and user authentication, were tested in isolation to verify accuracy and reliability.
- **Integration Testing:** Ensured seamless data flow and interaction between the frontend React application, the backend Node.js API, the MongoDB database, and the Socket.IO real-time service.
- **User Acceptance Testing (UAT):** Conducted with selected administrators, teachers, and parents from Systems Plus Computer College Caloocan. This testing assessed real-world usability, facial recognition accuracy under classroom conditions, and the clarity of notifications and reports. The feedback gathered was instrumental in refining the user interface and optimizing system performance.

4. Deployment and Monitoring

a. Deployment Strategies

- **Phased Rollout:** The system was deployed in a pilot phase to a select number of

grade school classes to monitor performance and gather initial user feedback before a full-scale institutional rollout.

- **Cloud-Based Hosting:** The frontend React application was deployed on Netlify for high availability. The backend Node.js server was deployed on Heroku, and the database was hosted on MongoDB Atlas, ensuring scalability and reliable cloud infrastructure.
- **Phased Rollout:** The system was deployed in a pilot phase to a select number of grade school classes to monitor performance and gather initial user feedback before a full-scale institutional rollout.
- **Cloud-Based Hosting:** The frontend React application was deployed on Netlify for high availability. The backend Node.js server was deployed on Heroku, and the database was hosted on MongoDB Atlas, ensuring scalability and reliable cloud infrastructure.

b. Monitoring and Support

- **Performance Monitoring:** Application performance and error tracking are monitored to ensure system uptime, response times, and functional reliability during critical school hours.
- **Continuous Improvement:** Support and feedback are managed through direct communication with the school administration and teachers. This feedback is essential for guiding future updates and maintenance, adhering to the principles of continuous improvement central to the Agile methodology.

RESULTS AND DISCUSSION

Testing

Testing was conducted among forty (40) teachers and sixty (60) parents/guardians from Systems Plus Computer College Caloocan to assess the functionality, usability, and effectiveness of the Web-Based Attendance System with Facial Recognition. The evaluation focused on system reliability, user interface design, accuracy of the facial recognition feature, and the effectiveness of email notifications. Respondents participated through a structured survey using a five-point Likert scale, covering aspects such as ease of navigation, timeliness of notifications, data security, and overall satisfaction. Results showed that most participants found the system efficient, reliable, and effective in improving the attendance monitoring process and school-to-parent communication. Feedback gathered from this testing phase was used to validate the system's core features and its readiness for implementation.

Description of Prototype

The prototype of the Web-Based Attendance System was developed using a modern MERN stack (MongoDB, Express.js, React, Node.js), integrating client-side facial recognition for automated student identification. The system features multiple modules including real-time attendance logging, automated email notifications for parents, and comprehensive reporting dashboards. The frontend was built using React with React Router for seamless single-page application navigation, and Material-UI (MUI) for a consistent and responsive user interface. Facial recognition capabilities were implemented directly in the browser using face-api.js, with camera access enabled through react-webcam. Real-time updates between the client and

server was facilitated using Socket.IO-client, and data visualization for attendance reports was achieved with Chart.js.

The backend utilized Node.js with Express.js to create a robust server architecture. MongoDB served as the primary database, with Mongoose as the Object Data Modeling (ODM) library to manage data schemas and relationships. Automated email notifications were handled through Nodemailer, while Socket.IO enabled bidirectional real-time communication between the server and connected clients. File upload functionality, such as document submissions for absence requests, was managed using Multer.

The prototype underwent iterative testing and evaluation to ensure that all core features including admin user management, facial recognition attendance logging, teacher dashboards, parent portals, and real-time notifications functioned as intended. Evaluation results confirmed that the prototype met its goal of creating a secure, efficient, and automated attendance management solution using contemporary web technologies.

Implementation Plan

The implementation of the Web-Based Attendance System follows a structured approach to ensure that it can be readily deployed and utilized within the school setting. The primary steps include:

- **Deployment Phase** – Installation and configuration of the backend server on a cloud platform and deployment of the frontend to a web server. Setup of classroom stations with webcam-equipped devices connected to the school's network.
- **Data Enrollment Phase** – Collection of student facial data and parent email addresses, following the signed consent forms, to populate the system database.

- **Orientation and Training** – Conducting sessions for teachers on using the attendance dashboard and for parents on accessing the portal and understanding notifications.
- **Pilot Testing** – Initial rollout in selected grade school classrooms to gather feedback and identify technical or usability issues prior to full-scale deployment.
- **System Evaluation and Optimization** – Refinement of features based on user feedback to ensure smooth performance and accuracy.

Full Implementation – Integration of the system across all grade school classes, with ongoing technical support and monitoring.

Resources Needed:

Hardware:

- **Server:** A cloud-based virtual private server (e.g., AWS EC2, DigitalOcean Droplet) with minimum 2 vCPUs and 4GB RAM.
- **Classroom Devices:** Laptops or desktop computers with modern web browsers (Chrome recommended) and built-in or external HD webcams.
- **Network:** A stable, high-speed internet connection to support real-time facial recognition and data synchronization.

Software:

- **Frontend:** React, React Router, Material-UI (MUI), face-api.js, react-webcam, Socket.IO-client, Chart.js.
- **Backend:** Node.js, Express.js, MongoDB, Mongoose, Nodemailer, Socket.IO, Multer.
- **Development Tools:** Visual Studio Code, Git, and Postman.

Third-Party Services:

Cloud Database: MongoDB Atlas cloud database service.

- **Email Service:** A transactional email service (e.g., Gmail SMTP, SendGrid, Mailgun) for Nodemailer.
- **Hosting:** A platform for hosting the frontend React application (e.g., Netlify, Vercel).

Special Requirements:

- **Data Privacy Compliance:** Signed parental consent forms for the collection and processing of biometric data.
- **User Training:** Comprehensive guides and sessions for administrators, teachers, and parents.

Implementation Results

To determine the level of acceptability of the "Web-Based Attendance System with Facial Recognition, Notifications, and Data Report," the researchers developed a 5-point questionnaire checklist with 10 questions for teachers and 10 for parents. This checklist was designed to be answered by the primary users of the system. This tool enabled the researchers to gather systematic feedback from users, focusing on the system's usability, functionality, efficiency, and overall impact. The responses were carefully analyzed to identify areas for improvement, ensuring the system aligns with the client's expectations.

Scale	Range	Verbal Interpretation
5	4.50 – 5.00	Strongly Agree
4	3.50 – 4.49	Agree
3	2.50 – 3.49	Neutral
2	1.50 – 2.49	Disagree
1	1.00 – 1.49	Strongly Disagree

Table: 5 – Point Likert Scale

$$\text{Weighted Mean (WM)} = \frac{\sum(f \times X)}{\sum f}$$

Where: **f** = frequency of each scale point

X = value of the scale point

$\sum (f \times X)$ = sum of the products of each frequency and scale point

$\sum f$ = total number of responses

Pre-Survey Questionnaire (Teacher)

Questions	5	4	3	2	1	Mean	Interpretation
1. The current manual attendance process takes significant time from instructional activities.	28	10	2	0	0	4.65	Strongly Agree
2. Manual attendance records are often inaccurate or incomplete.	22	14	4	0	0	4.45	Agree
3. Tracking student punctuality is challenging with current methods.	25	12	3	0	0	4.55	Strongly Agree
4. I would benefit from an automated system that records attendance instantly.	20	15	5	0	0	4.38	Agree
5. Receiving immediate notifications when students are marked present would help my classroom management.	18	16	6	0	0	4.30	Agree
6. I am comfortable with using facial recognition technology for student identification.	15	14	9	2	0	4.05	Agree
7. Having a digital system to generate attendance reports would save me time.	26	12	2	0	0	4.60	Strongly Agree
8. I would use a feature to review and approve student absence requests online.	19	16	5	1	0	4.35	Agree
9. I believe an automated attendance system would improve communication with parents.	22	13	5	0	0	4.43	Agree
10. Overall, I think a digital attendance system would make my work more efficient.	24	14	2	0	0	4.55	Strongly Agree

Pre-Survey Questionnaire (Parents)

Questions	5	4	3	2	1	Mean	Interpretation
1. I would like to receive immediate notification when my child arrives at school.	42	15	3	0	0	4.65	Strongly Agree
2. Currently, I often don't know if my child arrived at school on time.	35	18	7	0	0	4.47	Agree
3. I am comfortable with the school using facial recognition to track my child's attendance.	25	10	12	3	0	4.12	Agree
4. I would use an online portal to view my child's attendance history.	32	22	6	0	0	4.43	Agree
5. I would prefer submitting absence requests online rather than sending paper notes.	38	17	5	0	0	4.55	Strongly Agree
6. Immediate attendance notifications would give me peace of mind about my child's safety.	45	12	3	0	0	4.70	Strongly Agree
7. I trust the school to handle my child's data securely and privately.	28	22	8	2	0	4.27	Agree
8. Better attendance tracking would help me monitor my child's school habits.	34	20	6	0	0	4.47	Agree
9. I believe digital attendance systems improve school-home communication.	36	18	6	0	0	4.50	Strongly Agree
10. Overall, I would support implementing an automated attendance system.	40	16	4	0	0	4.60	Strongly Agree

Pre-Survey Responses of the Teachers in Each Questions

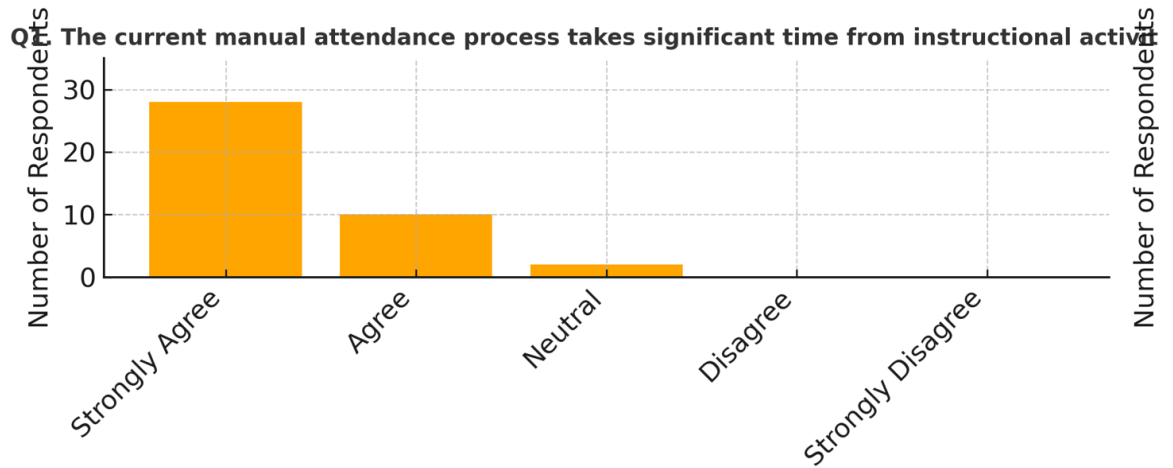


Figure 33.

Question 1: This shows that 70% strongly agreed and 25% agreed that the current manual attendance process takes significant time from instructional activities, while 5% were neutral.

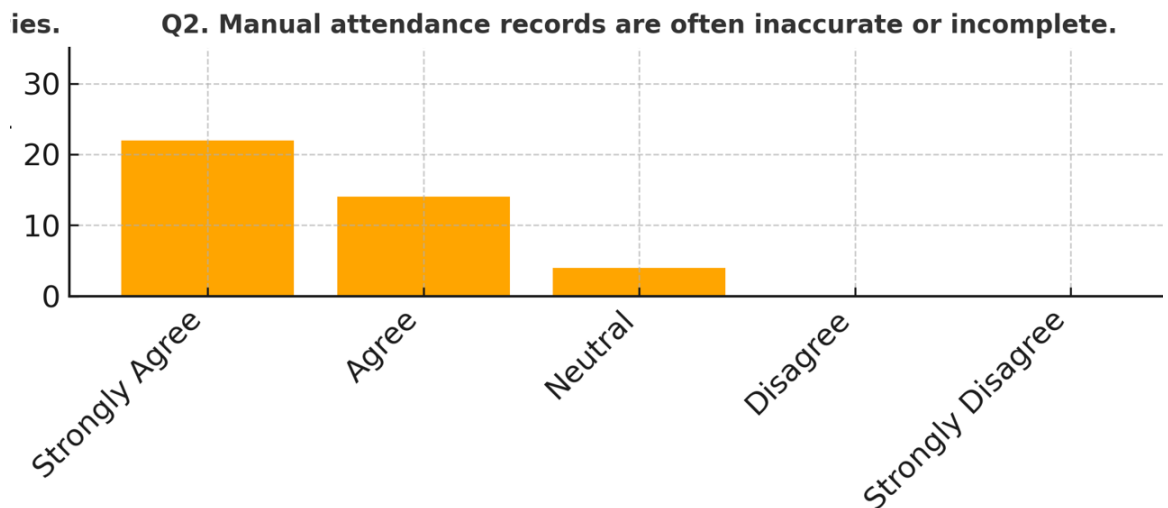


Figure 34.

Question 2: This shows that 55% strongly agreed and 35% agreed that manual attendance records are often inaccurate or incomplete, while 10% were neutral.

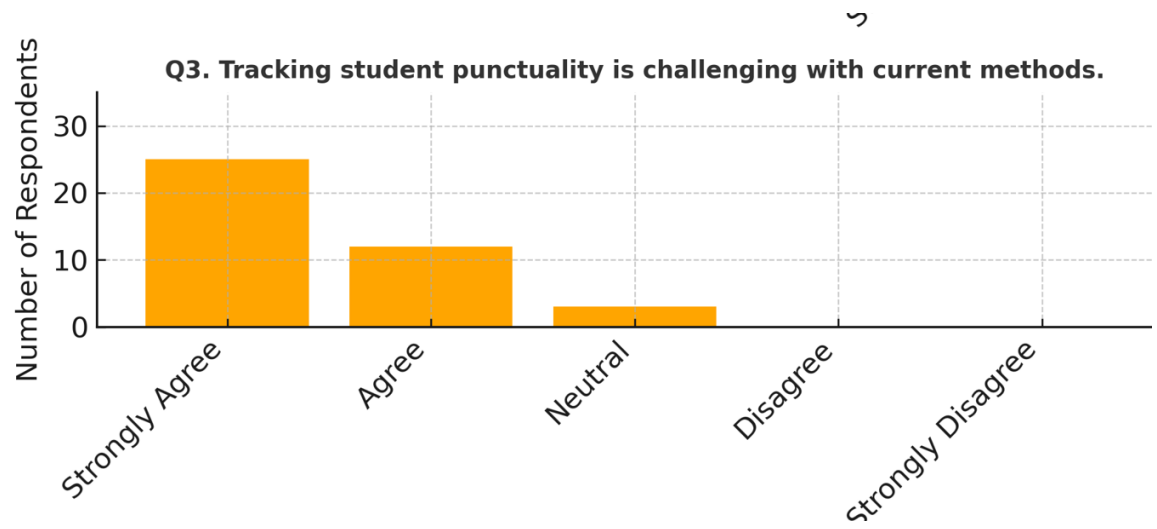


Figure 35.

Question 3: This shows that 62.5% strongly agreed and 30% agreed that tracking student punctuality is challenging with current methods, while 7.5% were neutral.

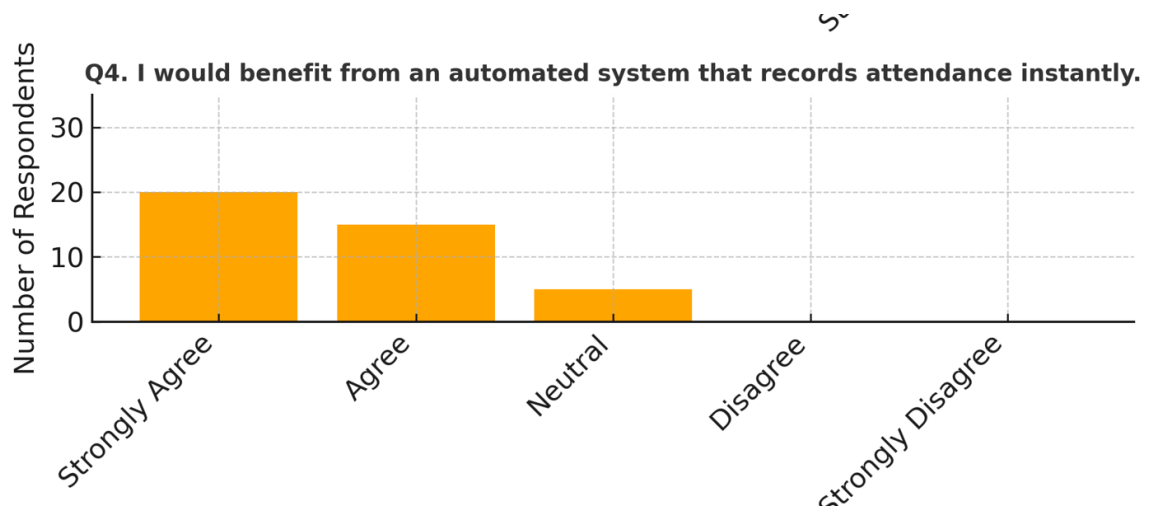


Figure 36.

Question 4: This shows that 50% strongly agreed and 37.5% agreed that they would benefit from an automated system that records attendance instantly, while 12.5% were neutral.

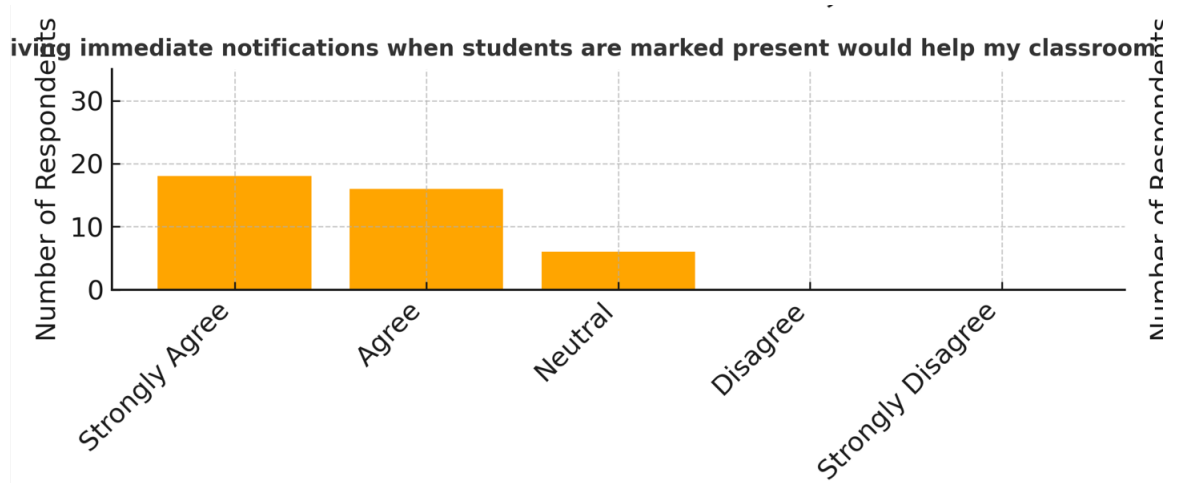


Figure 37.

Question 5: This shows that 45% strongly agreed and 40% agreed that receiving immediate notifications when students are marked present would help in classroom management, while 15% were neutral.

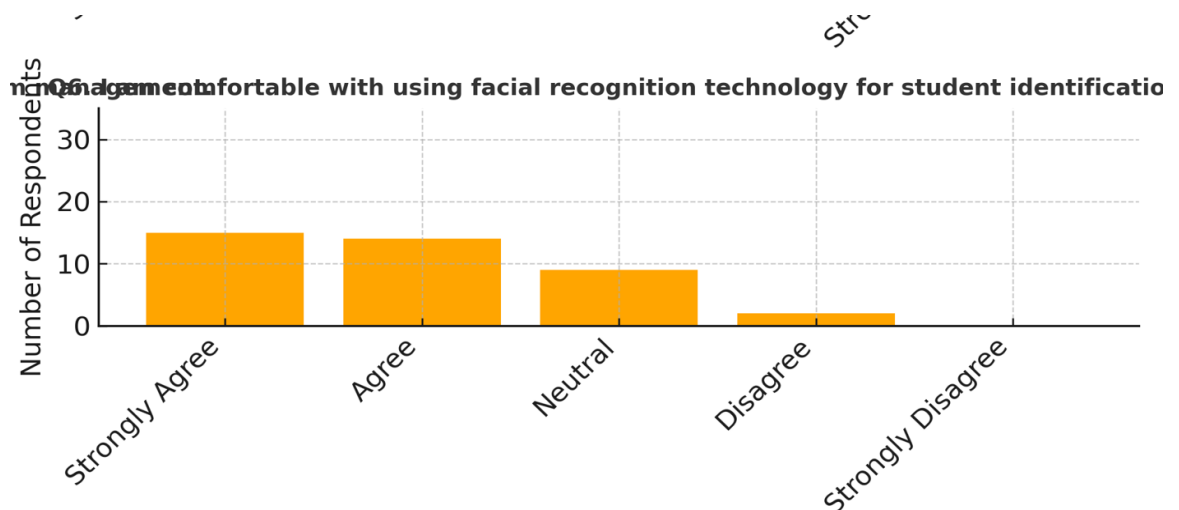


Figure 38.

Question 6: This shows that 37.5% strongly agreed, 35% agreed, and 22.5% were neutral that they are comfortable using facial recognition technology for student identification, while 5% disagreed.

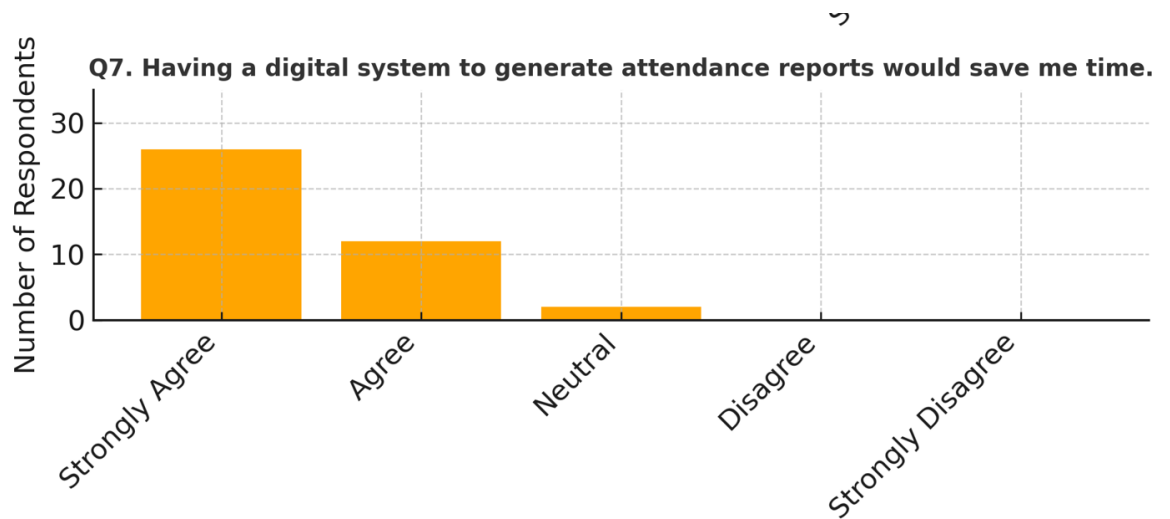


Figure 39.

Question 7: This shows that 65% strongly agreed and 30% agreed that having a digital system to generate attendance reports would save time, while 5% were neutral.

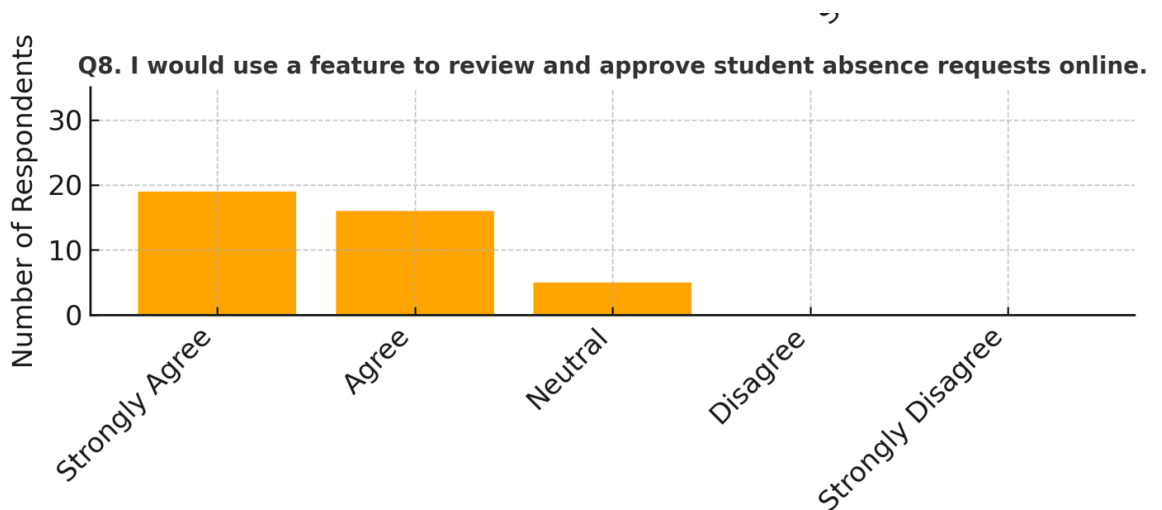


Figure 40.

Question 8: This shows that 47.5% strongly agreed and 40% agreed that they would use a feature to review and approve student absence requests online, while 12.5% were neutral

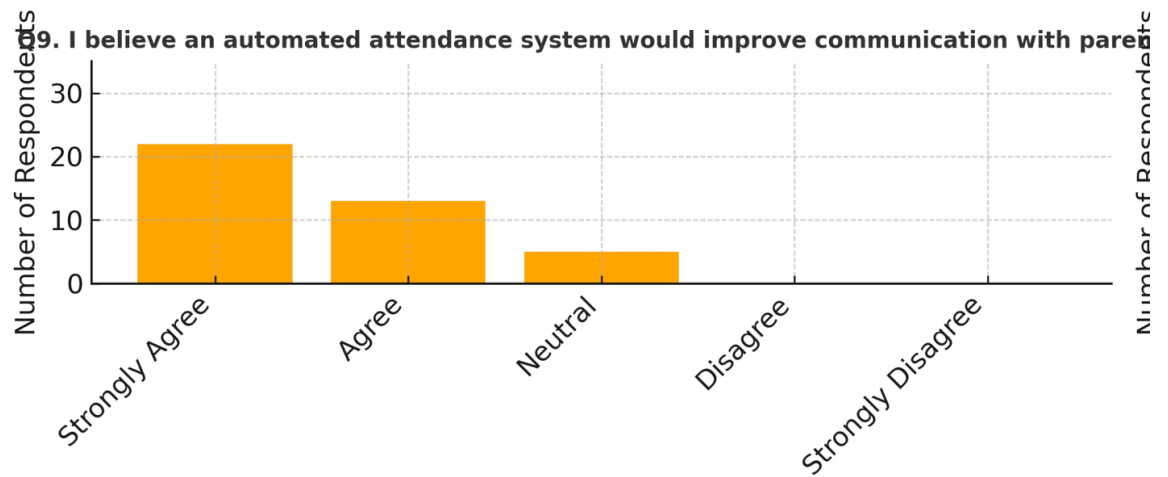


Figure 41.

Question 9: This shows that 55% strongly agreed and 32.5% agreed that an automated attendance system would improve communication with parents, while 12.5% were neutral.

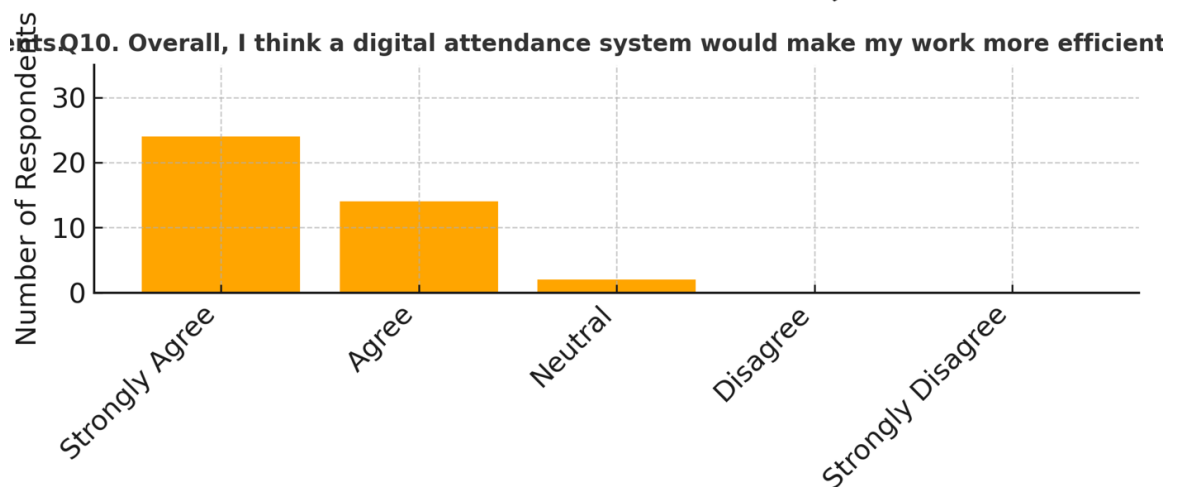


Figure 42.

Question 10: This shows that 60% strongly agreed and 35% agreed that a digital attendance system would make their work more efficient, while 5% were neutral.

Pre-Survey Responses of the Parents in Each Questions

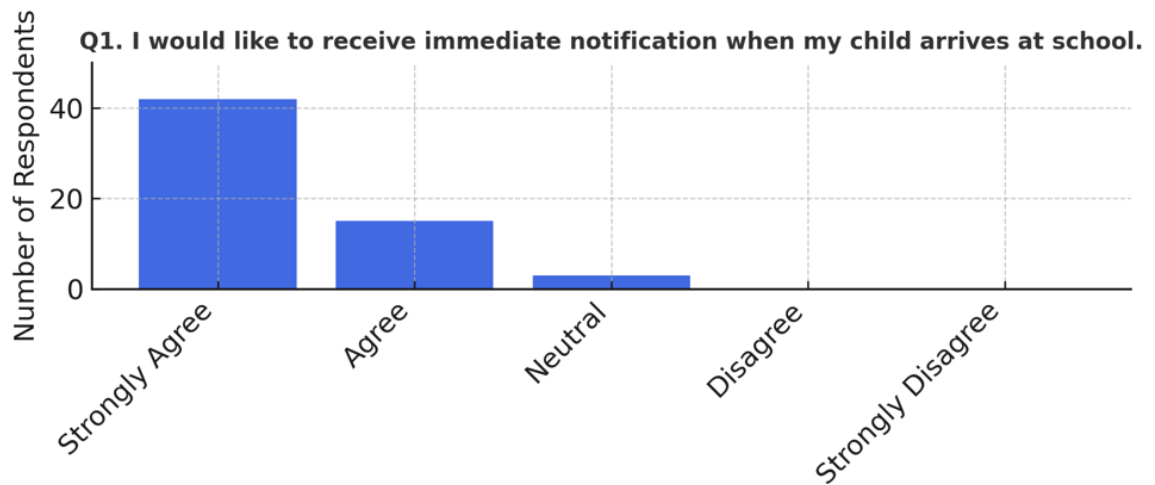


Figure 43.

Question 1: This shows that 70% strongly agreed and 25% agreed that they would like to receive immediate notification when their child arrives at school, while 5% were neutral.

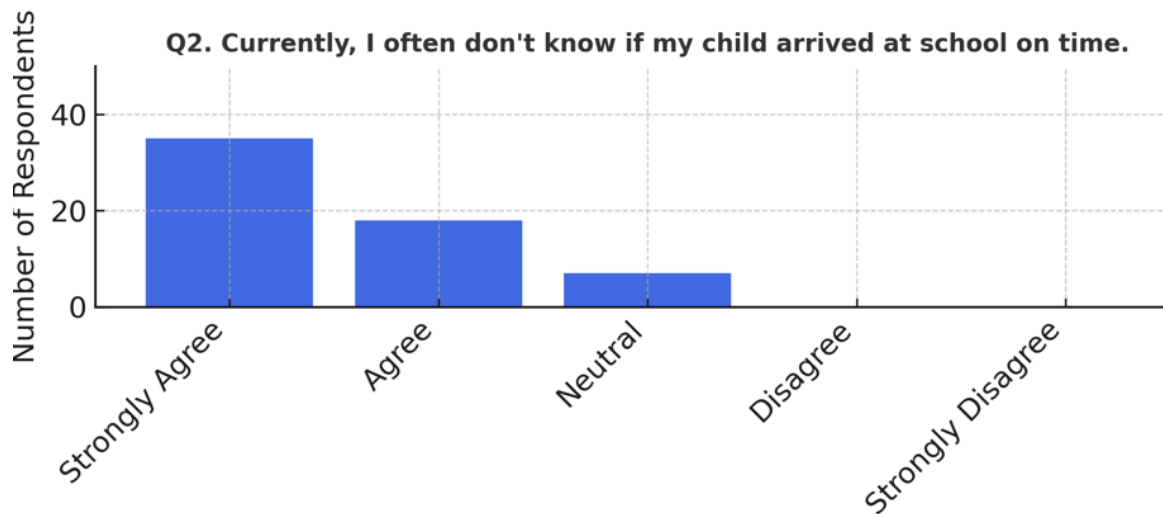


Figure 44.

Question 2: This shows that 58.3% strongly agreed and 30% agreed that they often don't know if their child arrived on time, while 11.7% were neutral.

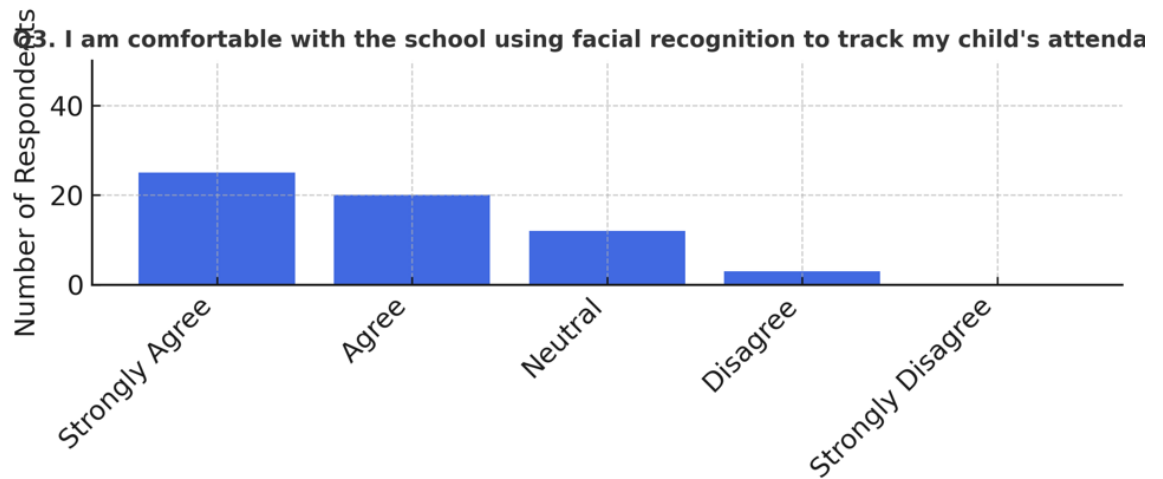


Figure 45.

Question 3: This shows that 41.7% strongly agreed, 33.3% agreed, and 20% were neutral that they are comfortable with the school using facial recognition, while 5% disagreed.

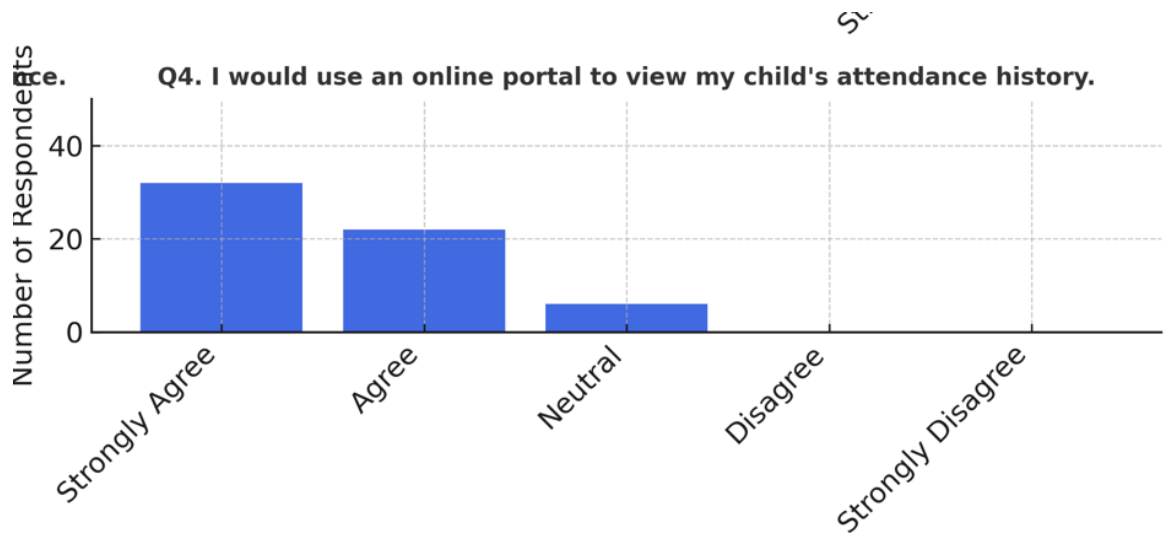


Figure 46.

Question 4: This shows that 53.3% strongly agreed and 36.7% agreed that they would use an online portal to view their child's attendance history, while 10% were neutral.

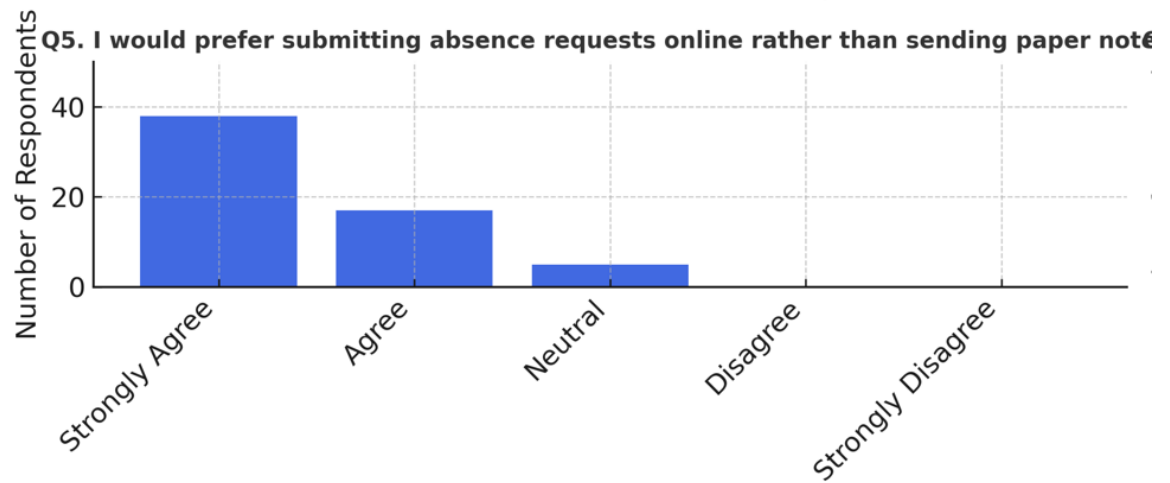


Figure 47.

Question 5: This shows that 63.3% strongly agreed and 28.3% agreed that they would prefer submitting absence requests online rather than sending paper notes, while 8.3% were neutral.

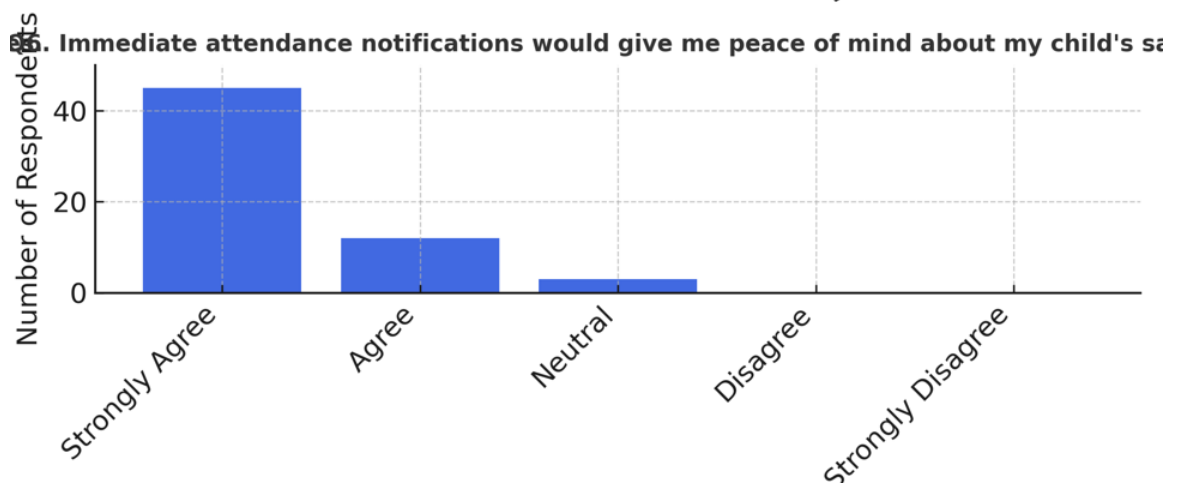


Figure 48.

Question 6: This shows that 75% strongly agreed and 20% agreed that immediate attendance notifications would give them peace of mind about their child's safety, while 5% were neutral.

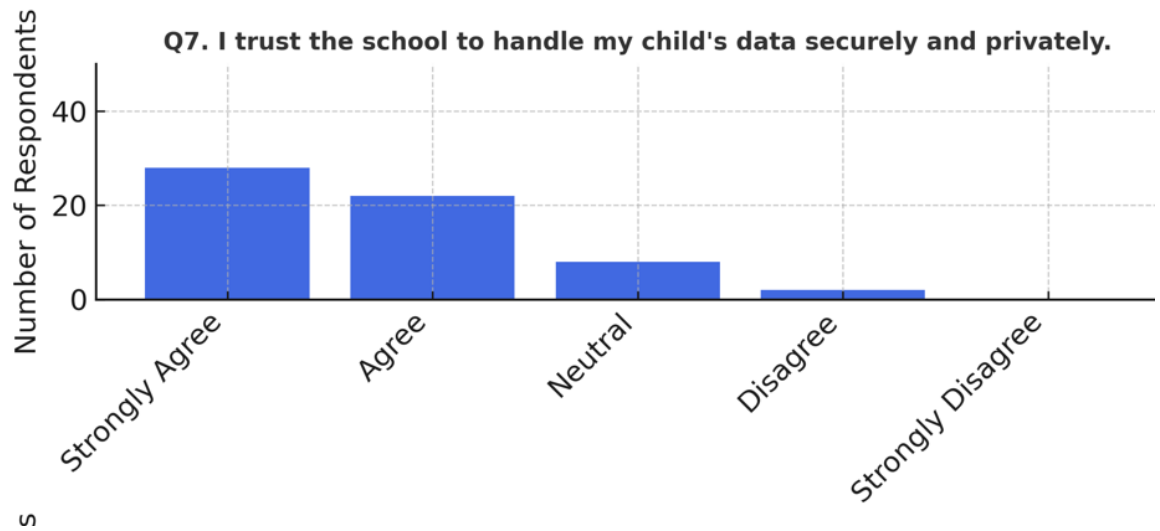


Figure 49.

Question 7: This shows that 46.7% strongly agreed, 36.7% agreed, and 13.3% were neutral that they trust the school to handle their child's data securely, while 3.3% disagreed.

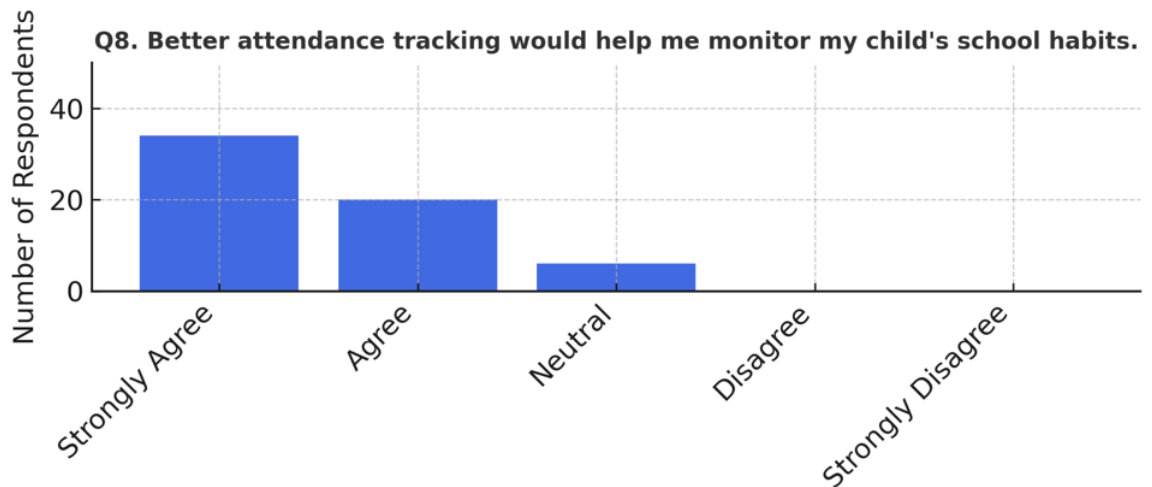


Figure 50.

Question 8: This shows that 56.7% strongly agreed and 33.3% agreed that better attendance tracking would help them monitor their child's school habits, while 10% were neutral.

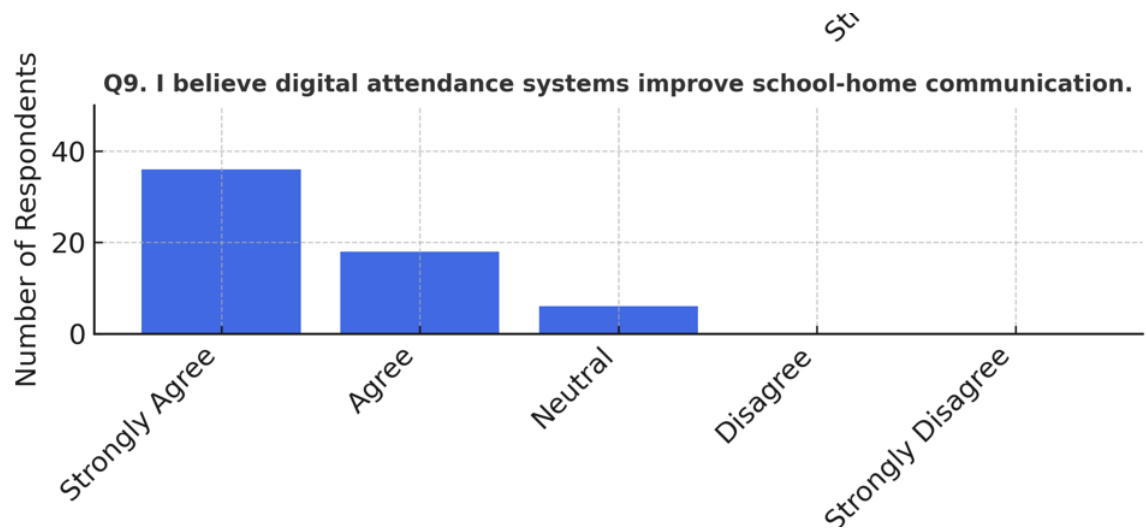


Figure 51.

Question 9: This shows that 60% strongly agreed and 30% agreed that digital attendance systems improve school-home communication, while 10% were neutral.

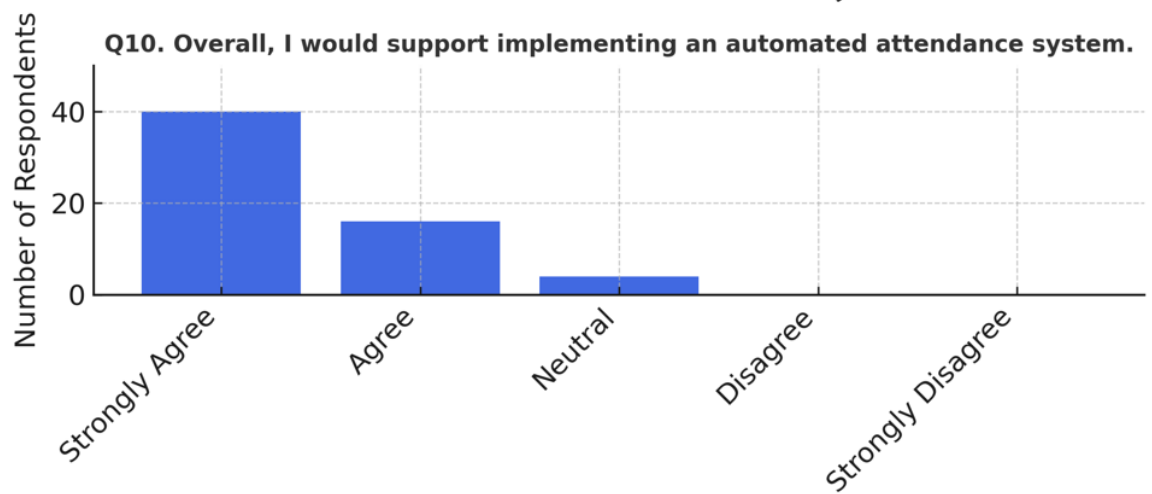


Figure 52.

Question 10: This shows that 66.7% strongly agreed and 26.7% agreed that they would support implementing an automated attendance system, while 6.7% were neutral.

Post -Survey of Web-Based Attendance System with Facial Recognition, Notifications, and Data Report for Grade School Students at Systems Plus Computer College Caloocan. (Teachers).

Questions	5	4	3	2	1	Mean	Interpretation
1. The system dashboard is easy to navigate.	25	13	2	0	0	4.58	Strongly Agree
2. The facial recognition feature provides a faster way to take attendance.	28	10	2	0	0	4.65	Strongly Agree
3. The system accurately identifies students.	22	15	3	0	0	4.48	Agree
4. Generating attendance reports is straightforward.	20	16	4	0	0	4.40	Agree
5. Reviewing absence requests within the system is efficient.	26	12	2	0	0	4.60	Strongly Agree
6. The real-time attendance monitoring feature is useful.	30	8	2	0	0	4.70	Strongly Agree
7. I feel the system securely handles student data.	18	15	7	0	0	4.28	Agree
8. The system performs reliably without frequent errors.	15	18	6	1	0	4.18	Agree
9. This system meets my expectations for a digital attendance solution.	24	14	2	0	0	4.55	Strongly Agree
10. I would recommend implementing this system.	26	12	2	0	0	4.60	Strongly Agree

Post -Survey of Web-Based Attendance System with Facial Recognition, Notifications, and Data Report for Grade School Students at Systems Plus Computer College Caloocan. (Parents).

Questions	5	4	3	2	1	Mean	Interpretation
1. The parent portal is easy to access and use.	35	20	5	0	0	4.50	Strongly Agree
2. The email notifications are timely and accurate.	40	18	2	0	0	4.63	Strongly Agree
3. Submitting an absence request online is convenient.	38	19	3	0	0	4.58	Strongly Agree
4. The system provides clear attendance records.	32	23	5	0	0	4.45	Agree
5. I trust that my child's facial data is handled securely.	28	22	10	0	0	4.30	Agree
6. The system improves communication with the school.	42	16	2	0	0	4.67	Strongly Agree
7. The system helps me stay informed about my child's attendance.	45	13	2	0	0	4.72	Strongly Agree
8. The system functions reliably without technical problems.	30	22	8	0	0	4.37	Agree
9. The system meets my expectations for tracking attendance.	36	21	3	0	0	4.55	Strongly Agree
10. I would recommend this system to other parents.	39	18	3	0	0	4.60	Strongly Agree

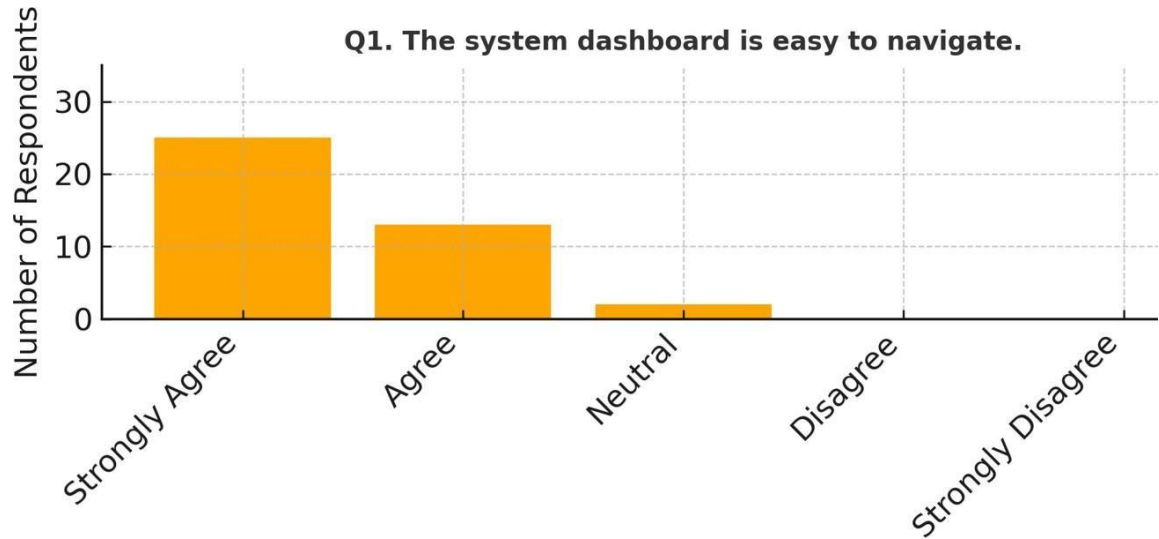


Figure 53.

Question 1 This shows that 62.5% strongly agreed and 32.5% agreed that the system dashboard is easy to navigate, while 5% were neutral.

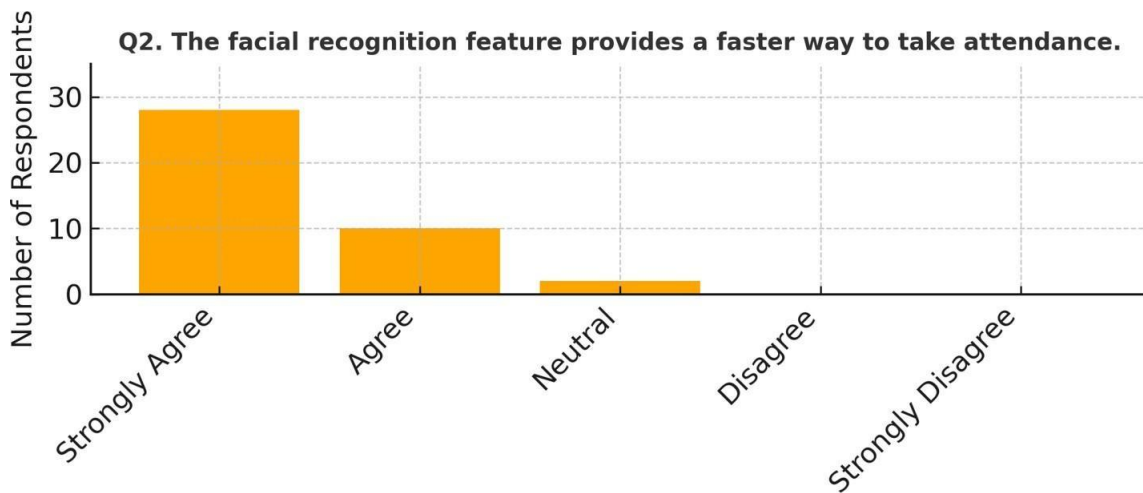


Figure 54.

Question 2 This shows that 70% strongly agreed and 25% agreed that the facial recognition feature provides a faster way to take attendance, while 5% were neutral.

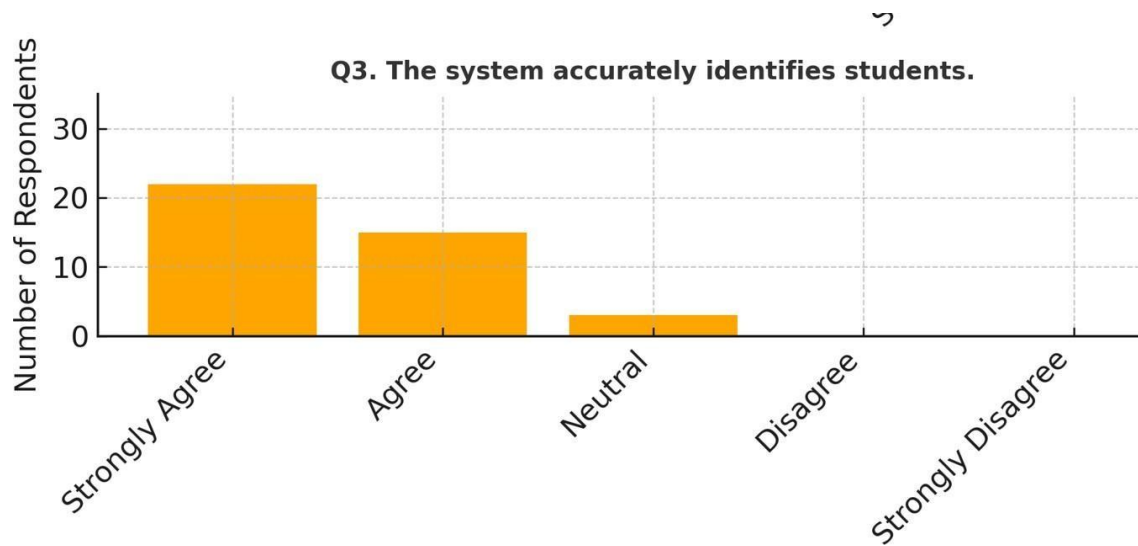


Figure 55.

Question 3: This shows that 55% strongly agreed and 37.5% agreed that the system accurately identifies students, while 7.5% were neutral.

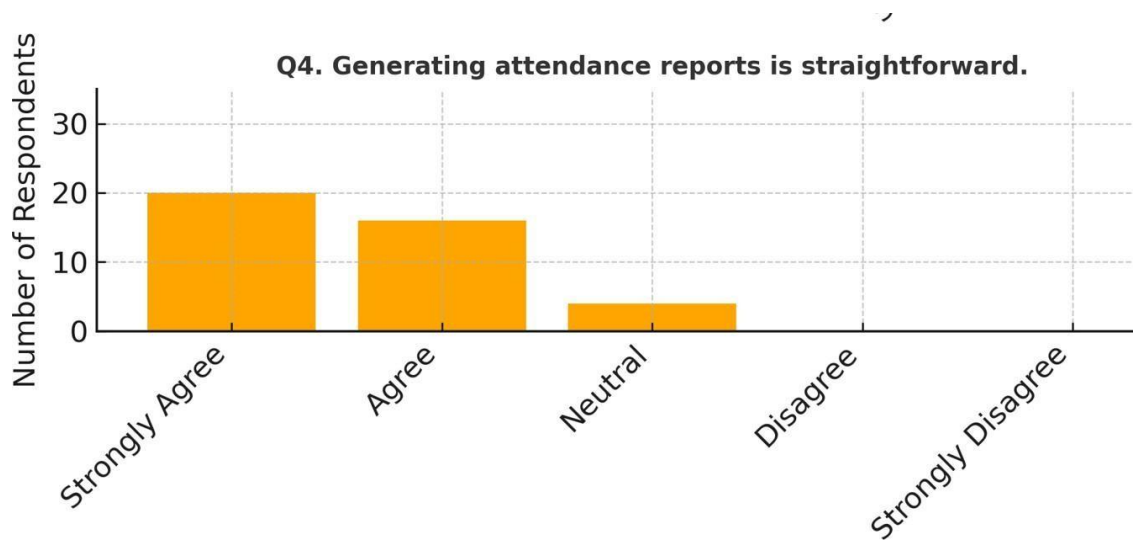


Figure 56.

Question 4: This shows that 50% strongly agreed and 40% agreed that generating attendance reports is straightforward, while 10% were neutral.

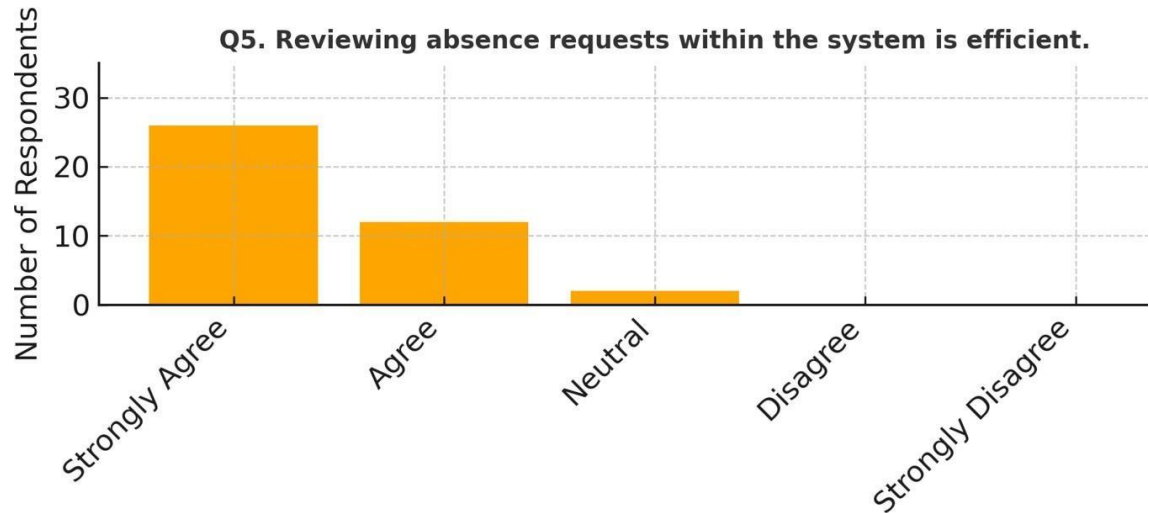


Figure 57.

Question 5: This shows that 65% strongly agreed and 30% agreed that reviewing absence requests within the system is efficient, while 5% were neutral.

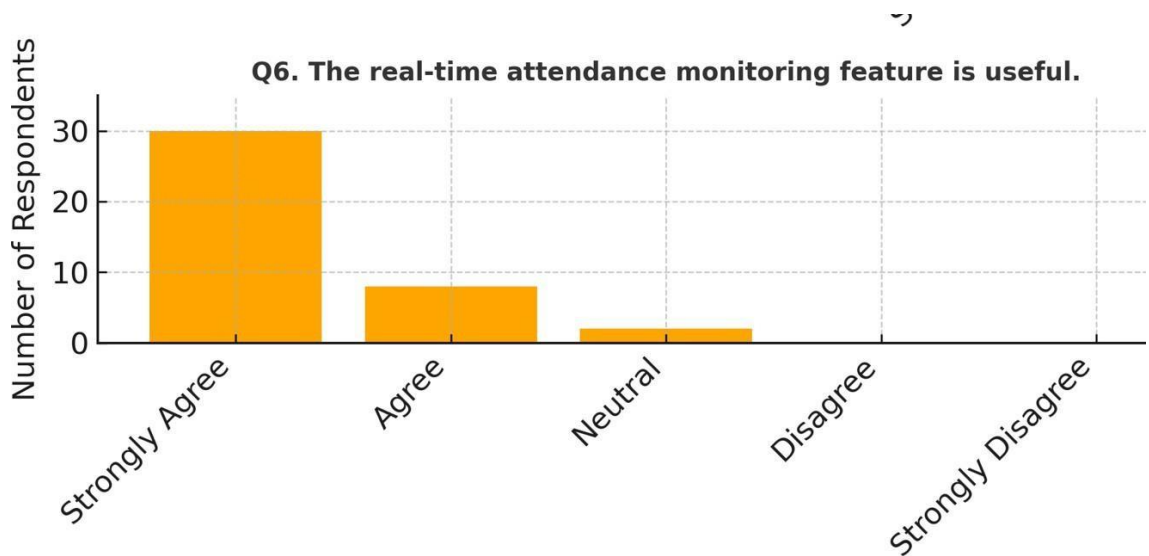


Figure 58.

Question 6: This shows that 75% strongly agreed and 20% agreed that the real-time attendance monitoring feature is useful, while 5% were neutral.

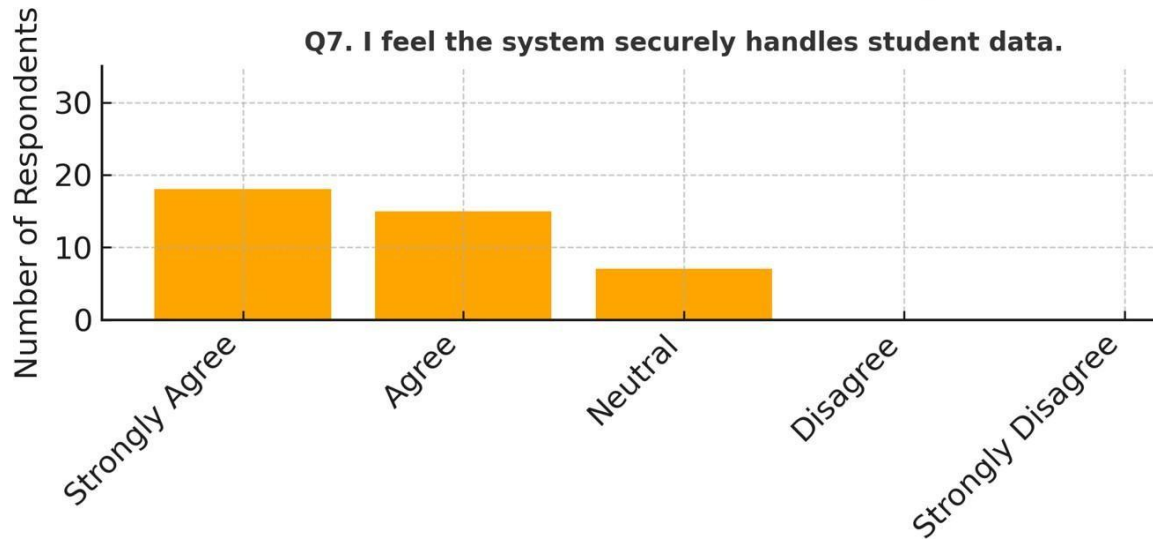


Figure 59.

Question 7: This shows that 45% strongly agreed and 37.5% agreed that the system securely handles student data, while 17.5% were neutral.

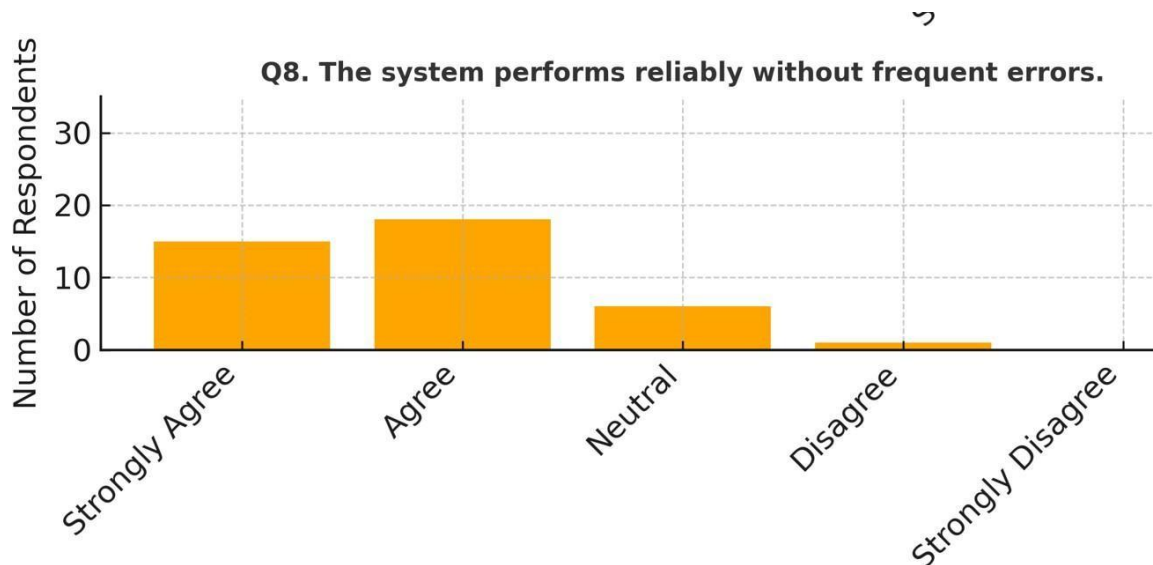


Figure 60.

Question 8: This shows that 37.5% strongly agreed and 45% agreed that the system performs reliably without frequent errors, while 15% were neutral and 2.5% disagreed.

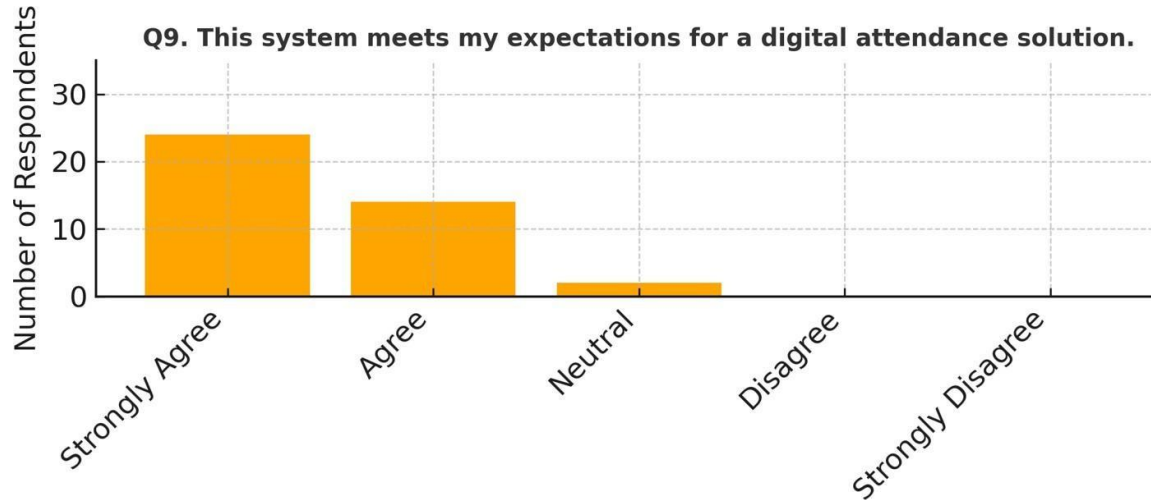


Figure 61.

Question 9: This shows that 60% strongly agreed and 35% agreed that the system meets their expectations for a digital attendance solution, while 5% were neutral.

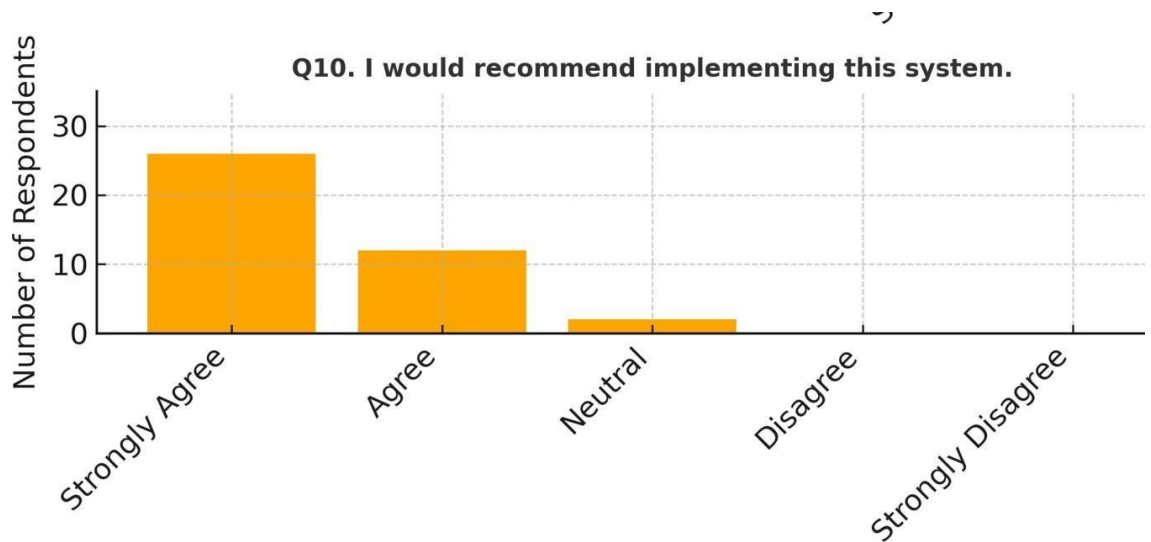


Figure 62.

Question 10: This shows that 65% strongly agreed and 30% agreed that they would recommend implementing the system, while 5% were neutral.

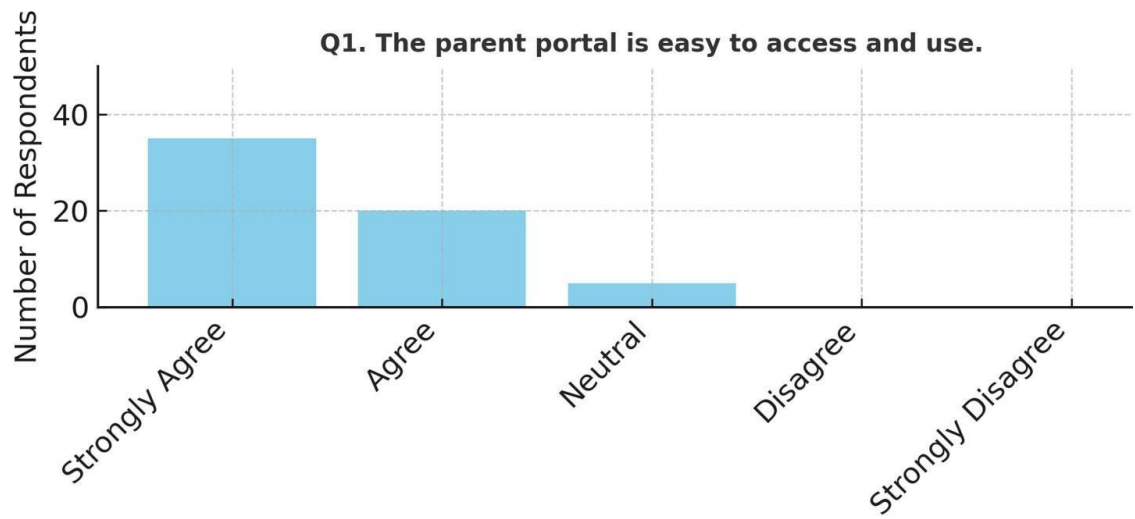


Figure 63

Question 1: This shows that 58.3% strongly agreed and 33.3% agreed that the parent portal is easy to access and use, while 8.3% were neutral.

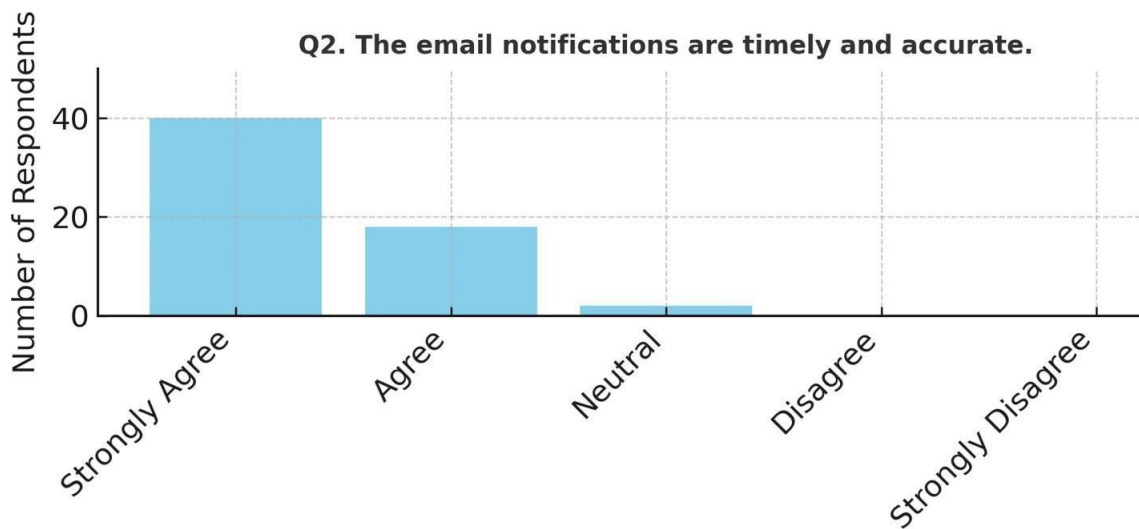


Figure 64.

Question 2: This shows that 66.7% strongly agreed and 30% agreed that the email notifications are timely and accurate, while 3.3% were neutral.

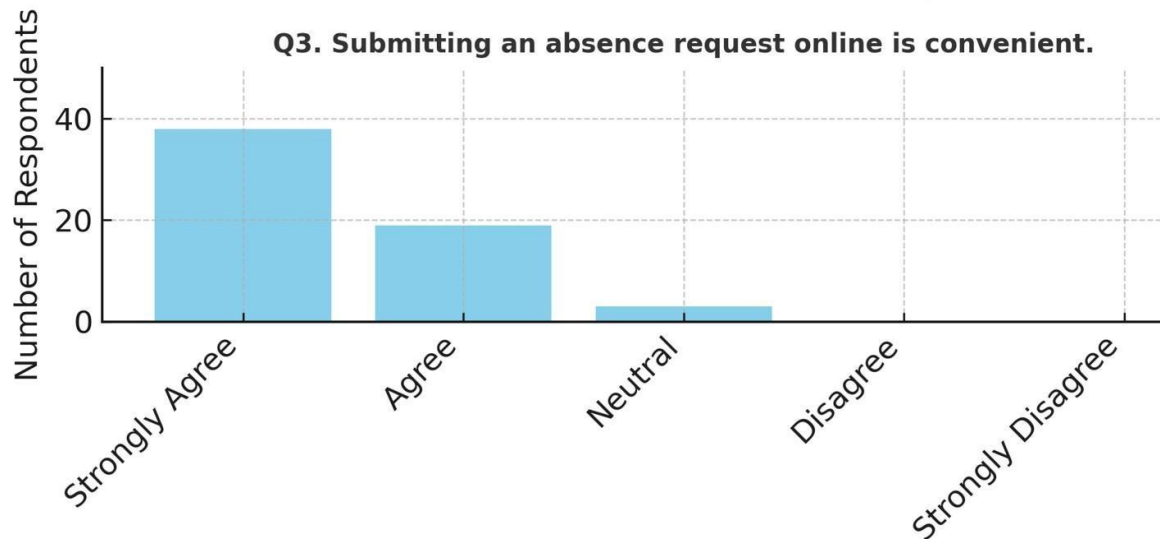


Figure 65.

Question 3: This shows that 63.3% strongly agreed and 31.7% agreed that submitting an absence request online is convenient, while 5% were neutral.

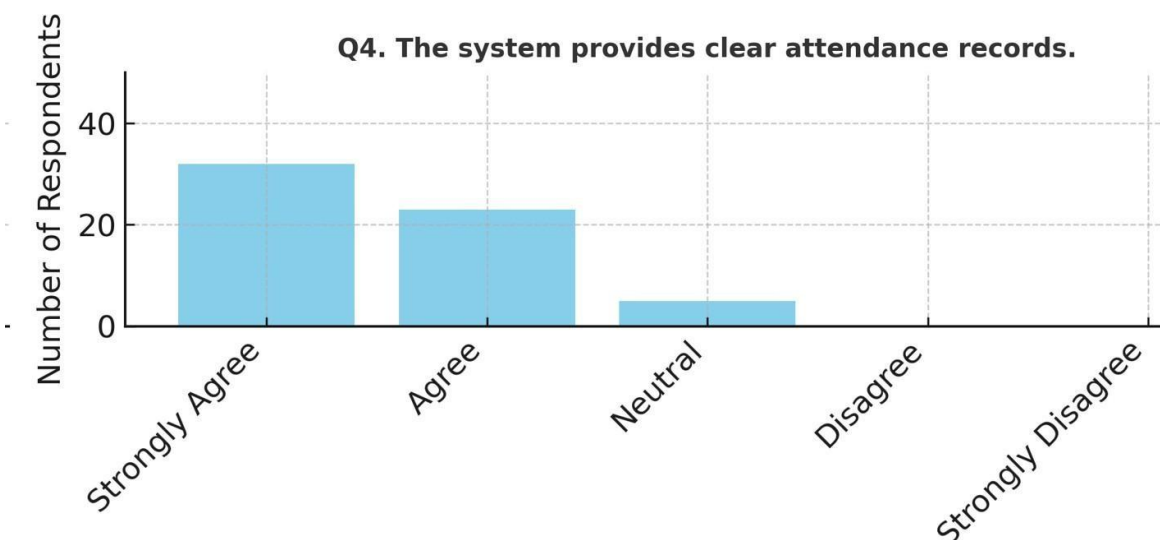


Figure 66.

Question 4: This shows that 53.3% strongly agreed and 38.3% agreed that the system provides clear attendance records, while 8.3% were neutral.

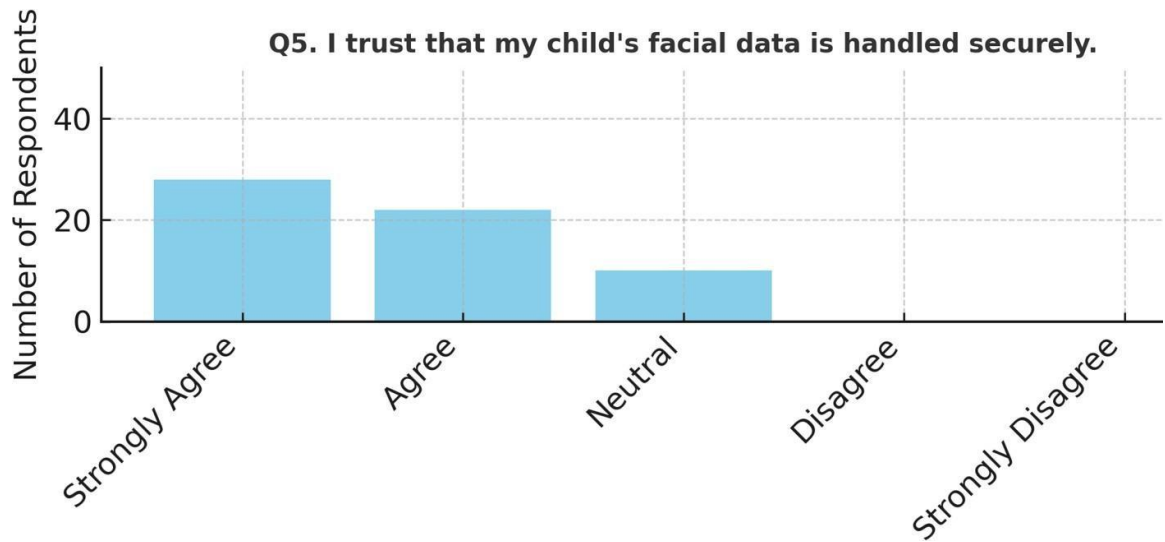


Figure 67.

Question 5: This shows that 46.7% strongly agreed and 36.7% agreed that they trust the system to handle their child’s facial data securely, while 16.7% were neutral.

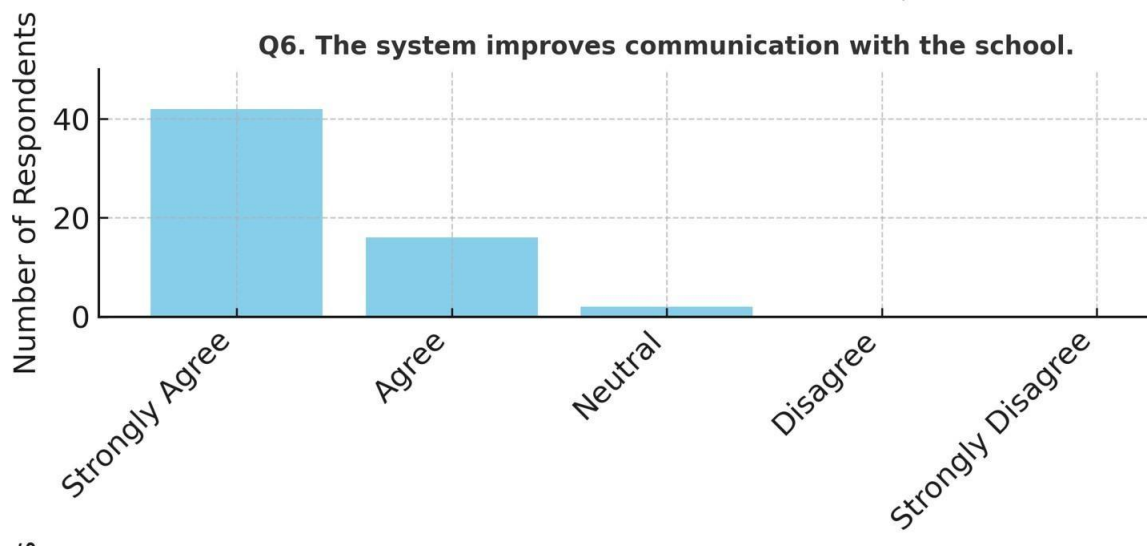


Figure 68.

Question 6: This shows that 70% strongly agreed and 26.7% agreed that the system improves communication with the school, while 3.3% were neutral.

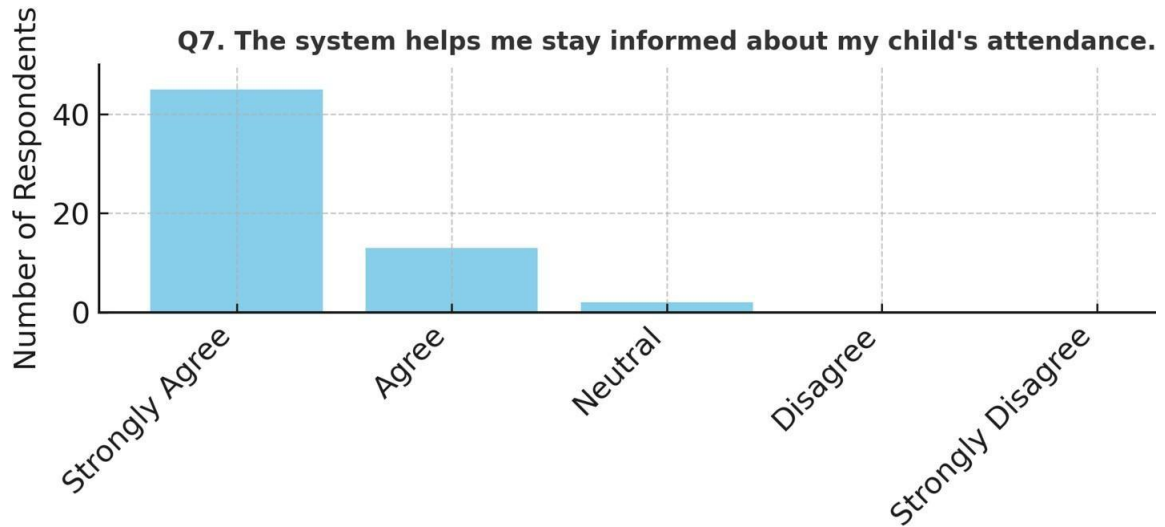


Figure 69.

Question 7: This shows that 75% strongly agreed and 21.7% agreed that the system helps them stay informed about their child's attendance, while 3.3% were neutral.

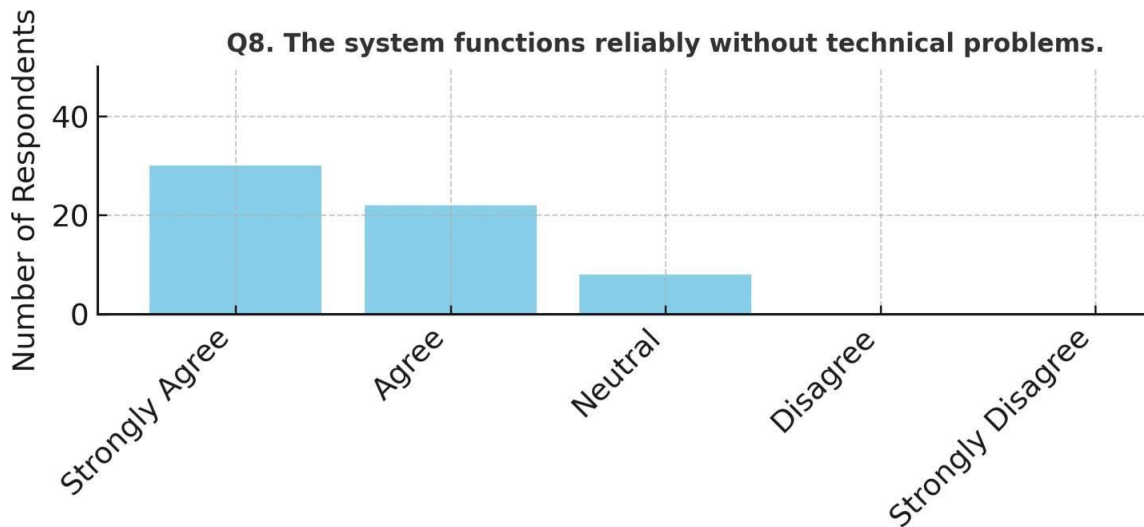


Figure 70.

Question 8: This shows that 50% strongly agreed and 36.7% agreed that the system functions reliably without technical problems, while 13.3% were neutral.

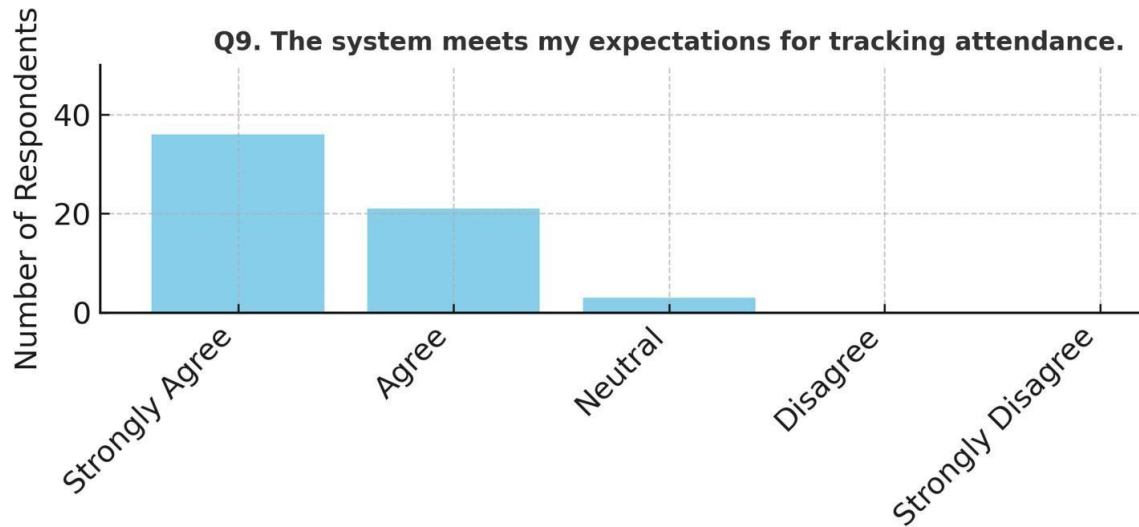


Figure 71.

Question 9: This shows that 60% strongly agreed and 35% agreed that the system meets their expectations for tracking attendance, while 5% were neutral.

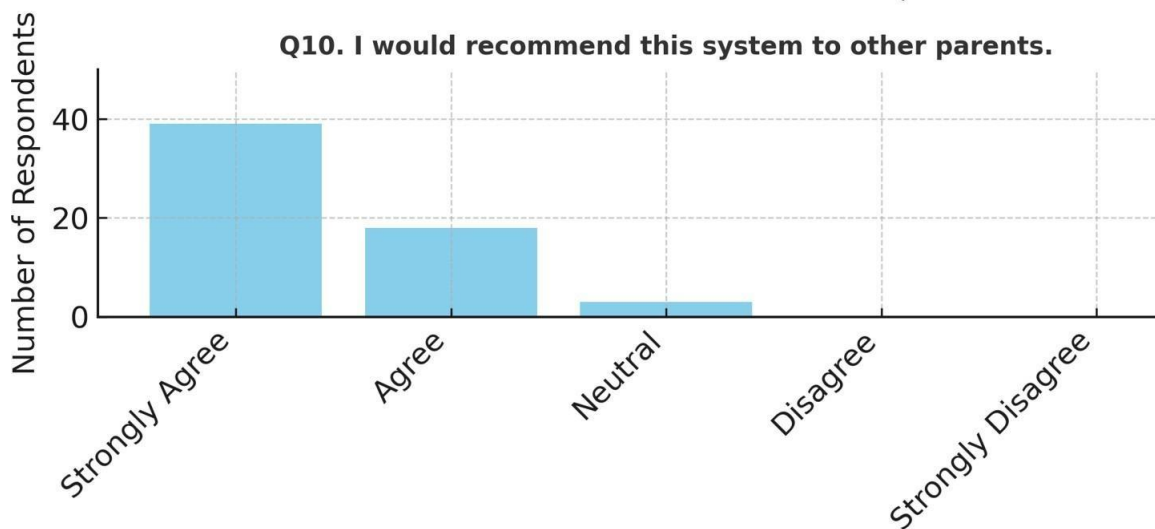


Figure 72.

Question 10: This shows that 65% strongly agreed and 30% agreed that they would recommend this system to other parents, while 5% were neutral.

CONCLUSION

In conclusion, the development and implementation of the Web-Based Attendance System with Facial Recognition, Notifications, and Data Report for Systems Plus Computer College Caloocan has proven to be a highly effective and innovative solution for modernizing student attendance management. By integrating facial recognition technology, real-time email notifications, and comprehensive data reporting, the system successfully transformed a traditionally manual, time-consuming, and error-prone process into an automated, secure, and efficient one.

Results from the pre-implementation survey revealed strong anticipation and support from both teachers and parents, with mean scores predominantly in the "Agree" to "Strongly Agree" range across all indicators. The post-implementation feedback confirmed the system's success, with users reporting high levels of satisfaction. Key features such as the real-time email notifications (Parent Mean = 4.63) and the efficiency of the facial recognition check-in (Teacher Mean = 4.65) were rated most highly, validating the system's core objectives of improving speed and communication. Teachers strongly agreed that the system saved valuable instructional time and provided useful monitoring tools, while parents strongly agreed that it kept them better informed and improved their connection with the school. The system's design and technical performance, built on the robust MERN stack with client-side face-api.js, were also positively received. The interfaces for both teachers and parents were found to be intuitive and easy to navigate. These findings demonstrate that a well-designed, automated system can significantly reduce administrative burdens, enhance the accuracy of records, and foster a more transparent and collaborative environment between schools and families.

Overall, the system successfully achieved its core objectives of creating a contactless,

accurate, and efficient attendance-tracking process that provides real-time insights for teachers and immediate peace of mind for parents. The overwhelmingly positive feedback from all user groups validates its effectiveness as a mission-critical tool for modern educational institutions. Future enhancements may include the development of a dedicated mobile application for increased accessibility and the integration of predictive analytics to identify and address student absenteeism trends proactively.

By leveraging cutting-edge web technologies, this attendance system contributes meaningfully to the school's digital transformation and demonstrates how strategic automation can create a more secure, efficient, and connected academic community.

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Nordin, N., & Mohd Fauzi, N. H. (2020). A Web-Based Mobile Attendance System with Facial Recognition Feature. *International Journal of Interactive Mobile Technologies (iJIM)*, 14(05), pp. 193–202. <https://doi.org/10.3991/ijim.v14i05.13311>

APPENDICES

Transcript of Interview

INTERVIEWER: Researchers: RJ, Jason, Raf and Jenesis

Principal: Dr. Chona

Researchers: Good Afternoon Maam Chona kami po ay from STI College Caloocan, napili po namin yung school nyo as a client po namin sa ginagawa namin na capstone po which is Web-based Attendance System po, sa Attendance System po is target namin yung elementary students nyo po from Grade 1-6 then dun po sa attendance system na yon di na po manual ang mangyayari katulad po nung example is tinatawag nyo pa po or roll calling kung tawagin which is kung nasa 40 mahigit po ang students nyo aabutin ng atleast 10-15mins yon bago matapos yung pag aattendance.

Principal: ay maganda yan syempre katulad nga ng sinabi mo ayun talaga ang gamit namin na attendance which yung traditional.

Researchers: aside from that po maam yung features pa ng system is after maattendance si student massent na thru email or SMS yung message sa parent na nakapag attendance na yung anak nila kase diba may mga instances na yung student e pinapahatid lang sa trike then meron din is yung kunwari nasa work si parent yung yaya or kasambahay yung naghahatid mga ganon po.

Principal: ay oo dito ganon nga ang nangyayari tyaka maganda rin yang naisip nyo na may massend sa magulang katulad nyan email sa SMS okay din sya pero baka naman mahal yan, kase isipin nyo kung araw araw may pasok edi magkano yun sa system e tapos ilan yung mga students natin from grade 1-6 diba.

Researchers: ayun nga po ang isa sa mga concern namin po si SMS naman po is parang alternative lang po lalo sa mga parents na di maalam sa mga internet internet ganon

Principal: sa amin naman dito sa SPCC mostly mga nanay sa grade school hands on pa sila kagaya ng mga announcements ganyan sinesend namin sa gc kase may mga gc gc kame dito ng mga sections, dun sinend then sasabihin ng teachers na may iniemail kami ganon kung may announcements or kung ano.

Researchers: ah so mostly po yung mga katulad ng mga announcements sinisend nyo po sa email? ah okay po so possible po ba na iimplement namin sa system namin is email lang?

Principal: ppwede naman siguro kase ayun yung pinaka ano namin pagdating sa pag inform pero mostly kapag kunwari message lang sa gc nalang nag uusap kung may concern ganyan.

Researchers: ah kung ganon po edi po pala email nalang ang gamitin namin ang concern po kase namin is yung pag ccommunicate kase with school to parent

Principal: oo ppwede naman yan email nalang ang gamitin nyo

Researchers: okay lang po ba maam humingi kami ng signature nyo katunayan na nag aagree kayo sa pag iimplement ng email nalang ang gamitin medium sa pag send ng attendance ng student

Principal: Oo ppwede naman

Researchers: Sige po maam gawa po kami ng letter then pagbalik namin po is papirmahan namin sainyo, meron pa palang features yung attendance aside po sa napag usapan namin meron din po kaming Attendance Records po na kung saan si teacher pwede po sya mag generate ng excel type na document para po sa mga students natin hindi na sya manually irrecord.

Principal: ay oo maganda yan para automatic na no hindi na manually irerecord sa excel yung attendance per section

Researchers: yes po maam kase po dinigitalized po kasi natin para hindi na po mahirapan

mag record ng mga teachers natin.

Researchers: aside po dun sa nabanggit meron di po sa side ni parent na absence request, dito naman po is kapag yung anak po nila is for example nagkasakit that day hindi nakapasok ppwede po sila mag send ng medical certificate or any katibayan kung bakit absent si student.

Principal: maganda din yan, alam nyo kasi ako teacher din sa grade 6 nangyayari kase kapag absent ang bata nagkasakit ganon hindi sila nagbibigay ng med cert or kahit letter baka siguro tinatamad lang ganon, kung may system kayo pwede sana natin yon makita kung papano sya nagwork

Researchers: ay oo nga po e kaso under work in progress pa po yung system namin kase last time po kase busy po kayo sabi ng guard kaya di kami makapasok.

Principal: ay oo pala andami kase ginagawa sa school puro meeting katulad non dumating si Along yung Mayor natin nagkaron ng meeting dyan sa court kaya hindi muna ako tumanggap ng mga bisita outside.

Researchers: baka po next week po maam meron na po kami maipakita sa inyong system or next visit namin kelan po ba kayo available maam or kahit po sa weekdays ano yung pinaka wala po kayong ginagawa

Principal: ang halos wala ako ginawa is friday 1-4pm ganyan kase 5pm uwian ko na e mga ganong araw lang kase kapag Monday to Thurs may tinuturuan ako then after non may minsan nagkakabiglaang meeting kaya di ako makaacomodate ng mga bisita katulad nung isa sa STI din yon ata yung sakanila is parang sa science ba yon

Researchers: oo ata po **Principal:** pumunta sila kaso busy naman ako kaya pinabalik ko nalang sila

Researchers: sige po maam ayun palang naman po yung mga ano namin baka by next week

or next next week bumalik po kami dito friday po ano

Principal: oo basta friday ang pinakafree ako basta hapon na ha

Researchers: sige po maam balik nalang po kame kasama po yung system kapag umokay napo, Thank po maam!

Principal: sige sige ingat kayo ha

APPENDIX A. RESOURCE PERSONS



Name: Dr. Chona B. Arce

Position: Principal of Systems Plus Computer College Caloocan

Address: 141, 143 10th Ave, Grace Park East,
Caloocan, 1400 Metro Manila



APPENDIX B. RELEVANT SOURCE CODE

APPENDIX C. EVALUATION TOOL/TEST DOCUMENTS

Sample Pre-Survey Questionnaire (Teacher)

This survey is part of our Capstone Project for the Bachelor of Science in Information Technology. The purpose of this project is to develop a Web-Based Attendance System with Facial Recognition, Notifications, and Data Report for Grade School Students at Systems Plus Computer College Caloocan. We are gathering feedback on various aspects of the proposed system to improve its functionality, reliability, and value for the school community

QUESTIONS	Strongly Agree (5)	Agree (4)	Neutral (3)	Disagree (2)	Strongly Disagree (1)
1. The current manual attendance process takes significant time from instructional activities.					
2. Manual attendance records are often inaccurate or incomplete.					
3. Tracking student punctuality is challenging with current methods.					
4. I would benefit from an automated system that records attendance instantly.					
5. Receiving immediate notifications when students are marked present would help my classroom management.					
6. I am comfortable with using facial recognition technology for student identification.					
7. Having a digital system to generate attendance reports would save me time.					
8. I would use a feature to review and approve student absence requests online.					
9. I believe an automated attendance system would improve communication with parents.					
10. Overall, I think a digital attendance system would make my work more efficient.					

Sample Pre-Survey Questionnaire (Parent)

This survey is part of our Capstone Project for the Bachelor of Science in Information Technology. The purpose of this project is to develop a Web-Based Attendance System with Facial Recognition, Notifications, and Data Report for Grade School Students at Systems Plus Computer College Caloocan. We are gathering feedback on various aspects of the proposed system to improve its functionality, reliability, and value for the school community.

QUESTIONS	Strongly Agree (5)	Agree (4)	Neutral (3)	Disagree (2)	Strongly Disagree (1)
1. I would like to receive immediate notification when my child arrives at school.					
2. Currently, I often don't know if my child arrived at school on time.					
3. I am comfortable with the school using facial recognition to track my child's attendance.					
4. I would use an online portal to view my child's attendance history.					
5. I would prefer submitting absence requests online rather than sending paper notes.					
6. Immediate attendance notifications would give me peace of mind about my child's safety.					
7. I trust the school to handle my child's data securely and privately.					
8. Better attendance tracking would help me monitor my child's school habits.					
9. I believe digital attendance systems improve school-home communication.					
10. Overall, I would support implementing an automated attendance system.					

Sample Post-Survey Questionnaire (Teachers)

This survey is part of our Capstone Project for the Bachelor of Science in Information Technology. The purpose of this project is to develop a Web-Based Attendance System with Facial Recognition, Notifications, and Data Report for Grade School Students at Systems Plus Computer College Caloocan. We are gathering feedback on various aspects of the proposed system to improve its functionality, reliability, and value for the school community.

QUESTIONS	Strongly Agree (5)	Agree (4)	Neutral (3)	Disagree (2)	Strongly Disagree (1)
1. The system's dashboard is easy to navigate.					
2. The facial recognition feature works quickly and accurately.					
3. The process for generating attendance reports is straightforward.					
4. The system makes recording attendance faster than manual roll call.					
5. The interface for reviewing student absence requests is clear.					
6. The system helps me monitor student punctuality more effectively.					
7. I feel the system protects student data privacy.					
8. The system runs consistently without errors.					
9. This automated system meets my expectations for attendance tracking.					
10. I would recommend this system to other teachers.					

Sample Post-Survey Questionnaire (Parent)

This survey is part of our Capstone Project for the Bachelor of Science in Information Technology. The purpose of this project is to develop a Web-Based Attendance System with Facial Recognition, Notifications, and Data Report for Grade School Students at Systems Plus Computer College Caloocan. We are gathering feedback on various aspects of the proposed system to improve its functionality, reliability, and value for the school community.

QUESTIONS	Strongly Agree (5)	Agree (4)	Neutral (3)	Disagree (2)	Strongly Disagree (1)
1. The parent portal is easy to access and use.					
2. The email notifications provide timely updates on my child's attendance.					
3. Submitting an absence request through the system is convenient.					
4. The system provides clear and understandable attendance records.					
5. I trust that my child's facial data is handled securely.					
6. The system improves communication between me and the school.					
7. The system helps me stay informed about my child's school attendance.					
8. The system functions reliably without technical problems.					
9. The system meets my expectations for tracking my child's attendance.					
10. I would recommend this system to other parents.					

APPENDIX F. PERSONAL TECHICAL VITAE

Curriculum Vitae of
Salazar, Jason R.
Block 27 Lot 59 PH1,
Golden Hills Marilao, Bulacan
 salazajasonr0159@gmail.com
Cell. No. +63 9665575694



EDUCATIONAL BACKGROUND

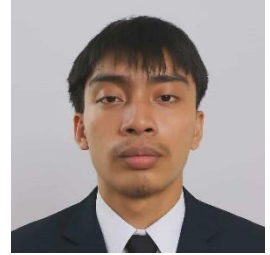
Level	Inclusive Dates	Name of school/ Institution
Tertiary	2020-Present	STI College Caloocan
Senior High School	2018-2020	STI College Caloocan
High School	2014-2018	Caloocan High School
Elementary	2008-2014	Caloocan Elementary School

SKILLS	Level of Competency	Date Acquired
Video Editor	Intermediate	2019
Computer Literacy	Advanced	2018
Interpersonal Skills	Advanced	2018
Critical Thinking	Advanced	2017

TRAININGS, SEMINARS, OR WORKSHOPS ATTENDED

Inclusive Dates	Title of Training, Seminar, or Workshop
June 2023	SAP University Alliances
June 2024	Java Fundamentals

Curriculum Vitae of
Jenesis B. Joaquin
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(+63) 9269117810
jenesisgracejoaquin9@gmail.com



CAREER OBJECTIVE

As an accomplished IT professional with extensive experience in cloud computing, cybersecurity, and data analytics, I, **Jenesis B. Joaquin**, am dedicated to leveraging my technical expertise to drive innovation, enhance operational efficiency, and support organizational growth. My goal is to secure a challenging position where I can apply my skills in strategic IT planning, data-driven decision-making, and security management to develop high-quality solutions, optimize system performance, and contribute to the achievement of business objectives.

CORE QUALIFICATIONS

- Proven expertise in **Cloud Computing** and **Infrastructure Management**
- Strong background in **IT Strategy**, **Project Management**, and **Cybersecurity**
- Skilled in **Database Administration** and **Data Analytics**
- Effective **Communicator** with excellent **Stakeholder Engagement** abilities

TECHNICAL SKILLS & EXPERIENCE

- Proficient in **Java Programming** and **Web Design**
- Experienced in **SQL Database Operations** and **PHP Database Organization**
- Skilled in **Software Installation**, **Configuration**, and **Troubleshooting**
- Experienced in **Hardware Diagnostics** and **Maintenance**
- Strong **Analytical** and **Problem-Solving** skills
- Excellent **Written** and **Verbal Communication**

Curriculum Vitae of
Arjay J. Arcangel
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CAREER OBJECTIVE

As an aspiring IT professional, I aim to pursue a position as IT Service Desk personnel where I can apply my technical knowledge and problem-solving skills. I am eager to learn and grow in the field of information technology while providing quality support and assistance to users and maintaining efficient IT operations.

CORE QUALIFICATIONS

- Basic understanding of IT support and troubleshooting
- Knowledgeable in computer hardware setup and maintenance
- Familiar with software installation and configuration
- Good communication and teamwork skills
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- Willingness to learn and adapt to new technologies

EXPERIENCE AND SKILLS

- Experienced in using Visual Studio Code for basic programming and web development
- Able to create simple games and small-scale projects
- Skilled in HTML website creation and design
- Basic knowledge of Java programming
- Proficient in hardware troubleshooting and basic computer repairs

Curriculum Vitae of
John Rafael S. Maulion
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CAREER OBJECTIVE

As an experienced IT professional with a strong background in cloud computing, cybersecurity, and data analytics, I, John Rafael Maulion, am committed to leveraging my expertise to drive technological innovation and operational efficiency. My objective is to secure a challenging position where I can utilize my skills in strategic IT planning, data-driven decision-making, and security management to contribute to organizational growth and success. I aim to apply my technical proficiency and problem-solving abilities to deliver high-quality solutions, enhance system performance, and support the achievement of business objectives.

CORE QUALIFICATIONS

- Expertise in Cloud Computing and Infrastructure Management
- Experienced in IT Strategy and Project Management
- Effective Communication and Stakeholder Engagement
- Background as an administrator of database operations

EXPERIENCE AND SKILLS

- Excellent Communication Skills
- Basic PHP Database Organizer
- Skilled Java Programming
- WEB Designing
- Skilled SQL Database Operation
- Computer Software Troubleshooting
- Hardware Troubleshoot